TEL AVIV UNIVERSITY
The Iby and Aladar Fleischman Faculty of Engineering
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A thesis submitted toward the degree of
Master of Science in Electrical and Electronic Engineering
In Tel Aviv University

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This research work was carried out at Tel-Aviv University
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Abstract

A relay carries traffic between a source and a destination that have no direct path between them. This thesis asks a single question: when is a *learned* relay worth using in place of a classical one, and what sets the boundary between them?

The comparison is run on one configuration, fixed in advance — a single-relay SISO link with i.i.d. Rayleigh fast fading on both hops, complex baseband, Gray-coded QPSK, uncoded transmission — leaving the relay function as the only variable. Eight strategies are evaluated, from amplify-and-forward (AF) and symbol-wise decode-and-forward (DF) to generative and sequence models, as Monte Carlo estimates with 95% confidence intervals.

On a matched, memoryless channel the classical relay wins. Learned relays beat AF at low SNR, but from 6 dB upward DF matches or exceeds every learned relay at no parameter cost. Capacity is not the binding constraint: a size sweep replicated over three initializations locates the smallest relay that costs nothing measurable, and *channel memory* rather than parameter count sets it. Four parameters suffice on the memoryless channel, while the evaluated three-tap channels need 73 to 145 and a window spanning the interference. Error does not rise beyond the spread across initializations as capacity grows.

The picture inverts once the channel is unknown. Against an unmodeled three-tap ISI filter both evaluated memoryless classical relays, AF and zero-threshold DF, fail outright, DF's error rate rising as transmit power grows, while a 170-parameter windowed network restores reliable relaying to within 1–1.5 dB of a Viterbi detector handed the exact taps. Two measured boundaries locate the crossover. The first is the reliability of the channel estimate: at a 10 dB operating point, pilot-aided classical detection stays ahead at twenty pilots and loses by ten. The second is arithmetic: sequence detection costs $2M^L$ multiply-accumulates per symbol against the fixed relay's $2WH + 4H$, the two equal near three and a half taps for the architecture deployed here and $158\times$ apart at seven.

The learned relay is therefore never better than a classical receiver that has been given the right model. Its value is that it still works when no such model, or no reliable estimate of one, is available, and that its cost does not grow with channel memory. Both statements are bounded to the classical pipelines implemented here and to the impairment family the relay was trained on.

Acknowledgments

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Contents

List of Symbols

Symbols are grouped as Latin capitals, Latin lowercase, Greek letters and other notation. Where a symbol carries more than one meaning, every meaning is listed and the context that disambiguates it is named.

Latin capitals

C	Ergodic Rayleigh (Shannon) capacity, bits per channel symbol
E_b	Energy per information bit
E_s	Average energy per transmitted symbol
G	Goodput, information bits per channel symbol; $G = \eta(1 - \text{FER})$; *also* AF amplification gain where explicitly stated
K	Constraint length of the convolutional code
L	FIR channel length (number of taps; memory order $L-1$); *also* block length in symbols where explicitly stated
L_{max}	Idealized buffer-readiness budget in the coded study, in symbols
M	Constellation order (points per constellation)
N_0	One-sided noise power spectral density
P_e	Bit error probability
$Q(\cdot)$	Gaussian Q-function
R	Code rate, k/n
S	Number of trellis states, 2^{K-1} for a rate-$1/n$ code

Latin lowercase

$f_\theta(\cdot)$	Relay processing function; the only block varied in the canonical comparison
g	Unknown channel gain in the flat-channel control
$h_1[i]$, $h_2[i]$	Hop-1 and hop-2 fading coefficients at symbol index i
i	Symbol (time) index
k	Information bits per code input group; *also* bits per symbol, $\log_2 M$
n	Coded output bits per input group
$n_1[i]$, $n_2[i]$	Additive noise on hops 1 and 2
w	Window half-width; the relay observes $2w+1$ samples
$x[i]$	Symbol transmitted by the source
$x_R[i]$	Symbol transmitted by the relay

$y_R[i]$, $y_D[i]$	Signal received at the relay and at the destination

Greek letters

$\alpha_k(s)$	BCJR forward recursion at trellis state s, step k
$\beta_k(s)$	BCJR backward recursion at trellis state s, step k
γ	Linear signal-to-noise ratio, $\gamma = 10^{\mathrm{SNR}_{\mathrm{dB}}/10} = E_s/N_0$
$\bar{\gamma}$	Average signal-to-noise ratio
$\gamma_k(s', s)$	BCJR branch transition-and-observation probability; distinct from the SNR γ above, disambiguated by its arguments
η	Spectral efficiency, $\eta = R \log_2 M$ bits per symbol; *also* learning rate in Appendix 9.2
θ	Learned relay parameters; *also* unknown carrier phase in the flat-channel control (Section 6.2.1)
σ^2	Real-noise variance in real AWGN sections; total complex-noise power where $n \sim \mathcal{CN}(0, \sigma^2)$, giving variance $\sigma^2/2$ per real axis

Other notation

$\mathcal{CN}(0, 1)$	Circularly-symmetric complex Gaussian distribution, unit variance
$\mathbb{E}[\cdot]$	Expectation
$\nVdash[\cdot]$	Indicator function
$\hat{\cdot}$	Estimate of the quantity beneath the hat

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Chapter 1

Introduction

Relay communication is a key technique in contemporary wireless systems, using intermediate nodes between a source and a destination to extend coverage, enhance reliability, and increase throughput. Classical relay strategies such as amplify-and-forward (`AF`) and decode-and-forward (`DF`) are extensively analyzed and widely used as baseline protocols; in several canonical relay-channel models, block-based `DF` is capacity-achieving or capacity-approaching under asymptotically long codewords [1, 2, 3, 4, 5]. As clarified in Remark 4.2, these results concern coded block operation and do not directly characterize the uncoded, symbol-wise setting studied in this thesis.

Recent machine-learning advances have motivated applying neural networks to physical-layer problems including detection, decoding, and end-to-end system optimization: models that can approximate analytically intractable nonlinear mappings and have been reported to offer advantages in low-SNR or non-ideal regimes.

This motivates the central question of this thesis: *under what conditions can neural network–based relays improve upon or complement classical relay strategies in two-hop relay communication?*

This thesis answers that question within a unified simulation framework in which, for the canonical comparison, the relay function is the only variable. Nine relay methods are implemented (Section 4.2), of which eight are evaluated in the canonical comparison (Section 5.2): two classical relay strategies (`AF` and `DF`) and six neural relays spanning supervised learning (`MLP`, SNR-adaptive `Hybrid` relay), generative modeling (variational autoencoder `VAE`), and sequence-based architectures (`Transformer`, `Mamba-S6`, and `Mamba-2 SSD` with structured state-space duality); a conditional GAN (`cGAN` with `WGAN-GP`) is implemented but excluded from the canonical comparison on cost grounds.

The objective is not to propose a new relay design but to answer the question above with a boundary rather than a verdict: to say where learning helps, where it does not, and why. The question divides in two, and Chapter 3 poses it that way. Asked without a budget, it is a question about accuracy and about what each detector must know, and in the evaluated comparisons its boundary is governed principally by the *reliability of the channel estimate*. Asked under a latency and computational budget, it becomes a question about what can be afforded, and its boundary is *arithmetic*. The two do not fall in the same place, and the distance between them is one of this thesis's results: Section 6.5.1 measures a channel with memory under a latency constraint, the one setting where a trellis search growing as M^L and a relay that must buffer a whole codeword are both in play, and determines there which constraint actually binds. Existing studies typically vary several things at once, a set of architectures and a set of channel models together, making it difficult to attribute a difference in error rate to model structure, parameter count, or the channel.

Method: one assumption at a time. This thesis answers the question with a monotone sequence of four layers, each relaxing a further assumption about the channel; the learned relay and the metric are held fixed throughout, while the classical baseline is strengthened at each layer to remain the toughest available opponent, and the channel, modulation, and evaluation budget are adapted as needed. Read in order, their outcomes *are* the answer to the central question; each is established in the chapter that runs the layer, and none is anticipated here.

At **layer 1** the channel is memoryless and perfectly known, which is the canonical setup of Chapter 5; this layer asks whether learning can improve on a classical rule that is already the MAP symbol decision for that channel. **Layer 2** (Chapter 6) adds channel memory while holding channel knowledge fixed, so that what is isolated is the *function class* a relay must realize, together with the cost of realizing it, with a QPSK re-check of the same question in Section 6.2.2. **Layer 3** degrades the channel estimate to a finite pilot budget at one measured SNR, which turns the classical receiver's advantage into a quantity with a price rather than a property. **Layer 4** redraws a realization from the same impairment family for every block, removing per-block CSI while retaining prior knowledge of that family through training, and sets the learned relay against blind classical methods.

One methodological finding surfaces along the way and is reported where it was measured (Section 7.6): what looked like one architecture's instability across random initializations turns out to be a single silent optimization failure, an order of magnitude off in training loss yet invisible in the validation accuracy the training loop already monitors.

The shape of that sequence is the contribution. What is put under pressure from layer to layer is not the classical algorithms themselves but their *preconditions*: each classical baseline is used only under its stated model assumptions, and matched MLSE is optimal for sequence error rather than necessarily for BER. The question each layer asks is therefore which method's requirements remain met. Hypotheses H1–H4 are settled at layer 1, and H5 at layers 2–4.

Scope of this thesis. This thesis studies one communication problem, held unchanged throughout, the classical source–relay–destination (two-hop) link, under *one canonical setup*: SISO on both hops, i.i.d. Rayleigh fading, complex baseband, QPSK, uncoded BER. The only axis varied is the *relay function*. BPSK is never transmitted on the canonical channel; it is the transmitted alphabet only in the unknown-channel experiments of Chapter 6, where the one-dimensional alphabet keeps the trellis small enough for an exact Viterbi benchmark, and it otherwise appears only in the closed-form expressions those chapters quote. Two dedicated departures from it are included: a supplementary coded study (Section 5.5) that measures a block-DF relay, soft-information forwarding, and an adaptive modulation-and-coding scheme against this baseline, and the unknown-channel contribution (Chapter 6). Other departures from the canonical setup are identified as future work in Chapter 7:

Table 1.1: Scope of the thesis: the canonical setup (QPSK on Rayleigh), its supplementary coded-study departure, and the unknown-channel contribution. No other topology, relay protocol, or performance metric is evaluated in the main line of this work; all such variations are deferred to future work

Axis	**Canonical value (future work in parentheses)**
Topology	SISO on both hops *(future work: MIMO second hop)*
Channel	i.i.d. Rayleigh fast fading (canonical), applied identically on both hops; AWGN solely as the noise term of the canonical model, the simulator being calibrated instead against the closed-form two-hop DF composition on the canonical channel itself (Table 5.2); unknown, mismatched, and time-varying channels are studied in the dedicated contribution (Chapter 6) *(future work: Rician and other channel families)*
Relay Function *(the studied axis)*	*Classical:* AF, DF. *Learned:* MLP, Hybrid, VAE, Transformer, Mamba-S6, Mamba-2 SSD; cGAN is implemented but excluded from the canonical comparison on cost grounds
Modulation	QPSK is paired with the canonical Rayleigh channel throughout (Chapter 5) and re-checked under the unknown-ISI channel of Chapter 6, whose relay comparisons otherwise use BPSK; 16-QAM additionally appears, not as a relay-architecture question, as a rung of the coded modulation-and-coding study (Section 5.5) *(future work: 16-QAM and denser as a relay-architecture question, 16-PSK, and beyond)*
Performance Metric	Uncoded BER vs. SNR, Monte-Carlo averaged

How to read the scope of any statement in this thesis. Three studies are reported and they do not share a channel model, so a claim means nothing without knowing which one it belongs to. The rule is applied throughout and is not restated per chapter. Statements about channel memory, intersymbol interference, sequence detection, Viterbi MLSE, pilot-based estimation or blind equalization belong to the unknown-channel study of Chapter 6 alone. Statements about channel coding, block-DF, modulation-and-coding adaptation or latency budgets belong to the supplementary coded study of Section 5.5 alone. Everything else concerns the canonical setup, where the channel is memoryless and perfectly known, the noise white, and the transmission uncoded. One term is ambiguous across two of them: the coded study's soft-decision Viterbi decoding of a convolutional code is error-correction decoding, a different use of the algorithm from the channel-memory sequence detection the unknown-channel chapter means by Viterbi MLSE.

Table 1.2 collects the assumptions that bound every result reported here, together with what each one does and does not license. It is the single reference for scope; individual chapters state a qualification only where it is needed to read a specific number.

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Table 1.2: Assumptions as implemented, the claims they support, and the generalizations they do not

Dimension	As implemented	Does not support
Channel	Two-hop SISO, half-duplex, i.i.d. Rayleigh fast fading, complex baseband, Gray-coded QPSK, uncoded, with the relay function as the only varied axis. Chapters 5 and 6 depart from this deliberately and say so.	Multi-relay, full-duplex, MIMO, or constellations denser than 16-QAM. No direct source–destination path is modeled, so nothing here bears on cooperative diversity.
Relay channel knowledge	No per-block channel estimate and no online adaptation. The relay is trained on the impairment family it is later tested on; layers 3 and 4 redraw realizations from that family.	Family-agnostic or model-agnostic operation. The supported property is *realization-agnostic* inference amortized over a trained family; a structurally different family is untested.
Comparator knowledge	AF and symbol-wise DF are memoryless; Viterbi MLSE is run with genie taps or with an LS estimate from a stated pilot budget; CMA is blind. BCJR/APP is measured at QPSK only.	A claim against classical receiver design in general. The comparators are the specific pipelines implemented here; reduced-state sequence estimation, decision feedback and hybrid model-based detectors are not evaluated.

Dimension	As implemented	Does not support
Statistical basis	95% intervals over Monte Carlo channel and noise realizations at fixed trained weights; paired Wilcoxon tests at $\alpha = 0.05$ per comparison. Training-seed replication over three initializations in Sections 5.4 and 7.2.3 only.	Joint significance across the full set of tests (no multiplicity correction), confirmatory inference (comparisons were not prespecified), equivalence from a null result, or seed-robustness for any single-run result.
Cost accounting	Multiply-accumulates per symbol at fixed (W, H, M), plus wall-clock for reference implementations (vectorized NumPy against an interpreted trellis).	An implementation-independent speed advantage, or a universal crossover in channel memory: the $L^{\star}$ figure is arithmetic at one architecture and one counting convention.
Deployment	Parameter count and float32 footprint.	Embedded or IoT suitability. Quantized accuracy, throughput, memory traffic, energy per symbol and a hardware implementation are all unmeasured.

Of the classical relay protocols surveyed in the literature review, only those evaluated in this thesis are part of the studied system alongside the neural relays above: amplify-and-forward and symbol-wise decode-and-forward on the canonical comparison, the coded study's block-DF exception (Section 5.5), and the differential detection, pilot-aided Viterbi MLSE, and blind equalization baselines of the unknown-channel study (Chapter 6). Material outside that configuration, such as the general relay channel with a direct link, is background only.

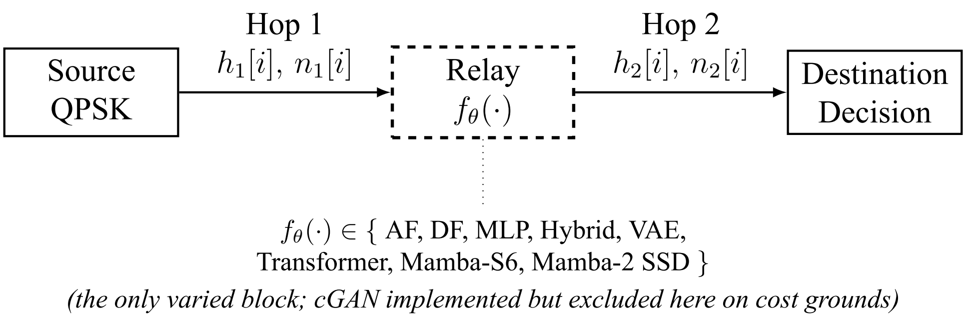


Figure 1.1: Canonical two-hop relay benchmark used throughout the thesis. The source, the channel model on both hops, and the destination are the same in every experiment of the canonical comparison; only the relay processing function $f_\theta(\cdot)$ (dashed) is varied among the eight candidate strategies compared there. The fading itself is not static: $h_1[i]$ and $h_2[i]$ are redrawn every symbol. Observed BER differences in the canonical comparison are therefore attributable to the relay strategy alone

1.1 Scope and Contributions

System model studied in this thesis. Throughout this work, a single canonical setup underlies the main line of argument: a half-duplex two-hop link, one relay, *no* direct source–destination path, SISO i.i.d. Rayleigh *fast* fading on both hops, complex baseband, Gray-coded QPSK, uncoded BER. Section 4.1 states it in full, once, and is the reference for every detail omitted here. The learned component of the system is confined to the *relay processing function* $f_\theta(\cdot)$, the block for which AF and DF are the classical alternatives; the source, the channel model, and the destination demodulator are held fixed in the canonical comparison, so observed BER differences there are attributable to the relay strategy alone; the coded and unknown-channel studies below deliberately vary the coding, channel, or destination processing as well. The simulator is calibrated on the canonical channel itself: symbol-wise DF's measured BER matches the closed-form two-hop composition closely, within 3.6% relative error at every tabulated SNR (Table 5.2; the largest gap is at 12 dB), once the QPSK constellation's E_s/N_0-to-E_b/N_0 conversion is applied. Beyond the canonical core the thesis makes two further contributions, of unequal weight, and this section states only what they are; their findings belong to the chapters that establish them. A smaller supplementary study (Sections 5.5–5.5.4) replaces the uncoded symbol-wise DF baseline with a practical finite-length coded block-DF relay, adds soft-decision counterparts of both the classical and the learned relay, and re-derives the comparison first at equal E_s/N_0, then at equal throughput, then under an adaptive code rate and a latency deadline. The principal experimental contribution (Chapter 6) studies learned relaying under *unknown and mismatched* channels that deliberately violate the canonical memoryless assumptions, and maps where a minimal learned relay adds value over the strongest classical baseline evaluated, and where it does not (H5). All other departures — other channel families, a MIMO second hop, learned constellations — are identified as future work in Chapter 7. Capacity-theoretic results and the cooperative-diversity order are quoted in Chapter 2 as classical *background only*; they are not the operating point evaluated here.

Figure 1.1 summarizes this benchmark. The "identical" components named there are identical as *models*, not as particular channel realizations: the fading is *fast*, so $h_1[i]$ and $h_2[i]$ are redrawn independently for every symbol. Nothing in this thesis is a static or quasi-static link.

Chapter 2

Background and Related Work

2.1 Classical Relay Strategies

The fundamental question that motivates this thesis is: can a learned relay function $f_\theta(\cdot)$ complement classical relay functions (AF scaling, DF regeneration) when their channel assumptions are mismatched or the relevant mapping is difficult to specify analytically? On the matched memoryless channel, the relevant symbol statistics are already captured by the classical MAP rule; the potential advantage of learning must therefore be established empirically in low-SNR or mismatched regimes rather than presumed.

2.1.1 Amplify-and-Forward (AF)

The AF relay amplifies the received signal by a gain factor G that normalizes the output power: (Equation 2.1)

$$G = \sqrt{\frac{P_{\text{target}}}{\mathbb{E}[|y_R|^2]}} \tag{2.1}$$

For CSI-assisted variable-gain AF, the standard dual-hop end-to-end SNR is [3, 6]: (Equation 2.2)

$$\text{SNR}_{\text{eff}}^{\text{AF}} = \frac{\text{SNR}_1 \cdot \text{SNR}_2}{\text{SNR}_1 + \text{SNR}_2 + 1} \tag{2.2}$$

This form assumes *CSI-assisted variable-gain* AF (Laneman et al.), in which the relay computes a fresh gain from its instantaneous first-hop channel state. That is different from the implementation in this thesis: its gain is a single block-adaptive scalar estimated from the empirical received power over the simulated block. It is therefore data- and noise-dependent and requires the block to be available before the gain is fixed; it is neither conventional statistical fixed-gain AF nor per-symbol CSI-assisted variable-gain AF. Equations 2.2–2.25 are included only as standard background and are not closed forms for the simulated AF results, which come entirely from Monte Carlo evaluation.

This reveals AF's fundamental limitation: the effective SNR is bottlenecked by the weaker hop even when the other has high SNR, because the relay amplifies the first hop's noise along with its signal, accumulating both at the destination.

2.1.2 Decode-and-Forward (DF)

The DF relay demodulates the received signal, recovers the transmitted bits, and re-modulates clean symbols:

$$\hat{b}_R = \text{demod}(y_R), \quad x_R = \text{mod}(\hat{b}_R) \tag{2.3}$$

The end-to-end BER for DF is: (Equation 2.4)

$$P_e^{\text{DF}} = P_{e,1}(1 - P_{e,2}) + (1 - P_{e,1}) \cdot P_{e,2} \tag{2.4}$$

where $P_{e,1}$ and $P_{e,2}$ are the BER of the first and second hops, respectively: an end-to-end bit error occurs exactly when *one* of the two hops flips the bit (two flips cancel), so this expands to $P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2}$, the same expression used in the AWGN analysis (Equation 2.27). For BPSK modulation over AWGN (real-valued noise with variance $1/\text{SNR}$), each hop's BER is $P_e = Q\left(\sqrt{\text{SNR}}\right)$. DF provides clean regeneration at high SNR but suffers from error propagation when the first-hop BER is non-negligible. As discussed in Remark 4.2, the "DF" baseline used here is the uncoded symbol-wise slicer, not practical finite-length coded block-DF; its optimality claims are therefore confined to the uncoded setting.

2.1.3 Other Classical Relay Techniques

AF and DF are two canonical relay strategies: AF linearly forwards an analog observation, whereas DF regenerates a detected symbol or decoded block. Compress-forward, estimate-forward, and compute-forward address different network models and objectives and are not intermediate points on a single AF–DF scale. The focus on AF and symbol-wise DF is deliberate but limited: they are representative relay-function baselines for the canonical benchmark, not strict performance bounds or an exhaustive account of classical relaying.

2.2 Machine Learning in Wireless Communication

Applying machine learning to physical-layer wireless communication has gained significant momentum, driven by neural networks' ability to learn complex non-linear mappings directly from data where analytical solutions are intractable or suboptimal: channel estimation [7], signal detection [8], autoencoder-based end-to-end communication [9], and resource allocation [10].

2.2.1 Theoretical Basis: Universal Approximation and Denoising

The theoretical justification for applying neural networks to relay signal processing rests on two pillars. First, **universal approximation** results [11, 12, 13] show that feedforward networks with suitable non-polynomial activations can approximate continuous functions on compact domains to arbitrary accuracy, given sufficient width. This existence result does not guarantee finite-sample learning, optimization, or generalization. For relay denoising, the target function maps noisy observations to clean transmitted symbols:

$$f^* : \mathbb{R}^{2w+1} \to [-1, 1], \quad f^*(y_{i-w}, \ldots, y_{i+w}) = \mathbb{E}[x_i \mid y_{i-w}, \ldots, y_{i+w}] \tag{2.5}$$

This conditional expectation f^* is the Bayes-optimal denoiser: it minimizes the mean squared error (MSE) over all possible estimators. For BPSK symbols corrupted by AWGN, f^* reduces to the posterior mean:

$$f^*(y) = \tanh\left(\frac{y}{\sigma^2}\right) \tag{2.6}$$

which is a smooth sigmoid approaching the hard-decision signum function as $\sigma^2 \to 0$. A single tanh unit with the right input scaling represents this posterior *exactly* for a fixed σ^2. The deployed relay, however, is trained at SNR $\in \{5, 10, 15\}$ dB and evaluated over 0–20 dB, so one set of weights must approximate a whole *family* of posteriors indexed by σ^2, and on the fading channel a mixture over random effective SNRs. That a 169-parameter network approximates that family closely enough to track the classical baselines is an empirical finding of this thesis, not a consequence of the exact single-σ^2 representation.

Second, the **bias-variance decomposition** provides a framework for understanding model complexity:

$$\mathbb{E}[(\hat{x} - x)^2] = \text{Bias}^2(\hat{x}) + \text{Var}(\hat{x}) + \sigma^2_{\text{irreducible}} \tag{2.7}$$

For the denoising problem considered here the irreducible term is the Bayes risk $\mathbb{E}[\mathrm{Var}(X \mid Y)]$, the conditional uncertainty that remains after observing Y, rather than the channel-noise variance σ^2 itself. The decomposition is used as a conceptual framework; the irreducible term should be read that way. Increasing capacity can change both bias and variance, but it does not necessarily reduce one or increase the other in modern regularized networks. Since the target f^* is a smooth low-dimensional mapping in the memoryless special case, the working hypothesis is that additional capacity may buy little. Whether that occurs is an empirical question; Section 7.2 reports only what was measured and does not infer overfitting without train–test-gap evidence.

2.2.2 Prior Work in Deep Learning for Physical-Layer Processing

Ye et al. [7] demonstrated joint DNN-based channel estimation and signal detection in OFDM systems using received pilot and data information. Their approach replaces separately implemented estimation and equalization stages and was reported to be robust under several modeled impairments; it does not eliminate pilots.

Dorner et al. [9] proposed treating the modulator, channel, and demodulator as an autoencoder trained end-to-end with a differentiable message-reconstruction objective and evaluated by error rate. This approach jointly optimizes the communication chain rather than directly minimizing the nondifferentiable BER.

In relay-assisted systems specifically, most existing work applies neural networks to *relay-selection* and *resource-allocation* rather than to the relay's own signal processing: learning-based classifiers pick the best relay from CSI, SNR or energy constraints while the relay operation itself stays conventional [14].

A line of work closer to this thesis's benchmark places the learned component *inside* an optimal detector rather than beside it: Tsai et al. [15] replace the channel-model-dependent branch likelihoods of the BCJR recursion with a neural network, and Karanov et al. [16] extend that construction to ISI channels with bursty impulsive noise, both to avoid needing full channel state information. What matters here is the condition under which the learned variant wins: Chen et al. [17] report it outperforming the conventional algorithm specifically when channel knowledge is imperfect, the same boundary this thesis draws in Chapter 6. That literature also fixes a different comparator, BCJR, the bit-error-minimizing detector, rather than the sequence-optimal MLSE this thesis benchmarks against throughout. That comparator is added at QPSK in Section 6.2.3, which finds it never worse than MLSE and not separated from it above 16 dB; it is unmeasured at BPSK (Section 7.5).

A broader perspective on machine learning's role across wireless protocol layers, including cooperative communications, is given in [18]. A separate line of work exploits the structural similarity between AF relay networks and neural networks, treating relay nodes as neurons and tuning relay gains with deep-learning optimization to exploit hardware nonlinearities [19, 20]; the gains reported there are on analog gain optimization, not symbol-level detection.

Classical AF/DF frameworks remain well understood and widely adopted [5] but rely on fixed linear forwarding or hard-decision rules. Learning the relay's symbol-level input-output mapping itself has drawn comparatively little attention, motivating the data-driven alternative studied here.

2.2.3 Neural Network Relay Processing

For relay processing specifically, a supervised learning approach trains a neural network f_θ to minimize the empirical mean-squared error: (Equation 2.8)

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N}(\hat{x}_i - x_i)^2, \quad \hat{x}_i = f_\theta(y_{i-w:i+w}) \tag{2.8}$$

where $\hat{x}_i = f_\theta(y_{i-w:i+w})$ is the network output based on a sliding window of $2w+1$ received symbols, and x_i is the clean transmitted symbol. This window-based approach provides temporal context that enables the network to exploit statistical dependencies in the noise-corrupted signal.

A well-known example of $f_\theta(\cdot)$ is the *multilayer perceptron (MLP)*, a feedforward artificial neural network composed of an input layer, one or more fully connected hidden layers with nonlinear activation functions, and an output layer, trained using backpropagation to model nonlinear input–output relationships. The MLP equation is shown in equation 2.9:

$$\mathbf{h} = \text{ReLU}(\mathbf{W}_1\mathbf{w} + \mathbf{b}_1), \quad \hat{x} = \tanh(\mathbf{W}_2\mathbf{h} + \mathbf{b}_2) \tag{2.9}$$

The sliding window formulation ($i \in [i-w : i+w]$) is retained for architectural consistency with the unknown-ISI experiments of Chapter 6. Under the canonical i.i.d. per-symbol fading model, adjacent observations carry no additional information about the current symbol, and the window is redundant there by construction. Under the canonical model (i.i.d. fast fading, white noise), adjacent symbols are statistically independent, so the equalized sample $\tilde{y}_R[i]$, formed with receiver-side CSI, is already a sufficient statistic for $x[i]$; a single-symbol estimator ($w = 0$) therefore loses nothing and a window confers no theoretical benefit. The window is therefore not motivated by temporal correlation in the canonical setting; it is retained because it is required by the unknown-channel study of Chapter 6, where a 3-tap ISI channel couples adjacent symbols and the relay must span that memory. Keeping a fixed windowed architecture across all experiments isolates the effect of the channel assumptions from the network structure.

Multi-SNR training is a key design choice: training on data generated at multiple SNR levels (5, 10, 15 dB here) yields a single estimator whose denoising function generalizes across operating conditions rather than specializing to one noise level. In decision-theoretic terms the training objective averages the loss over the sampled SNRs, so what it minimises is *Bayes* risk under the empirical SNR distribution, not the worst-case (minimax) risk over that range: nothing in the objective protects the SNR at which the network happens to do worst.

A critical question in applying neural networks to relay processing, essential for deployment on resource-limited relay nodes, is the relationship between model complexity and performance: prior work has generally assumed larger models perform better, an assumption not rigorously tested in the relay context. The bias–variance framework predicts the opposite for a low-complexity target function like relay denoising – performance should plateau or degrade beyond a modest model size – a prediction investigated empirically in this thesis.

2.3 Generative Models for Signal Processing

Generative models offer an alternative to directly learning a denoising function (the discriminative approach above): they instead learn the distribution of clean signals $p(\mathbf{x})$. Bayesian denoising provides the general probabilistic framing:

$$p(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x}) \tag{2.10}$$

The generative approach is theoretically appealing because it separates the modeling of the signal prior from the noise model, potentially enabling better generalization to unseen noise conditions. The VAE implemented in this thesis does not compute that posterior explicitly. It is a latent-variable generative architecture trained supervised, with noisy observations at the encoder and the clean symbol as the reconstruction target, under a KL regulariser; Eq. (2.10) frames the problem it addresses rather than describing the operation it performs.

2.3.1 Variational Autoencoders

Variational Autoencoders (VAEs) [21] learn a latent representation by maximizing the evidence lower bound (ELBO). The generative model posits that data $\mathbf{x}$ is generated from a latent variable $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ through a decoder $p_\theta(\mathbf{x}|\mathbf{z})$. Since the true posterior $p(\mathbf{z}|\mathbf{x})$ is intractable, an encoder network $q_\phi(\mathbf{z}|\mathbf{x})$ approximates it. The ELBO objective is derived from the log-evidence decomposition: (Equation 2.11)

$$\log p_\theta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi}\left[\log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]}_{\text{ELBO}(\theta,\phi;\mathbf{x})} + \underbrace{D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p_\theta(\mathbf{z}|\mathbf{x}))}_{\geq 0} \tag{2.11}$$

Since the KL divergence is non-negative, the ELBO is a lower bound on the log-evidence. Maximizing the ELBO simultaneously trains the encoder to approximate the true posterior and the decoder to reconstruct the data. The ELBO decomposes into a reconstruction term and a regularization term:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{Reconstruction quality}} - \underbrace{\beta \cdot D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))}_{\text{Latent space regularity}} \tag{2.12}$$

The reconstruction term encourages the decoder to accurately reproduce the input from the latent code, while the KL term regularizes the latent space to be close to the standard Gaussian prior $p(\mathbf{z})$. The β parameter (β-VAE) [22] in Equation 2.12 controls the trade-off between reconstruction quality and latent space smoothness: $\beta < 1$ prioritizes reconstruction (beneficial for the relay task where accurate signal recovery is paramount), while $\beta > 1$ encourages disentangled latent representations.

The *reparameterization-trick* enables gradient-based optimization through the stochastic sampling layer:

$$\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2)) \tag{2.13}$$

Instead of sampling from equation 2.13, the sample is expressed as:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon} \tag{2.14}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. This moves the stochasticity outside the computational graph, allowing backpropagation through the encoder.

The implemented relay is a supervised denoising VAE rather than the unconditional model above: its encoder receives the noisy observation window, while reconstruction is evaluated against the clean target. The latent regularization supplies a generative inductive bias, but the denoising property follows from this noisy-input/clean-target training objective rather than from the standard VAE derivation alone. Sampling $\mathbf{z}$ at inference also injects variance into an otherwise deterministic estimation task; Section 2.3.2 reports the measured outcome without attributing a causal effect to that sampling.

2.3.2 Generative vs. Discriminative Paradigms for Relay Processing

The application of generative models to relay processing has, to the best of our knowledge, not been extensively studied. The key question is whether the generative inductive bias (learning the data distribution rather than just the input-output mapping) provides any advantage for the relay denoising task. For BPSK, the clean signal distribution is trivially simple (a discrete distribution on $\{-1, +1\}$), suggesting that the generative overhead may not be justified. This thesis compares VAE-based relay processing against classical and supervised learning methods on a common benchmark, and the results show that the generative paradigm provides no significant advantage for this particular task. A conditional GAN relay is implemented and described in

Chapter 4 but is excluded from the reported comparison on cost grounds, so no claim is made about it here. On the canonical setup the VAE tracks the supervised MLP and symbol-wise DF rather than trailing them, reaching 0.00972 at 20 dB against DF's 0.00972 and the MLP's 0.00992 (Table 5.2), and it stays inside the feedforward group at every SNR in the equal-parameter comparison as well (Table 5.3). The generative bias is therefore neither an advantage nor a handicap here: it costs parameters and training time to arrive where a 169-parameter regressor already is. One qualification remains, unchanged by this: the relay samples $\mathbf{z} \sim q_\phi(\mathbf{z} \mid \mathbf{x})$ stochastically at inference, and the deterministic variant that substitutes the posterior mean $\boldsymbol{\mu}$ at test time was not evaluated, so nothing here bears on whether that variant would do better.

2.4 Sequence Models: Transformers, State Space Models

Recent advances in sequence modeling have produced two competing paradigms with distinct computational properties and inductive biases. Both have achieved state-of-the-art results in natural language processing, but their relative merits for physical-layer signal processing (where sequences are real-valued temporal signals rather than discrete tokens) have, to the best of our knowledge, not been previously investigated.

2.4.1 Transformers and the Attention Mechanism

Transformers [23] use multi-head self-attention to capture global dependencies in sequences. The core attention mechanism computes a weighted combination of value vectors, where the weights are derived from the compatibility of query and key vectors:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{2.15}$$

where $\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{n \times d_k}$ and $\mathbf{V} \in \mathbb{R}^{n \times d_v}$ are obtained from the input via learned linear projections W^Q, W^K, W^V. The scaling factor $1/\sqrt{d_k}$ prevents the dot products from growing too large in magnitude, which would push the softmax into saturation regions with vanishing gradients.

Multi-head attention extends this by running h parallel attention heads, each with independent projections, and concatenating their outputs: (Equation 2.16)

$$\text{MultiHead}(\mathbf{X}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O, \text{head}_i = \text{Attention}(\mathbf{X}W_i^Q, \mathbf{X}W_i^K, \mathbf{X}W_i^V) \tag{2.16}$$

Each head can attend to a different aspect of the input – one to immediate neighbors for local denoising, another to longer-range patterns. The total parameter count for multi-head attention is:

$$3d_{\text{model}}^2 + d_{\text{model}}^2 = 4d_{\text{model}}^2 \tag{2.17}$$

for Q, K, V projections and the output projection. The attention matrix:

$$\mathbf{A} = softmax(\frac{\mathbf{QK}^T}{\sqrt{d_k}}) \in \mathbb{R}^{n\times n} \tag{2.18}$$

computes pairwise interactions between all n positions, yielding a time and memory complexity of: $O(n^2)$. For the 11-symbol relay window used in this thesis, this is negligible ($11^2 = 121$ entries). However, the quadratic cost becomes prohibitive for longer sequences (e.g., $n = 1000$ symbols), motivating linear-time alternatives.

Positional encoding is necessary because the attention mechanism is permutation-equivariant (it treats input positions symmetrically). This thesis uses sinusoidal positional encoding: (Equation 2.19)

$$PE_{(pos,2k)} = \sin(\frac{pos}{10000^{2k/d}}), \quad PE_{(pos,2k+1)} = \cos(\frac{pos}{10000^{2k/d}}) \tag{2.19}$$

which injects position information into the embeddings, enabling the model to distinguish between symbols at different positions in the window.

2.4.2 Structured State Space Models

Structured state space models (SSMs) [24] treat a sequence model as a discretized linear dynamical system. A continuous system $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$, $y(t) = \mathbf{C}\mathbf{x}(t) + Du(t)$ is discretized with a step size Δ (zero-order hold, $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$) to give the recurrence

$$\mathbf{x}_k = \bar{\mathbf{A}}\mathbf{x}_{k-1} + \bar{\mathbf{B}}u_k, \quad y_k = \mathbf{C}\mathbf{x}_k + Du_k \tag{2.20}$$

which is a linear recursive filter with a learned state transition. Two ingredients make this competitive with attention. The HiPPO framework supplies a principled initialization for $\mathbf{A}$ under which the state maintains a compressed polynomial approximation of the input history, and S4 [24] shows that a structured (diagonal or diagonal-plus-low-rank) $\mathbf{A}$ makes the recurrence cheap to evaluate. The resulting model is linear-time in sequence length, in contrast to attention's quadratic cost.

2.4.3 Mamba and Mamba-2

Mamba [25] makes the SSM parameters *input-dependent*, turning the time-invariant system into a time-varying one:

$$\Delta_k = \mathrm{softplus}(W_\Delta u_k + b_\Delta), \quad \mathbf{B}_k = W_B u_k, \quad \mathbf{C}_k = W_C u_k \tag{2.21}$$

With $\mathbf{A}$ diagonal and negative, Δ_k acts as a learned gate: a large Δ_k drives $\bar{\mathbf{A}}_k = \exp(\Delta_k \mathbf{A})$ toward zero, resetting the state so the new input dominates, while a small Δ_k leaves $\bar{\mathbf{A}}_k \approx \mathbf{I}$, preserving the state and ignoring the input. This selectivity is the property of interest for relay processing: in principle the model can retain signal structure while suppressing noise, deciding per sample which of the two an observation carries. The selective layer is wrapped in a gated block with an expand-factor-2 projection and a short causal convolution.

Remark 2.1. These adaptive-filtering connections become directly relevant only in the unknown-channel study of Chapter 6; on the canonical memoryless channel the relay reduces to a per-symbol function and there is no temporal structure for a state to track.

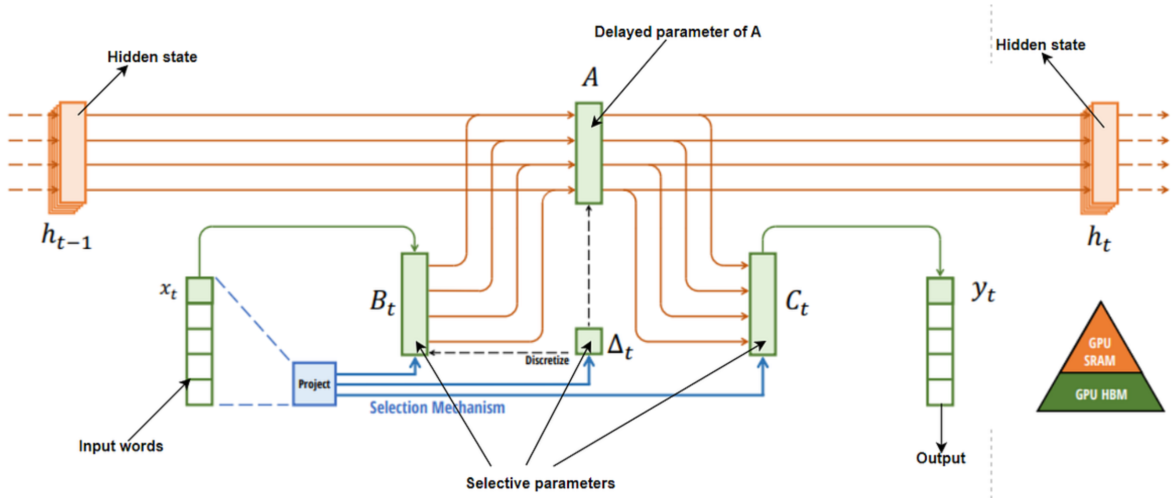


Figure 2.1: Structured SSMs map each channel independently through a higher-dimensional latent state; the selective variant makes the transition input-dependent so the state is reset or held per sample

Mamba-2 (SSD) [26] reformulates the same computation through an algebraic duality between the linear recurrence and a structured matrix multiplication: the output can be written $\mathbf{y} = M \cdot (\mathbf{B} \odot \mathbf{u})$ for a lower-triangular semi-separable matrix M. The practical consequence is that the sequence can be processed in chunks, with a batched matrix multiply inside each chunk and a single state hand-off between chunks, so the number of sequential steps falls from one per sample to one per chunk. Mamba-2 also forms the state-space parameters at the start of the block rather than sequentially along it. Both variants are evaluated here; at the short relay windows this thesis uses, the distinction is architectural rather than a source of measured advantage.

2.4.4 State Space Models for Signal Processing

The application of state space models as the *relay-side* processing function is, to the best of our knowledge, largely unexamined. Transformers and state-space models have already been applied to a range of wireless receiver and signal-processing tasks; what the literature reviewed here did not turn up is a controlled comparison of these architectures against one another, at a matched parameter budget, in the two-hop relay benchmark studied in this thesis. Classical signal processing relies extensively on LTI state space models (Kalman filters, Wiener filters), and the SSM framework provides a natural neural extension of these classical tools. The connection is direct: a trained SSM with fixed (input-independent) parameters implements a learnable linear filter, while the selective Mamba variant implements a learnable *adaptive* filter. For the relay denoising task, this means Mamba can potentially learn the optimal filter structure from data, without requiring manual specification of filter order, cutoff frequencies, or adaptation algorithms.

Within the literature reviewed for this thesis, no prior controlled comparison of Mamba-S6, Mamba-2 SSD, and Transformers for the same relay-processing task was identified. The relay window used throughout is short ($n = 11$), which is precisely the regime where attention's quadratic cost is negligible and a linear-time recurrence has nothing to recover; the throughput advantages reported for the SSD formulation at longer context lengths [26] are therefore not exercised by this application and are not measured here.

2.5 Theoretical Foundations: Channel Models

This section presents the theoretical BER analysis for each channel model used in this study, derives the closed-form expressions, and validates them against Monte Carlo simulation. The theoretical-simulative comparison serves two purposes: (i) it verifies the correctness of the simulation framework, and (ii) it establishes the baseline performance that AI relays must beat. This matters beyond internal bookkeeping. Every classical baseline the learned relays are measured against in the chapters that follow is itself checked against a closed form here, so those comparisons set a learned relay against a verified classical detector rather than against an unaudited simulation of one.

2.5.1 AWGN Channel: Theoretical Analysis

The AWGN channel adds zero-mean Gaussian noise to the transmitted signal: (Equation 2.22)

$$y = x + n, \quad n \sim \mathcal{N}(0, \sigma^2), \quad \sigma^2 = \frac{P_s}{\text{SNR}_{\text{linear}}} \tag{2.22}$$

where $P_s = \mathbb{E}[|x|^2] = 1$ for unit-power BPSK symbols. The noise variance is inversely proportional to the linear SNR.

2.5.1.1 Single-hop BER

For BPSK ± 1 symbols and real-valued AWGN with variance $\sigma^2 = 1/\text{SNR}$, an error occurs when the noise pushes the received sample past the decision boundary at the origin. The theoretical BER is: (Equation 2.23)

$$P_e^{\text{AWGN}} = Q\left(\frac{1}{\sigma}\right) = Q\left(\sqrt{\text{SNR}}\right) = \frac{1}{2}\,\text{erfc}\left(\sqrt{\frac{\text{SNR}}{2}}\right) \tag{2.23}$$

where $Q(x) = \frac{1}{2}\text{erfc}(x/\sqrt{2})$ is the Gaussian Q-function and $\text{SNR} = P_s/\sigma^2$ is the ratio of signal power to noise variance.

Remark 2.2 (Noise convention)**.** The standard textbook result $P_e = Q(\sqrt{2E_b/N_0})$ assumes that the one-sided noise PSD is N_0, giving a baseband noise variance of $N_0/2$ per real dimension. In our AWGN implementation the noise is purely real with variance $\sigma^2 = P_s/\text{SNR}$, so $N_0 = 2\sigma^2 = 2P_s/\text{SNR}$ and $E_b/N_0 = \text{SNR}/2$. Substituting yields $Q(\sqrt{2 \cdot \text{SNR}/2}) = Q(\sqrt{\text{SNR}})$, which is the expression used throughout this work.

By contrast, the fading channels add **complex** Gaussian noise with total power $1/\text{SNR}$ (i.e. $\sigma_I^2 = \sigma_Q^2 = 1/(2\,\text{SNR})$) and extract the real part after ZF equalization, halving the effective noise variance per decision dimension. This naturally introduces a factor of 2 inside the Q-function argument for all fading-channel BER expressions (the fading-channel analysis above), making them consistent with the standard textbook forms.

2.5.1.2 Two-hop AF BER

For amplify-and-forward with equal-SNR hops, the effective end-to-end SNR is:

$$\text{SNR}_{\text{eff}}^{\text{AF}} = \frac{\text{SNR}_1 \cdot \text{SNR}_2}{\text{SNR}_1 + \text{SNR}_2 + 1} = \frac{\gamma^2}{2\gamma + 1} \tag{2.24}$$

where $\gamma = \text{SNR}$ is the per-hop SNR. The resulting BER is

$$P_e^{\text{AF}} = Q\left(\sqrt{\text{SNR}_{\text{eff}}}\right) \tag{2.25}$$

under the same real-noise convention as the single-hop case (Remark 2.2). At high SNR,

$$\text{SNR}_{\text{eff}} \approx \frac{\gamma}{2} \tag{2.26}$$

confirming the well-known 3 dB penalty of AF relaying. This penalty is a property of the analytical CSI-assisted variable-gain model above. The AF relay implemented in this thesis uses a single block-adaptive scalar estimated from the empirical received power, so Eq. (2.26) is background rather than a closed-form prediction of the simulated AF curve, and the 3 dB figure should not be read as explaining it.

2.5.1.3 Two-hop DF BER

For a decode-and-forward relay with equal-SNR hops, an error occurs at the destination if exactly one hop introduces an error:

$$P_e^{\mathrm{DF}} = P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2} \tag{2.27}$$

With equal hops (i.e. $P_{e,1} = P_{e,2} = P_e^{\mathrm{AWGN}}$), this becomes

$$P_e^{\mathrm{DF}} = 2P_e(1 - P_e) \tag{2.28}$$

At high SNR, $P_e \ll 1$ and

$$P_e^{\mathrm{DF}} \approx 2P_e \tag{2.29}$$

i.e., the two-hop penalty is approximately a factor of 2 in BER.

2.5.2 Rayleigh Fading Channel: Theoretical Analysis

The Rayleigh fading channel models non-line-of-sight (NLOS) propagation where the signal undergoes multiplicative fading:

$$y = hx + n, \quad h \sim \mathcal{CN}(0, 1), \quad n \sim \mathcal{CN}(0, \sigma^2) \tag{2.30}$$

The fading coefficient h is a circularly-symmetric complex Gaussian random variable, so its magnitude $|h|$ follows a Rayleigh distribution:

$$f_{|h|}(r) = 2r \cdot e^{-r^2}, \quad r \geq 0 \tag{2.31}$$

with $\mathbb{E}[|h|^2] = 1$ (unit average power). The instantaneous SNR after equalization ($\hat{x} = y/h$)

becomes $\gamma_h = |h|^2 \cdot \text{SNR}$, which is exponentially distributed.

2.5.2.1 Single-hop BER

Averaging the conditional BER $P_e(\gamma_h) = Q(\sqrt{2\gamma_h})$ over the exponential distribution of γ_h yields the closed-form [27, Eq. 14-4-15] (Equation 2.32):

$$P_e^{\text{Rayleigh}} = \frac{1}{2}\left(1 - \sqrt{\frac{\bar{\gamma}}{1+\bar{\gamma}}}\right) \tag{2.32}$$

where $\bar{\gamma} = \gamma_b = \bar{E}_b/N_0$ is the average SNR *per bit*. This distinction is load-bearing, because the canonical experiments plot $\gamma_s = E_s/N_0$ per *symbol*: for QPSK, $\gamma_b = \gamma_s/2$, a 3.01 dB offset. Substituting gives the per-bit Rayleigh expression in terms of the plotted axis, $P_b = \frac{1}{2}\big(1 - \sqrt{\gamma_s/(\gamma_s+2)}\big)$, which is the form the DF-theory column of Chapter 5 evaluates. At high SNR ($\bar{\gamma} \gg 1$):

$$P_e^{\text{Rayleigh}} \approx \frac{1}{4\bar{\gamma}} \tag{2.33}$$

This $1/\text{SNR}$ decay is fundamentally slower than the exponential decay of AWGN ($Q(\sqrt{\gamma}) \sim e^{-\gamma/2}$), explaining why Rayleigh fading is significantly more challenging. The channel is **diversity-limited**: deep fades (where $|h| \approx 0$) cause errors regardless of the average SNR. Combating this diversity limitation is the primary motivation for the relay architectures studied in this thesis.

2.5.2.2 Two-hop DF BER

The two-hop DF relay over Rayleigh fading follows the same composition rule as AWGN:

$$P_e^{\text{DF,Rayleigh}} = 2P_e^{\text{Rayleigh}}(1 - P_e^{\text{Rayleigh}}) \tag{2.34}$$

2.6 Channel Coding: Convolutional Codes and Trellis Decoding

The channel models above assume an uncoded link. Chapter 5's supplementary coded study (Section 5.5) departs from that assumption, adding a convolutional code and a block-DF relay that decodes it. This section gives the minimal background needed to follow that study: what a convolutional code's rate and constraint length mean, how a receiver recovers the transmitted sequence from a noisy one, and how a single code is stretched across several rates.

2.6.1 Convolutional Encoding: Rate and Constraint Length

A convolutional encoder maps each group of k input bits to n coded output bits using a shift register of *constraint length* K – the number of k-bit input groups (including the current one) that influence each output. The code *rate* [27] is

$$R = \frac{k}{n} \tag{2.35}$$

so a rate-$1/2$ code, the mother code used throughout Chapter 5, emits two coded bits per information bit. This thesis uses $k = 1$ throughout, one information bit per trellis step; the encoder's output then depends on the current bit and the $K - 1$ preceding ones held in the shift register, so its evolution is a finite-state machine with [27]

$$S = 2^{K-1} \tag{2.36}$$

states, and the resulting state diagram unrolled over time is the code's *trellis*: every valid coded sequence corresponds to exactly one path through it. A larger K gives the code more memory to spread redundancy across, at the cost of a larger trellis to search; Chapter 5 measures this trade-off directly by sweeping $K \in \{3, 5, 7\}$, i.e. 4, 16 and 64 states (Section 5.5). A frame is conventionally *zero-tail terminated*: $K - 1$ zero bits are appended after the information bits to drive the encoder back to the all-zero state, so the trellis both starts and ends at a known point rather than an unknown one.

2.6.2 Maximum-Likelihood Sequence Decoding: The Viterbi Algorithm

Because every valid coded sequence is one path through the trellis, recovering the most likely transmitted sequence from a noisy received one is a shortest-path problem: find the trellis path whose hypothesized coded symbols are collectively closest to what was received. A brute-force search is infeasible – an L-bit information frame admits 2^L trellis paths. The Viterbi algorithm nonetheless solves the problem exactly, in time linear in L, by a dynamic-programming recursion over the trellis states: at each step it extends every surviving path by one branch and keeps only the lowest-cost path into each of the S states (add-compare-select), so the cost per step depends on S rather than on the number of paths [28, 27]. For a memoryless channel with an appropriate branch metric – squared Euclidean distance to the received sample under AWGN, as used by Chapter 5's coded block-DF relay [28] – the recovered sequence is

$$\hat{c} = \arg\min_{c \in \mathcal{C}} \sum_k |y_k - x(c_k)|^2 \tag{2.37}$$

where $\mathcal{C}$ is the set of trellis-valid coded sequences and $x(c_k)$ the modulated symbol for coded bits c_k – exactly the maximum-likelihood sequence estimate for the channel, with no approximation. Optimality is over the sequence, however, not over each bit independently, and this shapes how the decoder fails. When noise makes an incorrect path cheaper than the transmitted one, that path typically diverges from the correct path for several trellis steps before merging back, so a single decoding failure corrupts a *run* of information bits rather than an isolated one. Residual errors therefore arrive in bursts, and below the code's operating threshold these events become frequent enough that the burst penalty outweighs the coding gain. This is the mechanism behind the low-SNR coding penalty Chapter 5 measures (Section 5.5), where a fixed-rate code performs worse than no coding at all – a memoryless per-symbol slicer has no trellis to diverge along, and so cannot fail this way.

2.6.3 Soft-Decision Decoding: The Forward-Backward (BCJR) Algorithm

Viterbi decoding returns one hard path through the trellis and discards how close the alternatives were. The BCJR algorithm [29] instead computes, for every trellis transition, its posterior probability given the *entire* received sequence, using both a forward recursion $\alpha_k(s)$ (the likelihood of reaching state s at step k from everything received so far) and a backward recursion $\beta_k(s)$ (the likelihood of everything still to come, given state s at step k):

$$P(s_{k-1} = s',\, s_k = s \mid \mathbf{y}) \propto \alpha_{k-1}(s')\, \gamma_k(s', s)\, \beta_k(s) \tag{2.38}$$

where $\gamma_k(s', s)$ is the branch's transition-and-observation probability. Where Viterbi minimizes the probability of choosing the wrong *sequence*, BCJR minimizes the error probability of each *symbol* individually [29]; the two optimize different criteria, which is why neither dominates the other in Chapter 5's measurements. Marginalizing these joint probabilities over the transitions consistent with each coded bit or symbol gives a *soft* estimate – a posterior mean or log-likelihood ratio – in place of Viterbi's hard decision. This is the read-out Chapter 5 calls *soft block-DF* (Section 5.5): the relay forwards the BCJR posterior mean instead of committing to the single most likely codeword.

2.6.4 Puncturing, Spectral Efficiency, and Rate Adaptation

A code's rate and the modulation's constellation size M together set how much information it carries per channel use, its spectral efficiency [27]:

$$\eta = R \cdot \log_2 M \quad \text{[bits/symbol]} \tag{2.39}$$

so a rate-$1/2$ code halves the information rate of whatever modulation carries it, and comparing a coded link against an uncoded one at the same SNR without accounting for this is not a like-for-like comparison (Section 5.5.2 revisits Chapter 5's coded-versus-uncoded comparison on this basis). Rates other than a code's native k/n are obtained by *puncturing*: deleting a fixed, periodic subset of the encoder's output bits according to a puncturing pattern, and at the decoder re-inserting those positions as erasures (soft value zero) before running the identical trellis search – so one Viterbi or BCJR decoder implementation serves every punctured rate from the same mother code. This is how Chapter 5's rate-$1/2$ mother code is stretched to rates $2/3$ and $3/4$ for its link-adaptation study (Section 5.5.3).

Spectral efficiency alone is not the quantity a link adapts on, because it counts bits that were transmitted rather than bits that arrived. A higher-rate choice buys more bits per symbol but raises the frame error rate, and any real protocol discards a frame with residual errors. The operative objective is therefore *goodput*, the information rate actually delivered:

$$G = \eta \cdot (1 - \mathrm{FER}) \quad \text{[information bits/symbol]} \tag{2.40}$$

Maximizing G over a grid of modulation-and-coding schemes at each SNR – the standard adaptive modulation and coding (AMC) problem – traces the link-adaptation envelope that Section 5.5.3 measures. The maximization is unconstrained: the FER penalty in Equation 2.40 already prices in unreliability, so no separate reliability constraint is imposed.

Chapter 3

Research Objectives

3.1 Main Objective

To determine *under which conditions* a learned relay surpasses classical relay processing in two-hop communication, and where it does not. A single canonical setup (SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER) is the controlled baseline on which classical and AI-based strategies are compared as a function of SNR regime and computational constraints. Two deliberate departures from that baseline then locate the boundary the canonical comparison cannot reach: a coded chain, which tests whether the conclusion survives when the classical relay is allowed to decode a channel code (Section 5.5), and channels whose structure lies outside the model class the classical relays assume, which is where the learned relay's value is actually established (Chapter 6). The canonical setup is therefore the control, not the scope.

3.1.1 Two Research Questions

The main objective resolves into two questions that differ only in whether a budget is imposed, and which for that reason have different answers. Throughout, the classical reference is named explicitly rather than left as "classical processing": symbol-wise DF and AF where the channel is memoryless, Viterbi MLSE where it has memory, and block DF, a channel code decoded and re-encoded at the relay, where the relay is permitted to spend a frame of delay.

O1 (unconstrained). On a channel with memory, and with no budget on latency or computation, in which regime and under what conditions can a learned relay complement the classical method? This is a question about accuracy and about what each detector needs to know. It is answered by the cumulative controlled relaxations of Chapter 6: channel memory is introduced first, per-block CSI is then reduced, and finally the channel realization is redrawn within a family represented during training.

O2 (constrained). The same question when the relay must also meet a latency budget and a computational budget. Here the classical method may be the more accurate detector and still be unavailable, because a trellis search that grows as M^L eventually cannot be afforded and a relay that decodes a codeword cannot decide before the codeword ends.

Whether the two boundaries coincide is not settled in advance, and there is reason to expect they do not. O1's turns on *channel-estimate reliability*, how well the channel must be known before the classical detector can be run as specified at all, while O2's turns on *arithmetic and delay*, which the channel's memory governs whether or not the channel is known. Two different resources may well run out at two different points. If they do, the distance between them is a result in its own right rather than a restatement of either, and this thesis is structured to measure it: Section 6.3 locates the first by sweeping the pilot budget until the classical estimate fails, and Section 6.5.1 locates the second by measuring a channel with memory under a latency constraint.

Latency is singled out for measurement because it is more often assumed than measured. A sequence detector is commonly treated as latency-expensive on the grounds that it searches a trellis, and a windowed relay as cheap on the grounds that its window is short; neither follows from the structure of the algorithms, since a trellis search need not run to the end of the block and a window must be filled on both sides of the symbol it decides. Section 6.5.1 therefore measures the decision delay each detector actually imposes, rather than inferring it.

3.2 Research Gap and Motivation

Despite growing interest in AI for wireless communication, the literature reviewed for this thesis did not identify a controlled comparison of the particular relay-processing paradigms evaluated here. Because this was not a preregistered systematic review, the gap statements below are scoped to the cited and reviewed literature rather than asserted as exhaustive:

1. **Gap 1: Limited cross-paradigm relay comparison.** The cited work studies related receiver or end-to-end tasks, generally one architecture family at a time (Section 2.2.2 reviews it): the reviewed literature did not identify a comparison of supervised feedforward networks, variational autoencoders, attention-based Transformers, and state space models (Mamba [25], Mamba-2 SSD [26]) on the same relay denoising task under controlled conditions. The cGAN is implemented here but excluded from the reported comparison, so the adversarial paradigm is not quantitatively covered. State space models in particular have begun to be applied to receiver-side physical-layer tasks, such as channel state prediction [30], but not, to the best of our knowledge, to the relay processing function itself, where the node must re-transmit rather than merely decide.

2. **Gap 2: Limited characterization of the relay complexity–performance relationship.** Model-size trade-offs have been examined for other physical-layer tasks, including automatic modulation classification [31], but the reviewed literature did not identify a corresponding controlled size sweep for this relay task. In the memoryless BPSK-AWGN

special case, recovering one symbol is a smooth, one-dimensional estimation problem, so it is not evident that additional capacity helps. An equal-parameter comparison can remove raw parameter count as one explanation, although architecture, depth, width, state dimension, window, and optimization remain confounded.

This thesis addresses both gaps through a unified framework that implements nine relay strategies and compares eight of them on the canonical setup, with both original and normalized parameter counts, full statistical analysis, and a dedicated complexity study; Gap 1's adversarial paradigm (cGAN) is implemented but, being excluded from the canonical comparison on cost grounds, is addressed qualitatively rather than quantitatively. The following table summarizes the positioning of this work relative to the literature:

Table 3.1: Research positioning: comparison of this thesis against prior work in deep-learning-based relay communication

Aspect	Prior Work	This Thesis
Relay strategies compared	1–2 (typically AF vs. DF, or one AI method)	8 compared (2 classical + 6 AI); a ninth, cGAN, is implemented but excluded on cost grounds
AI paradigms	Single (e.g., supervised only)	4 implemented (supervised, generative, adversarial, sequential); adversarial (cGAN) excluded from the canonical comparison on cost grounds
Evaluation discipline	Single ad-hoc configuration	One canonical setup (SISO, Rayleigh, QPSK) for the eight-relay comparison, chosen in advance and never varied there, calibrated against closed-form theory on that channel; a supplementary coded study deliberately varies the code, rate, and modulation against this baseline
Complexity analysis	None	Normalized $\sim$3K-parameter comparison + complexity study (does BER rise again with model size?)

Aspect	Prior Work	This Thesis
State space models	Applied to receiver-side tasks; none as a relay processor	Mamba-S6, Mamba-2 SSD evaluated as relay processors
Statistical reporting	Varies among the cited studies	Paired Wilcoxon tests reported per comparison without multiplicity correction; 95% intervals and at least 10 trials per point

Relationship between the canonical study and the extension study. Hypotheses H1–H4 concern the canonical relay scenario introduced in Chapter 1: a memoryless two-hop channel with white noise and uncoded transmission. Hypothesis H5 studies a deliberately different regime in which the canonical model no longer represents the true channel. It is therefore a robustness extension rather than a direct continuation of the canonical experiments.

3.3 Research Hypotheses

Based on the theoretical background of Chapter 2, this thesis tests the following hypotheses. H1–H4 are posed and answered on the canonical setup. H5 is the central hypothesis of the thesis's principal contribution (Chapter 6), posed on channels *outside* the canonical family:

1. **H1 (AI advantage over AF at low SNR).** On the canonical setup, some AI-based relay strategies achieve lower BER than AF at low SNR (0–4 dB), where the non-linear denoising learned by the neural network provides an advantage over AF's noise amplification.

   Rationale: The scalar BPSK-AWGN posterior mean, $f^*(y) = \tanh(y/\sigma^2)$, provides qualitative intuition for a smooth nonlinear relay map, but it is not the exact estimator for the canonical Gray-QPSK Rayleigh channel. There, the per-axis posterior depends on the channel-compensated observation and its fading-dependent reliability. For BPSK in AWGN, moreover, hard detection of the posterior mean is identical to the MAP decision, since $\text{sign}(\mathbb{E}[X \mid Y]) = \text{sign}(Y)$; any relay-level benefit therefore arises from preserving soft reliability information for the second hop rather than from improving the single-hop hard decision. In either case, the local MMSE estimator is not necessarily the end-to-end BER-optimal relay function, so H1 is an empirical hypothesis rather than a theorem.

2. **H2 (DF leads the evaluated symbol-wise classical relays at high SNR).** The symbol-wise DF relay (Remark 4.2) achieves the lowest BER at medium-to-high SNR (≥ 6 dB), outperforming all AI methods.

   Rationale: In the scalar AWGN limit the posterior mean approaches a sign decision; on the canonical QPSK channel the corresponding high-SNR rule is nearest-neighbor slicing after channel compensation. This motivates DF as the strongest evaluated hard-regeneration baseline, not as a proof that it is end-to-end BER-optimal among all power-constrained per-symbol relay mappings.

3. **H3 (Overfitting at excess capacity).** There exists an optimal model size for relay denoising beyond which performance degrades due to overfitting, so that BER rises again once capacity exceeds what the mapping requires. Specifically, models with $\sim$100–200 parameters achieve performance comparable to models with 10–100$\times$ more parameters. Only the second, weaker part of this hypothesis was testable with the protocol originally adopted: establishing *overfitting* as the mechanism behind any upturn would require seed replication and train/test-gap measurement. The seed replication is performed for the MLP size axis by the sweep of Section 5.4, and the hypothesis is adjudicated there and in Section 7.2.3; the train/test-gap measurement, and any replication of the *cross-architecture* comparison, remain outstanding and are recorded in Section 7.4.

   Rationale: The low-dimensional target in the memoryless special case motivates testing whether modest capacity is sufficient. The effect of excess capacity on predictive variance is not fixed a priori and must be established by replicated training and held-out evaluation.

4. **H4 (Architecture convergence at an equal parameter budget).** When all AI models are normalized to the same parameter count ($\sim$3,000), the performance differences between architectures narrow significantly, indicating that parameter count is a more important factor than architectural choice. This is an equal-*budget* comparison rather than an isolation of architecture from capacity: as Table 4.2 records, meeting the budget required adjusting width, depth, state dimension and input window size together, so those degrees of freedom co-vary with architecture and are not separately controlled.

   Rationale: The evaluated models have enough representational capacity for the tested mapping. Equalizing parameter count tests whether raw capacity explains observed differences, while leaving architecture and several hyperparameters confounded.

Unlike H1–H4, which assume the canonical memoryless channel throughout, H5 deliberately introduces impairments not present in the system model of Chapter 1, including channel memory, ISI, and nonlinear distortion.

5. **H5 (Principal contribution, unknown-channel mitigation).** On the evaluated channels with memory (intersymbol interference) or asymmetry (biased nonlinearity), the *memoryless* classical relays (AF, symbol-wise DF) exhibit error floors or severe degradation, whereas the 170-parameter MLP mitigates the impairment after training on the relevant channel family, without per-block CSI or online adaptation.

   Rationale: AF forwards the impairment and symbol-wise DF applies a memoryless zero-threshold decision; neither explicitly compensates channel memory. For a known matched

FIR channel, Viterbi is the maximum-likelihood *sequence* detector; with estimated taps, LS-plus-Viterbi is a plug-in detector rather than an optimal one. The hypothesis is not that learning beats a BER-optimal classical receiver, but that a fixed learned architecture can approach matched-MLSE performance on an unseen realization drawn from the impairment family represented during training, without per-block CSI, pilots, explicit identification, or online adaptation. Its arithmetic is fixed only for a fixed deployed window; spanning longer memory generally widens that window and raises cost roughly linearly, whereas full-state MLSE evaluates M^L branches per symbol for an L-tap channel. Chapter 6 determines the empirical scope of this claim.

3.4 Specific Objectives

1. **Implement nine relay strategies and compare eight of them** spanning four learning paradigms: no learning (AF, DF), supervised learning (MLP, Hybrid), generative modeling (VAE), and sequence modeling (Transformer, Mamba-S6, Mamba-2 SSD); the ninth, a conditional GAN (cGAN), is implemented but excluded from the comparison on cost grounds.

2. **Evaluate on the canonical setup:** SISO Rayleigh fast fading with QPSK, with symbol-wise DF checked against the closed-form two-hop composition on that same channel as the analytical calibration.

3. **Investigate the complexity–performance trade-off** by testing model architectures ranging from 0 parameters (classical) to 26,179 parameters (Mamba-2 SSD), and by conducting a normalized comparison at approximately 3,000 parameters.

4. **Test whether the canonical conclusion survives a coded chain** by replacing the uncoded symbol-wise DF baseline with a genuine block-DF relay, a convolutional code decoded and re-encoded at the relay, and comparing it against coded-aware learned relays at equal E_s/N_0 and again at equal throughput, since a code that halves the information rate changes what "better" means (Section 5.5).

5. **Delineate the conditions under which a learned relay surpasses classical processing** on channels whose structure the classical relays do not model — unknown memory, nonlinear asymmetry, and composite cascades of both — by locating the boundary rather than demonstrating a capability: a memoryless control where learning is expected to buy nothing, a matched-receiver bound (Viterbi MLSE with genie CSI), a pilot-budget sweep to find the crossover, and a blind regime where no channel posterior exists (Chapter 6).

6. **Price the two objectives against each other** by measuring a channel with memory *under* a latency constraint, the one cell in which MLSE's M^L growth and block DF's frame buffering bind simultaneously, with every relay placed inside one identical coded chain so the comparison is between relays and nothing else, and with cost reported on axes that

assume no processor: decision delay in symbols and multiply-accumulates per symbol (Section 6.5.1).

7. **Identify practical deployment recommendations** for selecting the appropriate relay strategy given specific operational constraints (SNR range, computational budget, channel environment).

3.5 Scope and limitations

The following limitations define the scope of the canonical comparison, confined throughout to the setup below (SISO, Rayleigh fast fading, complex baseband, QPSK, uncoded); the supplementary coded and unknown-channel studies depart from it deliberately, as noted where relevant:

- **Modulation:** QPSK for the canonical relay comparison; the unknown-channel study (Chapter 6) mainly uses BPSK, with a QPSK re-check of the unknown-ISI result (Section 6.2.2), and 16-QAM enters only as a modulation-and-coding rung in the link-adaptation study. Higher-order constellations as a relay design question, and the joint 2D N-class formulation they require, are deferred to future work.
- **Channel knowledge:** Perfect CSI in the canonical core; estimation error is not modeled there. Chapter 6 lifts this with finite-pilot estimation, a swept pilot budget, and a zero-CSI blind regime.
- **Topology:** A single relay, two hops, half-duplex, SISO on both hops. Multi-relay, full-duplex and multi-antenna (MIMO) configurations are deferred to future work.

These delimitations isolate the relay processing function from protocol and system effects.

The scope that bounds every result in this thesis, and the generalizations it does not license, is set out once in Section 1 and not restated per chapter. The delimitations below are the ones specific to the objectives above.

Chapter 4

Methods

4.1 System Model

The system under study is a two-hop relay network with a single relay node. All methods in this chapter are described for the canonical setup of Chapter 1: SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER. *Fast fading* here means the strict memoryless case, $h[i] \sim \mathcal{CN}(0, 1)$ drawn independently for every symbol — i.i.d. per-symbol Rayleigh fading, not a fading process with a finite Doppler spectrum. Every later use of the term in this thesis carries that meaning. QPSK on the canonical Rayleigh channel is the pairing used here and throughout the canonical comparison; the coded and unknown-channel studies additionally use the pairings set out in Table 4.1.

Why per-symbol processing suffices here. Under the canonical model, with receiver-side CSI used for one-tap zero-forcing equalization, the equalized current observation $\tilde{y}_R[i]$ is a sufficient statistic for detecting $x[i]$ — neighbouring observations add nothing because the fading and noise are i.i.d. across symbols. No relay comparison in the canonical study therefore requires sequence detection, and the classical AF and DF baselines operate per symbol. Some neural architectures nevertheless receive a finite window of neighboring observations (Section 4.2); this is an architectural input convention for the evaluated feedforward and sequence models, and under the canonical i.i.d. assumptions it neither introduces channel memory nor supplies additional information about $x[i]$.

Figure 4.1: Two-hop relay system model: source transmits through two channels with a neural network relay in between. Each hop applies the channel operator followed by one-tap zero-forcing equalization in the canonical setup

Source bits are QPSK-modulated on the canonical setup, passed through Hop 1 (SISO Rayleigh), processed by the relay (classical or neural), transmitted over Hop 2 (SISO Rayleigh), and demodulated at the destination after one-tap zero-forcing equalization at the corresponding receiver. Chapter 6 reuses this chain with BPSK (DBPSK where phase must be removed by differencing), replacing Hop 1 with unknown ISI, a composite cascade, or a per-block redraw; Hop 2 is AWGN for the unknown-ISI comparison, Rayleigh otherwise. In the canonical setup, both hop channels are known perfectly at their respective receivers (Chapter 1); the unknown-ISI departure keeps perfect CSI, while the finite-pilot and blind departures relax it.

$$\text{Source} \xrightarrow{\text{Hop 1}} \text{Relay} \xrightarrow{\text{Hop 2}} \text{Destination} \tag{4.1}$$

4.1.1 Modulation

QPSK carries data in every canonical relay comparison; BPSK carries data mainly in the unknown-channel study (Chapter 6), which also re-checks the unknown-ISI result with QPSK (Section 6.2.2), 16-QAM additionally carries data as a rung of the coded link-adaptation study (Section 5.5), and denser constellations appear in Figure 4.2 for orientation only. The real-valued base case is BPSK, which maps a single bit to a real-valued symbol,

$$x = 1 - 2b, \quad b \in \{0, 1\} \implies x \in \{-1, +1\} \tag{4.2}$$

with hard-decision demodulation $\hat{b} = \mathbb{1}[\hat{x} < 0]$ at the destination. QPSK is defined in Section 4.5. Both are normalized to unit average symbol energy, $E_s = 1$, so they differ in bits per symbol rather than in transmit power: $k = 1$ and 2 respectively, giving $E_b = E_s/k$. Higher-order constellations as a relay design question are future work; 16-QAM appears only as a link-adaptation rung.

4.1.2 Hop Model

In the half-duplex two-hop model studied in this thesis, the relay cannot transmit and receive simultaneously, and there is no direct source–destination link. The communication proceeds in two time slots. On the canonical Rayleigh fast-fading channel each hop applies an i.i.d. complex fading coefficient followed by complex AWGN:

Slot 1 (Source $\rightarrow$ Relay), listening phase:

$$y_R[i] = h_1[i]\, x[i] + n_1[i], \quad h_1[i] \sim \mathcal{CN}(0, 1), \quad n_1[i] \sim \mathcal{CN}(0, \sigma^2) \tag{4.3}$$

Slot 2 (Relay → Destination), transmission phase:

$$y_D[i] = h_2[i]\, x_R[i] + n_2[i], \quad h_2[i] \sim \mathcal{CN}(0,1), \quad n_2[i] \sim \mathcal{CN}(0,\sigma^2) \tag{4.4}$$

where $x[i]$ is the complex symbol transmitted by the source at time i (QPSK on this canonical channel; the real BPSK alphabet $\{-1,+1\}$ never rides this complex channel, and is used in the real AWGN special case below and in the Chapter 6 experiments), $y_R[i]$ is the signal received at the relay, $x_R[i]$ is the signal transmitted by the relay, and $y_D[i]$ is the signal received at the destination. The relay output is written generally as

$$x_R[i] = f_\theta(\mathcal{Y}_{R,i})\,,$$

where $\mathcal{Y}_{R,i}$ denotes the observation supplied to the relay-processing function; for AF and symbol-wise DF, $\mathcal{Y}_{R,i} = y_R[i]$.

Each hop applies a channel function followed by optional equalization (Equation 4.5):

$$y = \mathcal{H}(x, \text{SNR}) + n \tag{4.5}$$

where $\mathcal{H}(\cdot)$ denotes the canonical Rayleigh fast-fading channel operator (the AWGN special case, $|h| \equiv 1$, is retained because the closed forms of Chapter 2 are stated in it). The calligraphic symbol distinguishes the channel *operator* from the scalar fading *coefficient* h used throughout the rest of this thesis.

Both hops use the same channel, canonically. $\mathcal{H}(\cdot)$ applies identically to both hops there, so a configuration is named by the pair (constellation, channel). The unknown-channel study is the exception, pairing an impaired Hop 1 with an unrelated Hop 2 – AWGN for the unknown-ISI comparison, Rayleigh otherwise (Chapter 6).

One-tap equalization: which statistic is actually formed. Two one-tap operations are commonly conflated, and they are not interchangeable for anything downstream of a hard decision. Phase compensation, $y \cdot h^*/|h|$, leaves a real channel with a random gain and *unchanged* noise variance; zero forcing, y/h, removes the gain as well and leaves a *conditional* noise variance inflated by $1/|h|^2$. Hard decisions cannot tell them apart, since the two differ by a positive scale factor, but soft relay inputs, posterior probabilities, AF power normalization and BCJR metrics all can.

The canonical simulator (`relaynet/channels/fading.py`) applies **zero forcing**, so the observation each receiver acts on is

$$\tilde{y}_R[i] = x[i] + \frac{n_1[i]}{h_1[i]}, \qquad \tilde{y}_D[i] = x_R[i] + \frac{n_2[i]}{h_2[i]}, \tag{4.6}$$

with conditional complex-noise power $\sigma^2/|h[i]|^2$ given $h[i]$, hence variance $\sigma^2/(2|h[i]|^2)$ on each real decision axis — a random, heavy-tailed quantity, not the constant per-axis variance $\sigma^2/2$ that phase compensation would give. The unknown-channel study of Chapter 6 does not use this convention throughout, and the difference is worth naming once here because the same symbol $\tilde{y}$ denotes three different observations across this thesis. Its core unknown-ISI experiment has no fading at all: the hop is the deterministic filter plus noise, $\tilde{y}[i] = (h * x)[i] + n[i]$ at constant σ^2, which is what makes a trellis built from h a genuine genie-CSI detector there. Its QPSK subsection adds a per-symbol Rayleigh magnitude on top of that filter and retains it rather than dividing it out, $\tilde{y}[i] = g[i]\,(h * x)[i] + n[i]$ with $g[i] = |h[i]|$, again at constant σ^2 — phase compensation rather than zero forcing. All three are legitimate receivers; they are simply different observations, and each chapter's derivations refer to the one its own simulator forms. Where it matters — soft relay inputs, posterior means, and the branch metrics of a genie detector — the distinction is stated at the point of use rather than assumed. Because the channel is complex, the canonical constellation is two-dimensional: QPSK places one bit on each axis, so both dimensions the channel acts in are used and the relay processes the pair; the real reduction above then applies per axis rather than discarding one. BPSK is not transmitted over the complex canonical fading channel here — a choice of experimental design, not of validity, since coherent BPSK over complex flat Rayleigh is perfectly well posed and loses nothing by projecting onto the real decision axis (Section 4.1.3). Besides its use as the transmitted alphabet of the unknown-channel study (Chapter 6), it appears only in the real AWGN special case below and the closed-form expressions of Chapter 2. The *AWGN limit* referenced in those closed forms is the special case $|h[i]| \equiv 1$, i.e. $y_R[i] = x[i] + n_1[i]$ and $y_D[i] = x_R[i] + n_2[i]$.

The complex noise terms are independent and identically distributed with total power

$$\sigma^2 = \frac{P_s}{\gamma} \tag{4.7}$$

where $P_s = \mathbb{E}[|x|^2]$ is the transmit symbol power (unit for every constellation studied, per the normalization of Section 4.1.1) and γ denotes the per-hop linear signal-to-noise ratio (SNR), equal on both hops.

4.1.3 Evaluated Configurations

This thesis evaluates the primary (constellation, channel) configurations listed in Table 4.1, plus the QPSK/unknown-ISI generalization of Section 6.2.2 noted there.

Coherent BPSK over complex flat Rayleigh fading is theoretically valid and is not excluded on any principled grounds. With $y = hx + n$ and $x \in \{-1, +1\}$, and h known at the receiver, $z = \Re\{h^* y\}$ (or any positive scaling of it) is a sufficient statistic: once the channel phase is removed the entire BPSK component lies on the real decision axis, and the orthogonal component carries no information about x. Discarding it therefore discards neither signal energy nor half of anything useful.

QPSK is the canonical constellation here for reasons of experimental design rather than validity. It exercises a genuinely two-dimensional complex constellation, carries two bits per symbol, and matches the complex-signal neural processing this thesis sets out to study: the fading coefficient $h \sim \mathcal{CN}(0, 1)$ rotates as well as scales, and a QPSK constellation has a phase for it to act on. BPSK is retained in Chapter 6 for the opposite reason, to keep the ISI trellis exact and the controlled comparisons compact. A complex constellation on AWGN is likewise not wrong, but it exercises none of what makes the complex channel interesting, since there is no phase rotation to undo, and it is not reported here.

Table 4.1: The primary evaluated (constellation, channel) pairings. The canonical and coded rows use the same channel on both hops; the unknown-channel row's Hop-1 filter is fixed throughout (not redrawn per block) and unknown to AF/DF/MLP, though the MLP is trained on it, matching the genie-CSI/pilot-aided Viterbi baselines; Hop-2 is AWGN (Chapter 6). A QPSK generalization of the unknown-ISI study, with the same Hop-1/Hop-2 split, is reported separately in Section 6.2.2

Constellation	Channel	Role	Why this pairing
QPSK (complex)	Rayleigh (complex)	Canonical	The complex case, and the operating point this thesis draws its conclusions on. QPSK places one bit on each axis, so both dimensions the fading coefficient acts in carry information and none is discarded.
BPSK (real)	Unknown 3-tap ISI	Extension	Used only in Chapter 6, where the impairment under study is channel *memory* rather than fading, and the one-dimensional alphabet keeps the trellis small enough for an exact Viterbi benchmark.
16-QAM (complex)	Rayleigh (complex)	Coded MCS rung	Used only as one rung of the modulation-and-coding grid of the coded study (Section 5.5), where the object of study is the rate-and-modulation choice rather than the relay's ability to process a denser constellation.

Beyond its role as the noise term of every channel model and its use in the closed-form expressions quoted as analytical background (Chapter 2), AWGN also appears as the Hop-2 channel of one unknown-channel relay comparison (Chapter 6); no standalone AWGN-only relay study is run.

A consequence worth stating plainly. The SNR argument sets the noise power from the average *symbol* energy, so it denotes E_s/N_0 in every configuration. Since $E_b = E_s/k$, E_b/N_0 is 3.01 dB below the stated value for QPSK. Where a BER figure is read against a BPSK closed form, this conversion is stated explicitly at the point of use.

Power Normalization. All relay strategies normalize their output power to ensure a common empirical transmit-power constraint:

$$x_R \leftarrow x_R \cdot \sqrt{\frac{P_{\text{target}}}{P_{\text{current}}}} \tag{4.8}$$

Here P_{current} is estimated over the complete simulated relay-output block. The normalization is therefore block-adaptive and noncausal at the symbol level; it is appropriate to the block-buffered simulation but should not be read as an online gain known before the block has been observed.

4.1.4 Relay Processing

On Hop 1 the relay's single receive antenna observes, as shown in Equation 4.9:

$$\tilde{y}_R[i] = x[i] + \frac{n_1[i]}{h_1[i]}, \quad \frac{n_1[i]}{h_1[i]} \sim \mathcal{CN}\left(0, \frac{\sigma^2}{|h_1[i]|^2}\right) \tag{4.9}$$

matching Eq. (4.6): the effective observation follows one-tap zero-forcing equalization of the Rayleigh fading coefficient (Section 4.1), which removes the gain and leaves conditional complex-noise power $\sigma^2/|h_1[i]|^2$, or variance $\sigma^2/(2|h_1[i]|^2)$ per real axis, rather than the fixed per-axis variance $\sigma^2/2$ that phase compensation would leave. Note that the per-symbol effective SNR is $|h_1[i]|^2\gamma$ (a random, exponentially distributed quantity) not a constant — the same value either convention would give, which is why the two are easy to conflate — and the AWGN limit is recovered by setting $|h_1[i]| \equiv 1$. The relay applies its processing function to a sliding window of received samples and transmits a cleaner estimate $\hat{x}_R = f_\theta(\tilde{y}_{R,i-w:i+w})$; classical AF/DF use $w = 0$. This is purely a noise-removal task: in the canonical SISO setup there is no inter-stream interference at any stage, and the destination demodulates the Hop-2 output after that one-tap equalization with no further equalizer.

Remark 4.1 (Realizability of the symmetric window)**.** The canonical *channel* of this chapter is memoryless: after one-tap zero forcing, $\tilde{y}_R[i]$ is already a sufficient statistic for $x[i]$, so $w = 0$ loses nothing, and the classical AF and symbol-wise DF baselines do use $w = 0$ and are strictly causal. The learned relays nevertheless take a symmetric window $y_R[i-w:i+w]$ with $w>0$ everywhere they appear, the canonical experiments included: the MLP uses $w=2$ (a 5-sample window, Section 4.2) and the sequence models $w=5$. This is an *architectural input convention*, not a claim about the channel. Under the canonical i.i.d. assumptions the extra samples are redundant, they carry no additional information about $x[i]$, and the convention is adopted so that the same *input format*, a symmetric window of received samples, carries over to the three-tap ISI channel of Chapter 6, where the window is what spans the memory and is no longer redundant.

What carries over is the format, not the dimensions, and it is worth being exact about this because the two chapters are otherwise easy to read as using one network. The canonical relay of this chapter takes $w=2$ (a 5-sample window) into 24 hidden units, 169 parameters. The unknown-ISI relay of Chapter 6 takes $w=5$ (an 11-sample window, wide enough to span the three-tap response) into 13 hidden units, 170 parameters. Both are two-layer windowed perceptrons of essentially the same size, trained by the same protocol, and neither is tuned per channel; but the window had to widen for the memory, and the hidden layer was traded down to keep the parameter budget comparable. The constant held across the two settings is therefore the *class* of relay and its order of magnitude in parameters, not a single fixed network.

Two consequences follow, and both apply to the canonical experiments rather than only to Chapter 6. First, a symmetric window is not causal: it reads w samples ahead. It is admissible in the half-duplex protocol only because the relay buffers the whole listening-phase block before transmitting, and it costs a fixed w-symbol processing latency. Second, that non-causality is a limitation of the learned relays, recorded as such rather than justified: a strictly causal past-only window $y_R[i-2w:i]$ is what a streaming deployment would require, and it is not evaluated in this thesis. Because the window is redundant on the canonical channel, none of the canonical results depend on the look-ahead; the cost is real only where the window is doing work, which is Chapter 6.

4.2 Relay Strategies

Nine relay strategies were implemented, spanning four learning paradigms. The selection of these nine strategies is designed to systematically explore the relay design space along three axes:

1. the degree of processing (from no learning to deep generative models).
2. the type of inductive bias (feedforward, recurrent, attention, generative).
3. the model capacity (from 0 to 26K parameters).

This section describes each strategy's architecture, training procedure, and the design rationale motivating each choice.

4.2.1 Selection and Classification of Relay Strategies

The selected nine relay strategies are drawn from four distinct architectural concepts:

(i) classical methods (AF and DF); (ii) supervised-learning models (MLP and hybrid MLP–DF); (iii) generative models (VAE and cGAN); and (iv) sequence models (Transformer, Mamba-S6, and Mamba-2 SSD).

1. **Classical AF:** Gain amplification as shown in Equation 2.1. AF is the minimal-processing classical baseline. It performs no learned processing and simply rescales the received signal, preserving both signal and noise. Its end-to-end effective SNR is given in Equation 2.2.

2. **Classical DF:** Demodulates, recovers bits, and re-modulates clean symbols (QPSK on the canonical channel, applied per axis). DF is the maximal-processing classical baseline. It performs full signal regeneration through symbol-wise demodulation and re-modulation. On the canonical matched channel and in the uncoded setting studied here, DF achieves the lowest BER among the classical baselines at medium-to-high SNR (see Remark 4.2). Its theoretical BER follows the error-composition formula of Equation 2.4.

3. **Supervised MLP (Minimal):** A two-layer feedforward neural network (multilayer perceptron, MLP) with 169 parameters following the relay architecture defined in Equation 2.9,

   where $\mathbf{w} \in \mathbb{R}^5$ is a sliding window from one rescaled real decision axis ($w = 2$ neighbors on each side), and the hidden layer has 24 neurons. Parameters: $(5 \times 24 + 24) + (24 \times 1 + 1) = 169$.

   Real network on a complex channel. The network above is real-valued, while the canonical channel and its QPSK constellation are complex. The two are reconciled by operating *per axis*: the physical QPSK components are $\pm 1/\sqrt{2}$, and it is convenient to express each in normalized binary coordinates, $u_I = \sqrt{2}\Re\{\tilde{y}\}$ and $u_Q = \sqrt{2}\Im\{\tilde{y}\}$, whose clean targets are ± 1. The identical 169-parameter network is applied to both normalized streams and the outputs are recombined as $\hat{x}_R = (\hat{u}_I + j\hat{u}_Q)/\sqrt{2}$ before the common power-normalization step. The parameter count is unchanged by this — one set of weights is shared across both axes, which is what makes the two sub-problems the *same* problem — but the arithmetic cost per transmitted symbol is twice the per-window figure. Every learned relay in the canonical comparison of Chapter 5 (MLP, hybrid, VAE, cGAN and the sequence models) is applied this way; AF and symbol-wise DF operate on the complex sample directly.

   This is a design choice rather than the only option, and the coded study of Section 5.5 makes the opposite one for contrast: its relay interleaves the I and Q windows into a single input vector and classifies over the joint four-point QPSK alphabet, so it sees the constellation as a two-dimensional object rather than two independent axes. That choice is the main reason its parameter count is larger, and it is stated where it is used.

> **Architecture note:** The MLP relay is a standard discriminative two-layer perceptron trained with supervised learning (MSE loss on input–output pairs). In contrast, the generative models in this study are the VAE and cGAN (Section 4.2), which learn to *sample from* or *approximate* the data distribution. The MLP relay simply learns a deterministic mapping $f : \mathbb{R}^5 \to [-1, +1]$ from a noisy observation window to a denoised estimate of one axis: a classical regression task.

Design rationale: The tanh output activation naturally constrains the output to $[-1, +1]$, matching the binary ± 1 alphabet each axis carries under the per-axis application described above (and, in the real-valued unknown-channel study of Chapter 6, the BPSK symbol range directly). The ReLU hidden layer provides the non-linearity needed to approximate a soft threshold. The fixed-variance form $\tanh(y/\sigma^2)$ is qualitative motivation, not the exact target: under zero forcing, reliability varies with $|h_1[i]|^2$, so the CSI-conditioned posterior mean forms a family of fading-dependent thresholds. The network receives no channel coefficient of its own. The canonical pipeline as a whole is not CSI-free, however: the one-tap zero-forcing front end that produces $\tilde{y}_R = y_R/h_1$ uses h_1, so CSI enters before the learned stage rather than inside it. Only the unknown-channel study of Chapter 6 withholds a per-block estimate from the pipeline as a whole. Because $|h_1[i]|$ is not supplied explicitly to the network, the noise on $\tilde{y}_R$ is not fixed-variance AWGN but a scale mixture of Gaussians, its per-symbol variance $\sigma^2/|h_1[i]|^2$ randomised by the fading. The canonical relay function is therefore $f^*(\tilde{y}) = \mathbb{E}[X \mid \tilde{Y}]$ under that mixture, not the fixed-σ^2 AWGN denoiser $\tanh(y/\sigma^2)$ that Chapter 3 uses for intuition: the network instead learns an empirical estimator marginalized over the fading distribution and training SNRs. With 24 hidden neurons and a 5-dimensional input, the network has approximately 34 parameters per input dimension; whether this capacity is sufficient is determined experimentally, not from the parameter count alone. He initialization is used for ReLU layers to maintain proper gradient flow.

Training uses MSE loss with multi-SNR training data (i.e., 5, 10, and 15 dB), 25,000 samples, and 100 epochs with a learning rate of 0.01.

4. **Hybrid:** SNR-adaptive relay that switches between MLP (low SNR) and DF (high SNR) based on a learned threshold. Combines the AI advantage at low SNR with DF's low-BER, zero-parameter operation at high SNR. Same 169 parameters as MLP.

   Design rationale: The Hybrid relay is motivated by the prediction that the two families are strongest in different SNR regimes — hypotheses H1 and H2 of Section 3.3 state that expectation and Chapter 5 tests it. If it holds, neither family is the right choice across the whole range, and a switching threshold lets the relay select between them per operating condition. The threshold is determined empirically from the training data by finding the

SNR at which MLP and DF BER curves cross. This approach requires no additional parameters beyond those of the MLP sub-network.

Training points at the relay node: SNR = **5, 10, 15 dB** (same trained MLP sub-network as the Minimal MLP relay).

5. **Generative VAE:** Probabilistic relay with encoder $q_\phi(\mathbf{z}|\mathbf{x})$ mapping to a latent space and decoder $p_\theta(\mathbf{x}|\mathbf{z})$ reconstructing the signal. Architecture: encoder $(7 \to 32 \to 16 \to \mu, \sigma^2(8))$, decoder $(8 \to 16 \to 32 \to 1)$. Total: 1,777 parameters. Trained with β-VAE loss ($\beta = 0.1$) for 100 epochs.

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The low $\beta = 0.1$ prioritizes reconstruction quality over latent space regularization, appropriate for a task where accurate signal recovery is paramount. The 8-dimensional latent space provides sufficient representational capacity for the BPSK signal manifold while maintaining a tractable KL divergence. The encoder window of 7 symbols provides local context.

6. **Generative cGAN (WGAN-GP)** [32, 33]**:** Adversarial relay with a generator conditioned on the noisy signal and a critic providing the training signal. Generator: $(7 + 8 \to 32 \to 32 \to 16 \to 1)$, Critic: $(1 + 7 \to 32 \to 16 \to 1)$. Total: 2,946 parameters. Trained with Wasserstein loss, gradient penalty ($\lambda = 10$), and L1 reconstruction loss ($\lambda_{\text{L1}} = 100$) for 200 epochs.

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The high $\lambda_{\text{L1}} = 100$ ensures that the generator is primarily supervised by the reconstruction loss, with the adversarial term acting as a regularizer that encourages outputs to lie on the clean signal manifold. The noise vector $\mathbf{z} \in \mathbb{R}^8$ provides the stochastic input needed by the GAN framework. The 5:1 critic-to-generator update ratio follows the WGAN-GP recommendation for stable critic training. The doubled epoch count (200 vs. 100) compensates for the slower per-step convergence of adversarial training.

7. **Sequential models: Transformer:** Multi-head self-attention over a window of 11 symbols. Architecture: $d_{\text{model}} = 32$, 4 attention heads, 2 encoder layers, feedforward dimension 128. Total: 17,697 parameters. Trained for 100 epochs with Adam optimizer ($\text{lr} = 10^{-3}$).

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The Transformer is included as the dominant sequence architecture in modern deep learning. The 11-symbol window provides the same temporal context as the Mamba-S6/SSD models. With $d_{\text{model}} = 32$ and 4 heads, each head operates in an 8-dimensional subspace, which is sufficient for capturing the local noise structure. The feedforward dimension of 128 ($4 \times d_{\text{model}}$) follows the standard Transformer expansion ratio.

8. **Sequential models: Mamba-S6** Selective state space model with input-dependent state transitions. Architecture: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, 2 Mamba blocks with residual connections. Total: 24,001 parameters. Each block applies: LayerNorm → expand $(32 \to 64) \to$ split to Conv1D/SiLU/S6 main branch and SiLU gate → contract $(64 \to 32) \to$ residual. Trained for 100 epochs with Adam optimizer (lr $= 10^{-3}$).

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The expand factor of 2 doubles the internal dimension during the S6 scan, and the state dimension $N = 16$ makes each layer a 16th-order adaptive filter, more expressive than a classical Wiener or matched filter. LayerNorm, residual connections and SiLU gating are standard stabilizers carried over from the reference implementation.

9. **Sequential models: Mamba-2 SSD** Structured State Space Duality model that replaces the sequential S6 recurrence with a chunk-parallel structured matrix multiply. Architecture: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, chunk size 8, 2 Mamba-2 blocks with SiLU gating and residual connections. Total: 26,179 parameters. Each block applies: LayerNorm → parallel gate/SSD branches → SiLU gate → contract $(64 \to 32) \to$ residual. The SSD layer builds a lower-triangular causal kernel M per chunk and applies it via batched matmul, with inter-chunk state passing for continuity. Trained for 100 epochs with Adam optimizer (lr $= 10^{-3}$) and gradient clipping ($\|\nabla\| \leq 1$).

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The chunk size of 8 balances parallelism against the quadratic memory cost of the $L \times L$ chunk matrix, giving two chunks at the 11-symbol window. Gradient clipping at norm 1.0 and the S4D-style initialization $A_{\text{log}} = \log(1, 2, \ldots, N)$ are carried over from the reference implementation.

Remark 4.2 (Terminology: symbol-wise DF vs. information-theoretic DF)**.** The term "decode-and-forward" is used in two distinct senses in the literature, and the distinction matters for interpreting the results of this thesis. In its *information-theoretic* sense [1, 2], DF is a *block* strategy: the source protects each message block with a strong (ideally capacity-achieving) channel code, the relay decodes the entire block, and only then re-encodes and forwards the recovered message. This block-DF is inherently *not* memoryless and, over a single link, can operate reliably at any rate below the first-hop capacity.

The canonical system studied in the core of this thesis is *uncoded*: one QPSK symbol carries two information bits, one per axis, and no outer error-correction code is applied there. Consequently, the "DF" baseline throughout the canonical comparison refers to *symbol-wise slicing*: per-symbol Gray demodulation on the coherently equalized I and Q components followed by QPSK re-modulation, with $w = 0$. This is the appropriate classical counterpart to the learned relays under the uncoded BER metric used there, but it is strictly weaker than block-DF with coding; the reported DF results should not be read as bounds on coded block-DF performance. This caveat is measured directly, not left as a standing assumption: Section 5.5 adds a practical finite-length coded block-DF baseline (a rate-1/2 convolutional code with soft-decision Viterbi decoding, swept over constraint length on the QPSK/Rayleigh link) and reports where it does and does not beat the symbol-wise DF used elsewhere in this thesis.

4.3 Simulation Framework

4.3.1 Monte Carlo BER Estimation

BER is estimated through Monte Carlo simulation, which provides an unbiased estimate of the true BER at each SNR point. The simulation parameters are:

- **Bits per trial:** 10,000
- **Trials per SNR:** 10 (independent random seeds)
- **Total bits per SNR point:** 100,000
- **SNR range:** 0 to 20 dB (step: 2 dB), yielding 11 evaluation points
- **Random seed control:** Each trial uses a unique seed for bit generation and noise realization

BER Computation:

$$\hat{P}_e = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(b_i \neq \hat{b}_i) \tag{4.10}$$

where $\mathbb{1}(\cdot)$ is the indicator function and $N = 10{,}000$ is the number of bits per trial. By the central limit theorem, for large N and moderate BER (say $P_e \geq 10^{-3}$), the per-trial BER estimate is approximately normally distributed with variance $P_e(1 - P_e)/N$.

Confidence Intervals. The 95% confidence interval is computed from the $M = 10$ independent trial estimates as:

$$\mathrm{CI}_{95\%} = \bar{P}_e \pm t_{0.975,M-1} \cdot \frac{s}{\sqrt{M}} \tag{4.11}$$

where $\bar{P}_e$ is the sample mean BER across trials, s is the sample standard deviation, and $t_{0.975,9} = 2.262$ is the two-sided critical value of the Student's t-distribution with 9 degrees of freedom. At low BER ($\hat{P}_e \lesssim 10^{-4}$), the normal approximation may become unreliable due to the small number of observed errors; however, in this regime the relay methods converge and the relative ranking is stable.

BER resolution. With $N \cdot M = 100{,}000$ total bits per SNR point, the minimum detectable

BER is approximately $1/N = 10^{-4}$ per trial, or 10^{-5} for the aggregate. BER values below this threshold are reported as 0.

Per-experiment simulation budgets. The budget above ($M = 10$ trials $\times N = 10{,}000$ bits $= 100{,}000$ bits per SNR point) applies to the canonical relay comparison and the parameter-normalization/complexity study (E2–E3). Other experiments use different budgets, stated in their respective sections: the main unknown-ISI study, the flat-channel control, and the QPSK generalization check (Chapter 6, Sections 6.2, 6.2.1, 6.2.2) use $10 \times 100{,}000 = 1{,}000{,}000$ bits per SNR point, matching the project-wide $M = 10$ standard; the composite cascade (Section 6.2.4) likewise uses $10 \times 100{,}000$; and the blind and partial-posterior sub-studies (Sections 6.4–6.3) use $50 \times 20{,}000 = 1{,}000{,}000$ bits per SNR or operating point. The larger trial count in those last two is deliberate rather than incidental: their hop-1 channel redraws its ISI, amplifier and phase realization for every block, so a trial is a draw from the impairment family and the ensemble average is limited by the trial count, not by bits per trial. Increasing bits per trial sharpens the estimate for one particular channel; only increasing trials averages over the family the study is about. Every experiment therefore uses $M \geq 10$ trials, so the Wilcoxon signed-rank test is attainable throughout (at $M = 5$ its smallest two-sided p is $2^{-4} = 0.0625$, which could not reach $p < 0.05$).

4.3.2 Statistical Significance Testing

Differences between relay methods are assessed using the **Wilcoxon signed-rank test**, a non-parametric paired test. For each pair of relay strategies (A, B) at each SNR point, the test compares the $M = 10$ paired BER observations (Equation 4.12):

$$H_0 : \text{the paired-difference distribution is symmetric about } 0 \tag{4.12}$$

The Wilcoxon test is preferred over the parametric paired t-test because BER observations are bounded in $[0, 1]$ and may be skewed. Under a symmetric paired-difference distribution the test can be interpreted as testing a zero median difference; without that assumption it tests a zero pseudomedian. Failure to reject this null is not evidence of equivalence, which would require a prespecified practical margin and an equivalence test. The significance level is set at $\alpha = 0.05$ *per comparison*; no multiple-comparison correction (e.g. Holm or Bonferroni, or false-discovery-rate control) is applied across the many relay-pair and SNR-point comparisons made throughout this thesis, so the reported significance claims should be read at the per-comparison level rather than as holding jointly across the full set of tests performed. Two further limits bound how these tests should be read. First, the comparisons were not prespecified: the relay pairs and SNR points reported were chosen after inspecting the curves, so the analysis is exploratory and its p-values are descriptive rather than confirmatory. Second, the intervals quoted throughout are computed over Monte Carlo channel and noise realizations at a fixed set of trained weights. They therefore do not include training-seed variability, which is reported separately and only where it was measured — the minimum-size sweep of Section 5.4 and the equal-budget comparison of Table 9.6, each replicated over three initializations. Every other learned result rests on a single training run, and a difference smaller than the 0.180 dB across-initialization spread measured there should not be treated as resolved.

4.3.3 Training Protocol

In the original canonical comparisons, each AI relay is trained **once** at the beginning of the experiment using: - **Multi-SNR training data:** Signals generated at SNR = 5, 10, 15 dB in equal proportions - **Training samples:** 25,000 (supervised), 20,000 (generative), 10,000 (sequence models) - **Single training per channel:** The same trained model is evaluated across all SNR points

The multi-SNR training protocol is critical: training at a single SNR would produce a model that performs well at that SNR but poorly at others (overfitting to a specific noise level). By training at three representative SNR values spanning the low-to-moderate range, the network learns a robust denoising function that generalizes across operating conditions. The SNR values 5, 10, 15 dB were chosen to cover the regime where AI relays provide the most benefit (low-to-medium SNR).

Impact of sparse SNR training grid. Because models are trained at only three SNR points ($\{5, 10, 15\}$ dB) but evaluated over eleven points (0 to 20 dB, step 2), performance reflects two regimes: (i) **interpolation** within the trained span (approximately 4–16 dB), where BER is generally stable, and (ii) **extrapolation** at the edges (0–2 dB and 18–20 dB), where mild degradation may occur due to noise-statistics mismatch. In the present results, this sparse-grid effect is secondary: the relay ranking remains consistent across the full evaluated SNR range.

Training method (what is trained, where it is trained). Training is performed **offline** before BER sweeps. Only the seven AI relays are optimized (MLP, Hybrid's MLP sub-network, VAE, cGAN, Transformer, Mamba-S6, Mamba-2 SSD); AF and DF are analytical baselines and have no trainable parameters. Training data are synthetically generated source/channel pairs under the multi-SNR protocol, with model-specific losses (MSE for supervised relays, ELBO for VAE, WGAN-GP + L1 for cGAN, Adam-based supervised loss for sequence models). Compute placement follows implementation: NumPy models (MLP/Hybrid) train on CPU, while PyTorch models train on CPU or CUDA depending on device availability/configuration. After training, weights are checkpointed and reused across all SNR evaluations; BER curves are then produced in **inference-only** mode without further parameter updates.

Weight initialization. He initialization [34] is used for all layers with ReLU activation ($W \sim \mathcal{N}(0, 2/n_{\text{in}})$), while Xavier initialization is used for layers with tanh activation ($W \sim \mathcal{N}(0, 1/n_{\text{in}})$). These initialization schemes maintain proper gradient magnitude through the network layers, preventing both vanishing and exploding gradients during training.

Optimizer settings. All PyTorch-based models use the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ (defaults). The MLP and Hybrid relays use a simple SGD implementation in NumPy with constant learning rate $\eta = 0.01$.

4.3.4 Reproducibility and Weight Management

All experiments use controlled random seeds at both the source (bit generation) and noise (per-trial seeding) levels to ensure reproducible results. Specifically, for seed s and trial t, the random state is initialized as seed$(s \cdot 1000 + t)$, ensuring that each trial is independent while the overall experiment is deterministic.

A **weight checkpoint system** saves trained model parameters to disk after training completes, enabling experiment resumption without retraining. This is particularly important for the cGAN (2-hour training time) and sequence models (24–37 minutes). The checkpoint system supports: - Automatic save after training with architecture metadata - Resume from saved weights with architecture validation - Seed-specific weight directories (e.g., `trained_weights/seed_42/`)

The framework includes 187 automated tests (pytest) covering all modules: channels, modulation, coding, relay strategies, simulation, statistics, and weight management, with 100% pass rate.

Code and reproducibility archive. The complete simulation framework (`relaynet`), every experiment script, and the committed result files each figure and table is generated from are maintained under version control (git) in the `Gilzuk/relaynet2` repository; each numerical claim in this thesis traces to a specific committed script and result file rather than to an unarchived intermediate run, and `verify_thesis_tables.py` in that repository re-derives every numerical table from its source file and flags any mismatch. The exact commit this thesis was typeset from is recorded in the repository's history rather than reproduced here, since stating a fixed hash in the text it describes is self-referential; the repository's tags and commit log identify the revision corresponding to any given PDF build.

4.4 Normalized Parameter Comparison

A fundamental confound in comparing AI architectures is that different models have vastly different parameter counts: from 169 (MLP) to 26,179 (Mamba-2 SSD): a ratio of 155:1. Performance differences between these models could reflect either (a) the superiority of one architecture's inductive bias, or (b) the simple advantage of having more learnable parameters. To reduce this confound, all seven AI models were scaled to approximately 3,000 parameters, giving a comparison at an equal parameter budget. The control is over capacity alone: as Table 4.2 shows, meeting a common budget required changing width, depth, state dimension and, for the feedforward models, the input window size, so those factors co-vary with architecture and the comparison does not isolate architectural inductive bias from them. The MLP-3K in particular sees an 11-sample window against its canonical 5, so it is not fed the same context as the models whose windows were left unchanged.

Choice of target parameter count. The target of ~3,000 parameters was chosen as a compromise: it is large enough that all architectures can express a reasonable denoising function (avoiding underfitting), yet small enough that the normalization represents a meaningful reduction for the larger models (Transformer: 5.9× reduction, Mamba-S6: 7.9×, Mamba-2 SSD: 8.7×). The MLP model is scaled **up** from 169 to 3,004 parameters, testing whether additional capacity benefits the simple feedforward architecture.

Scaling methodology. Each architecture's hyperparameters were adjusted to reach the target while preserving its essential structure:

Table 4.2: Normalized 3K-parameter model configurations: parameter count and scaling method for each architecture

Model	Parameters	Scaling Method
MLP-3K	3,004	Increased window (5→11) and hidden (24→231)
Hybrid-3K	3,004	Same as MLP-3K + DF switching
VAE-3K	3,037	Increased window (7→11), adjusted latent/hidden
cGAN-3K	3,004	Increased window (7→11), adjusted hidden layers
Transformer-3K	3,007	Reduced d_model (32→18), heads (4→2), layers (2→1)
Mamba-S6-3K	3,027	Reduced d_model (32→16), d_state (16→6), layers (2→1)
Mamba-2-SSD-3K	3,004	Reduced d_model (32→15), d_state (16→6), layers (2→1)

All seven configurations use a window size of 11 (versus 5 for the original MLP/Hybrid and 7 for the original VAE/cGAN; the original sequence models already used 11), and every parameter count lands within ±1.2% of the 3,000 target.

Parameter counting convention. All learnable parameters are counted, including biases, normalization parameters (LayerNorm gain/bias), and projection matrices. Non-learnable buffers (e.g., positional encoding tables, fixed **A** matrices) are excluded. The counts are verified programmatically via `sum(p.numel() for p in model.parameters() if p.requires_grad)`

Unified input window. All 3K models use a window size of 11 (vs. 5 for original MLP/Hybrid, and 7 for original VAE/cGAN), providing a common input context. This ensures that differences in performance reflect architectural inductive biases rather than differences in the amount of input information available.

This normalization removes parameter count as an explanation for any residual gap, which is what makes the comparison informative: if performance converges at equal parameter budgets, the conclusion is that capacity (not architecture) was the dominant factor in the unnormalized comparison. If significant gaps persist, the conclusion is weaker than "architectural inductive bias provides an advantage", because the scaling changed more than the architecture label: it is that something other than raw parameter count accounts for the gap, with architecture, depth, width, state dimension and window size still confounded among themselves.

4.5 Modulation: QPSK

The experiments use QPSK, the canonical constellation. This section defines it and the I/Q-splitting technique that lets real-valued relays process a complex constellation.

4.5.1 QPSK: Gray-Coded Quadrature Phase-Shift Keying

Quadrature Phase-Shift Keying maps pairs of bits (b_0, b_1) to complex symbols [27]:

$$x = \frac{(1 - 2b_0) + j(1 - 2b_1)}{\sqrt{2}} \tag{4.13}$$

yielding four constellation points at $\{(\pm 1 \pm j)/\sqrt{2}\}$ with unit average power ($E_s = 1$). The Gray coding ensures that adjacent constellation points differ by exactly one bit:

Table 4.3: QPSK Gray-coded symbol mapping: bit pairs to complex symbols

Bit pair (b_0, b_1)	Symbol	Quadrant
00	$(+1 + j)/\sqrt{2}$	I
01	$(+1 - j)/\sqrt{2}$	IV
11	$(-1 - j)/\sqrt{2}$	III
10	$(-1 + j)/\sqrt{2}$	II

Demodulation applies independent sign decisions on each component: $\hat{b}_0 = \mathbb{1}(\text{Re}(\hat{x}) < 0)$ and $\hat{b}_1 = \mathbb{1}(\text{Im}(\hat{x}) < 0)$. The spectral efficiency is 2 bits/symbol, double that of BPSK.

Theoretical QPSK BER. For uncoded QPSK over an AWGN channel, the BER per bit equals the BPSK BER at the same E_b/N_0 because each I/Q component carries an independent BPSK stream [27]:

$$P_b^{\text{QPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = P_b^{\text{BPSK}} \tag{4.14}$$

The advantage of QPSK is doubled throughput for the same BER and per-bit energy.

4.5.2 Constellation Diagram Summary

Figure 4.2 presents the constellation diagrams for the modulation schemes appearing in this thesis. The progression from BPSK (2 points on the real axis) through QPSK (4 points on the unit circle) to denser grids illustrates the fundamental trade-off between spectral efficiency and noise robustness: higher-order constellations pack more bits per symbol but reduce the minimum Euclidean distance between points, increasing the BER at a given SNR. QPSK is the constellation used in every canonical relay-architecture comparison; 16-QAM appears only as a rung of the coded link-adaptation study (Section 5.5), and BPSK only in the unknown-channel study's relay comparisons (Chapter 6). The others are shown for orientation.

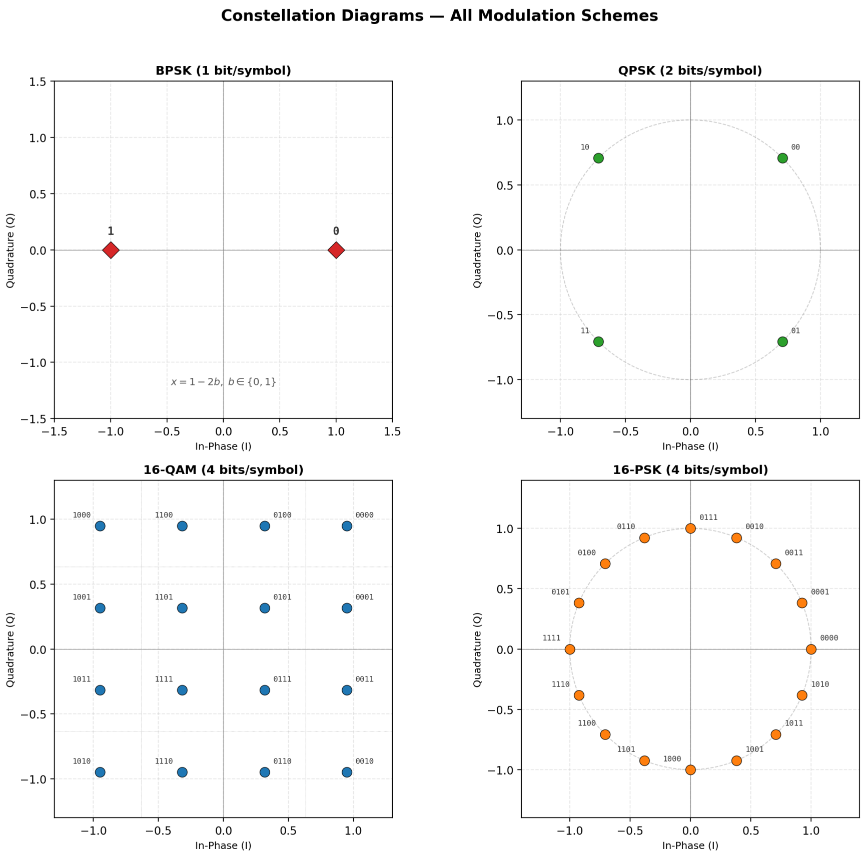


Figure 4.2: Constellation diagrams. (a) BPSK: 2 real-valued symbols at ± 1. (b) QPSK: 4 Gray-coded symbols on the unit circle. (c) 16-QAM: 16 Gray-coded symbols on a 4×4 rectangular grid with decision boundaries (dotted lines). (d) 16-PSK, shown for completeness only; it is not studied in this work. All constellations are normalised to unit average power

4.5.3 I/Q Splitting for AI Relay Processing of Complex Constellations

$$\hat{x}_R = \frac{f_\theta(\sqrt{2}\,\mathrm{Re}(y_R)) + j f_\theta(\sqrt{2}\,\mathrm{Im}(y_R))}{\sqrt{2}} \tag{4.15}$$

Justification and caveat. For rectangular constellations (QPSK, QAM), the I and Q components carry independent information and are corrupted by independent noise conditional on the channel, so per-axis denoising with a shared real-valued function is a natural and parameter-efficient decomposition. It is, however, a *restriction* of the hypothesis class to axis-separable functions, not an equivalence to joint 2D processing. In particular, when fading reliability is not supplied explicitly, a joint processor could use the shared I/Q magnitude as information about that reliability. No joint-QPSK ablation was run, so zero cost from splitting is not claimed; for denser constellations it becomes a structural limitation that a joint two-dimensional relay formulation would be needed to remove, which is left to future work.

Relay-specific handling:

Table 4.4: Relay-specific complex signal handling strategies and rationale

Relay type	Complex signal processing	Rationale
AF	Amplifies complex signal directly	Power normalization ($\|y\|^2$) is valid for complex vectors
AI relays	I/Q splitting (process Re and Im separately)	Real-valued networks; independence of I/Q in rectangular constellations

Limitation beyond binary-per-axis constellations. I/Q splitting works here because QPSK places a single binary decision on each axis, which is exactly what a tanh-output relay trained on $\{\pm 1\}$ represents. Constellations with more than two levels per axis (16-QAM and denser) break that correspondence and would require either modulation-specific retraining or a joint two-dimensional relay formulation; neither is evaluated here, and both are identified as future work.

Chapter 5

Core Experiments: The Canonical Setup

5.1 Experiments summary

Two experiments constitute the core of this chapter: the canonical relay comparison (E2) and the parameter-normalization and complexity study (E3), which together answer every canonical hypothesis (H1–H4). A supplementary coded study (Section 5.5) departs from the canonical setup within this same chapter, adding a rate-1/2 convolutional code, a block-DF relay and an adaptive modulation-and-coding scheme. The principal contribution (Chapter 6) is the unknown-channel study (E6), which tests H5. The core (E2, E3) and unknown-channel (E6) experimental configurations are summarised in the table below; the supplementary coded study's configuration is described separately in Section 5.5.

Each experiment subsection reports the key results inline (BER tables and figures); supporting material and model hyperparameters are collected in the appendices (Appendices 9.1–9.3).

Table 5.1: Summary of core and unknown-channel experimental configurations

Exp.	Topology	Modulation	Channel / Equalization	Focus
E2	SISO	QPSK	Rayleigh (canonical)	Canonical relay comparison (H1, H2)
E3	SISO	QPSK	Rayleigh (canonical)	Parameter normalization & complexity (H3, H4)
E6	SISO	BPSK, QPSK	Unknown (ISI; composite; blind; partial-posterior); control: Rayleigh	Extension: unknown-channel failure of AF/DF and learned mitigation (H5)

Exp.	Topology	Modulation	Channel / Equalization	Focus
E7	SISO	QPSK	Rayleigh (canonical), rate-1/2 convolutional code	Supplementary: coded block-DF against symbol-wise DF and coded-aware learned relays (Section 5.5)
E8	SISO	BPSK, QPSK	All nine families above, plus the coded scenario	Minimum relay size and the overfitting hypothesis H3, with three initializations per configuration (Section 5.4)

5.2 Canonical Relay Comparison (SISO, Rayleigh, QPSK)

Goal: Evaluate the classical and AI-based relay strategies on the canonical setup of this thesis.

Trials: - **Topology:** SISO (both hops) - **Modulation scheme:** QPSK - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** One-tap coherent (per hop); no ISI equalizer needed (memoryless) - **Reported learned relay architectures (six):** Supervised (MLP, Hybrid), Generative (VAE), Sequence (Transformer, Mamba-S6, Mamba-2 SSD) - **NN activation:** ReLU hidden, tanh output

Conclusion: On the canonical Rayleigh channel, AI relays match but do not surpass symbol-wise DF: DF achieves the lowest or tied-lowest BER at essentially every SNR point, including the low-SNR regime, while the best neural relays (MLP, Hybrid, VAE) track it within statistical uncertainty and clearly outperform AF. This confirms H1 (AI relays beat AF at low SNR) and H2 (DF is the best-performing evaluated symbol-wise classical relay at ≥ 6 dB, with zero trainable parameters) on the canonical channel.

5.2.1 Results Tables

Table 5.2: BER on the canonical setup: QPSK over i.i.d. Rayleigh fast fading, SISO on both hops, uncoded, $10 \times 10{,}000$ bits per SNR point. "DF th." is the closed-form two-hop composition of Eq. 2.34 (Section 2.5.2.2), included for direct comparison against the measured DF column. Column abbreviations: Transf. is the Transformer relay, Mamba2 the Mamba-2 SSD relay

SNR (dB)	AF	DF	DF th.	MLP	Hybrid	VAE	Transf.	Mamba-S6	Mamba2
0	0.4076	0.3337	0.3333	0.3349	**0.3329**	0.3359	0.4014	0.4071	0.4032
4	0.3129	**0.2219**	0.2216	0.2233	0.2220	0.2254	0.2758	0.2812	0.2772
8	0.1852	**0.1218**	0.1203	0.1229	0.1220	0.1235	0.1424	0.1448	0.1416

SNR (dB)	AF	DF	DF th.	MLP	Hybrid	VAE	Transf.	Mamba-S6	Mamba2
12	0.0822	0.0580	0.0560	0.0586	**0.0579**	0.0589	0.0628	0.0637	0.0621
16	0.0309	**0.0245**	0.0239	0.0247	**0.0245**	0.0247	0.0260	0.0259	0.0251
20	0.0110	**0.0097**	0.0098	0.0099	**0.0097**	**0.0097**	0.0101	0.0100	0.0098

Two features of Table 5.2 deserve comment. First, symbol-wise DF tracks the closed form (the "DF th." column above): reading the abscissa as E_s/N_0 and converting to E_b/N_0 for QPSK ($k = 2$, so 3.01 dB lower), the two-hop expression $2P(1 - P)$ with $P = \frac{1}{2}\big(1 - \sqrt{\bar{\gamma}/(1 + \bar{\gamma})}\big)$ gives agreement to within a few percent relative error at every tabulated point (1.1% at 20 dB, 2.4% at 16 dB, worst case 3.5% at 12 dB, each recomputable from the four-decimal values as printed), which provides a consistency check on the channel, equalization and hard-decision pipeline against the analysis of Section 2.5.2.2. Figure 5.1 plots this closed form alongside the measured curves, together with the single-hop floor P itself – the BER a genie that recovered hop 1 perfectly would still incur on hop 2 – which no relay may cross and none does at any SNR.

Second, the relays divide by family rather than by paradigm. The three feedforward relays, the MLP, the Hybrid and the VAE, are statistically indistinguishable from DF from 4 dB upward and from each other throughout; the VAE reaches 0.00972 at 20 dB, matching DF exactly at the resolution of this budget. The two sequence models are the ones that separate, trailing by roughly 0.07 in BER at 0 dB before converging by 16 dB. Capacity is not the explanation, since the sequence models are the larger of the two groups; the memoryless channel simply offers no temporal structure for their inductive bias to exploit, a reading the equal-parameter comparison of Section 5.3 tests directly and confirms.

Two comments on Table 5.2. First, symbol-wise DF is the best or tied-best relay at every tabulated point, and the learned relays sit within a few thousandths of it from 4 dB upward: on a matched, known channel the learned relay matches DF rather than beating it, at 169 parameters against DF's none. This is H2, and it is the result the rest of the thesis is measured against. Second, the two sequence models trail the feedforward relays at low SNR (0 dB row of Table 5.2) and converge with them by 16 dB, so their extra capacity buys nothing on a memoryless channel; Section 5.3 separates that from the parameter-count confound.

Why QPSK and not BPSK. QPSK places one bit on each axis the Rayleigh coefficient acts in, carries two bits per symbol, and is processed by the I/Q-split relays without modification, which makes it the constellation that exercises the complex-signal processing this thesis studies. Coherent BPSK over the same complex channel would be entirely valid (after channel-phase removal the whole BPSK component lies on the real decision axis, so projecting onto it discards no signal energy) and the choice here is one of experimental design rather than of correctness (Section 4.1.3). One consequence must be kept in view when reading these BER figures against BPSK closed forms: the SNR axis here is E_s/N_0, and QPSK carries two bits per symbol, so a given abscissa corresponds to an E_b/N_0 that is 3.01 dB lower than the same number on a

BPSK axis. The constellations are not on a common axis and their error rates are not directly comparable.

5.2.2 Results Figures

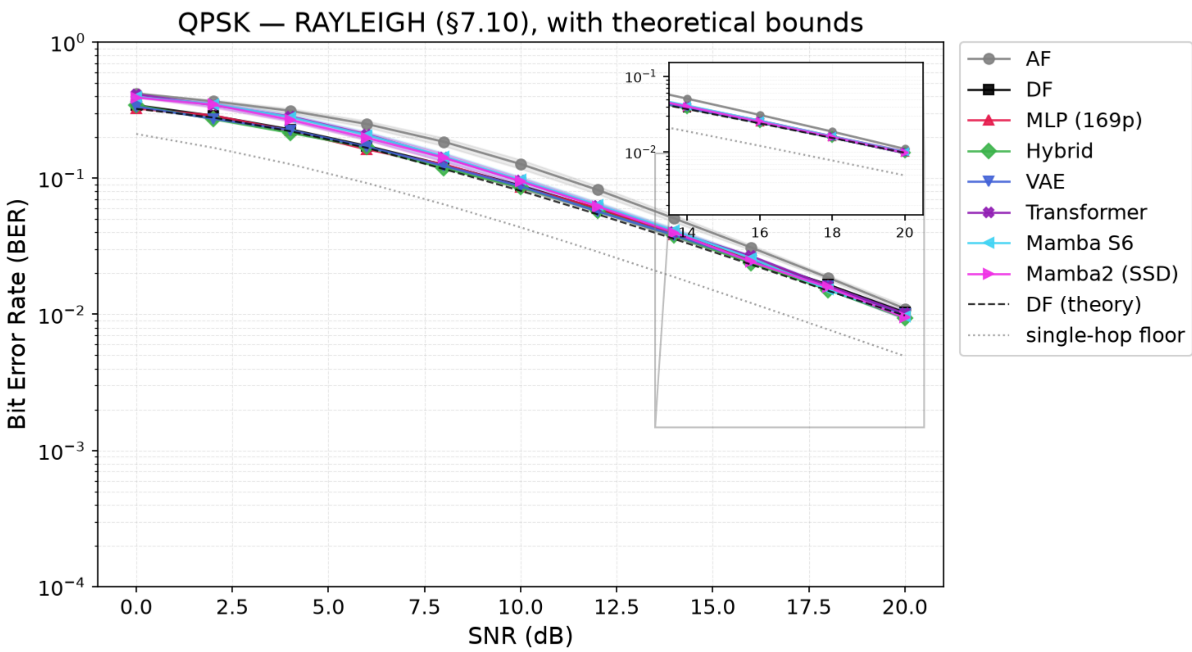


Figure 5.1: The canonical setup: QPSK over i.i.d. Rayleigh fast fading, BER vs. E_s/N_0 for the eight relays of Table 5.2, with 95% confidence intervals, plus two theoretical reference curves: "DF (theory)" (dashed), the closed-form two-hop composition, which the measured DF curve tracks closely, within a few percent relative error everywhere (worst case 3.6%, at 12 dB); and the "single-hop floor" (dotted), the BER a genie that recovered hop 1 perfectly would still incur on hop 2 – a reference floor for this fixed-rate, fixed-power relay chain. AF is uniformly worst; symbol-wise DF and the feed-forward learned relays are indistinguishable from 4 dB upward; the two sequence models trail at low SNR and converge by 16 dB

5.3 Parameter Normalization & Complexity Trade-off

This study tests H3 (complexity versus BER) and H4 (architecture convergence at an equal parameter budget) on the canonical setup.

Goal: Isolate architectural inductive biases from parameter count effects and characterize the complexity-performance trade-off for relay denoising.

Trials: - **Topology:** SISO (both hops) - **Modulation scheme:** QPSK - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** One-tap coherent (per hop); no ISI equalizer needed (memoryless) - **NN architecture:** All normalized to ~3,000 parameters, plus original sizes (169 to 26K) - **NN activation:** ReLU hidden, tanh output

Conclusion: A minimal 169-parameter MLP matches models roughly 140× larger, and no larger model evaluated here improves on it; whether the curve eventually turns upward, and whether overfitting would be the reason, is not tested by *these* experiments (Section 7.2). It is tested by the size sweep of Section 5.4, which replicates every configuration across three initializations and finds no upward arm above the initialization spread. At an equal budget of 3,000 parameters the gap between feedforward and sequence architectures is strongly SNR-dependent: it falls monotonically from 17.4% at 0 dB to 4.1% at 20 dB (Table 5.3). Parameter count therefore accounts for the comparison at high SNR, where the choice of model is nearly irrelevant, but not at low SNR, where the six equal-sized models separate cleanly into a feedforward group tracking DF and a sequence group tracking AF. Since all six carry the same budget, that separation is not explained by capacity; it is attributable to the remaining differences between the models, of which architecture is the most salient but not the only one, since depth, width, state dimension and window size were adjusted to meet the budget (Table 4.2). The generative VAE falls inside the feedforward group at every SNR rather than apart from it, ending at 0.0101 against MLP-3K's 0.0099, and it is ahead of all three sequence models from 0 to 16 dB; the generative paradigm is therefore not a handicap at this budget (Section 2.3.2).

5.3.1 Results Tables

Table 5.3: Equal-parameter-budget comparison on the canonical setup (QPSK over Rayleigh). All learned columns are the ∼3,000-parameter variants; every learned model is scaled to approximately 3,000 parameters, so parameter count is removed as an explanation for the residual differences; width, depth, state dimension and window size still co-vary with architecture (Table 4.2). AF and DF are the unparameterized references. Column abbreviations: Transf. is the Transformer relay, Mamba2 the Mamba-2 SSD relay

SNR (dB)	MLP	Hybrid	VAE	Transf.	Mamba-S6	Mamba2	AF	DF
0	0.3361	0.3391	0.3424	0.4068	0.4007	0.4001	0.4076	**0.3337**
4	0.2244	0.2271	0.2291	0.2818	0.2751	0.2742	0.3129	**0.2219**
8	0.1229	0.1229	0.1267	0.1458	0.1406	0.1405	0.1852	**0.1218**
12	0.0586	0.0580	0.0604	0.0640	0.0620	0.0622	0.0822	**0.0580**
16	0.0247	**0.0245**	0.0251	0.0259	0.0250	0.0251	0.0309	**0.0245**
20	0.0099	**0.0097**	0.0101	0.0100	0.0097	0.0098	0.0110	**0.0097**

5.4 Minimum Relay Size Across Channel Families

The complexity study of Section 5.3 establishes that no evaluated model improves on the 169-parameter MLP, but leaves two questions open: how far *below* 169 parameters the relay can be taken, and whether the answer is a property of the architecture or of the channel. This section answers both by sweeping relay size downward on every channel family used in this thesis.

Goal: Locate the smallest relay that costs nothing measurable against the thesis's own published relay, per channel, and determine which architectural axis — window, depth or width — sets that floor.

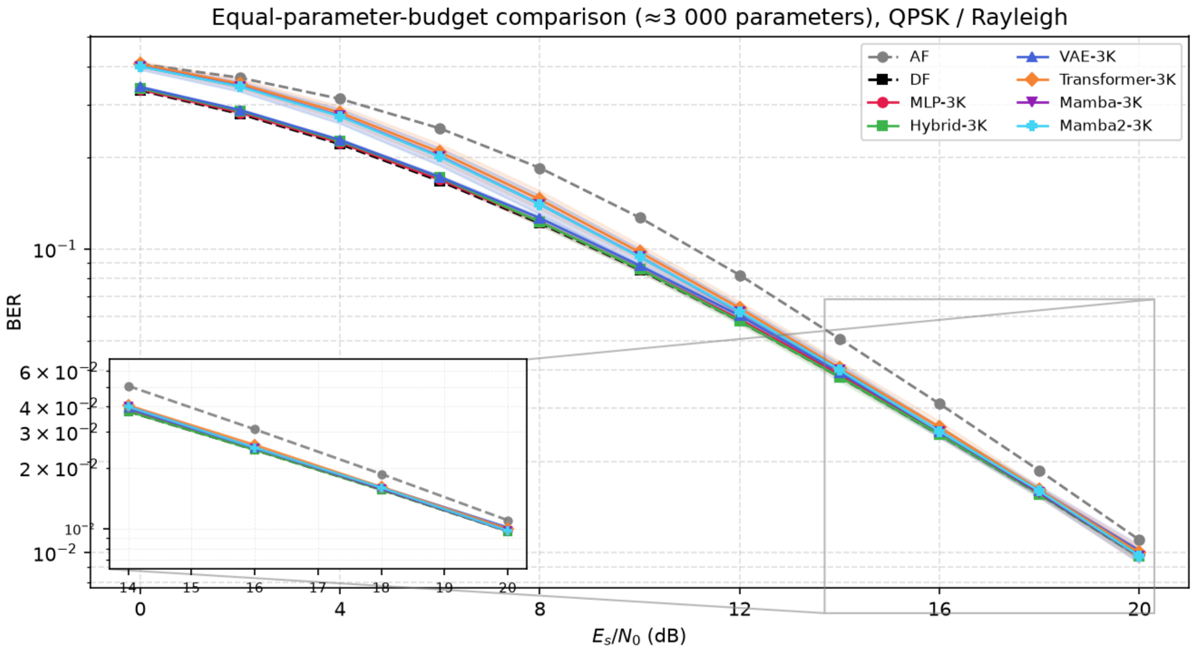


Figure 5.2: BER vs E_s/N_0 for all relay variants normalised to ≈3,000 parameters, QPSK over Rayleigh fading. AF and DF (dashed) are the unparameterised references. Shaded bands show 95% confidence intervals for the learned relays. The feedforward group (MLP-3K, Hybrid-3K, VAE-3K) tracks DF across the full range; the sequence models (Transformer-3K, Mamba-3K, Mamba2-3K) converge with them only at high SNR

Trials: - **Topology:** SISO, two hops, hop 2 per each channel family's own convention (AWGN or coherently-compensated Rayleigh) - **Modulation scheme:** BPSK or QPSK, per channel, as used elsewhere in this thesis - **Channels:** the nine families of Chapters 5 and 6, plus the coded scenario of Section 5.5 - **Comparators:** symbol-wise DF where the channel is memoryless, Viterbi/MLSE where it has memory, block DF for the coded scenario; each channel additionally compared against its own published relay (MLP-169, or MLP-756 coded) - **Relay architecture:** window $\in \{1, 3, 5, 7\}$, depth $\in \{1, 2, 3\}$, width from 1 to 48 - **Replication:** three independent initializations per configuration, each configuration held to its worst - **Metric:** additional SNR required to reach a given BER

Conclusion: The floor is set by channel memory, not by parameter count, and it is far lower than 169 parameters only where the channel is memoryless. Table 9.4 gives the smallest configuration meeting a range of degradation budgets against each scenario's own published relay. At zero degradation, three of the five memoryless channels are matched by four parameters (a single hidden unit with no window, $42\times$ smaller than MLP-169) whereas the three-tap channels require 73 to 145 parameters, only $1.2\times$ to $2.3\times$ smaller. Reading the window column rather than the parameter column explains why: every memoryless row settles at window 1 and every three-tap row needs window 5 or 7 until a full decibel of degradation is permitted. Figure 9.1 shows the same effect directly, widening the window is worth 8 to 11 dB on the three-tap channels and nothing at all on the memoryless ones, and Figure 9.2 plots the size required against the degradation the designer is willing to concede. Depth was varied from one to three hidden layers and improved no configuration; where a two-layer network matched, a single layer matched at fewer parameters. The complexity question is therefore not ”how many parameters” but ”how much window”, and the answer is fixed by the channel’s impulse response rather than by the architecture. Section 7.2.3 takes up what this implies for the overfitting hypothesis H3, and reports the across-initialization spread that bounds the resolution of these figures.

The full budget-by-budget table is Table 9.4 in Appendix 9.5, together with the two figures that show the window effect directly (Figures 9.1 and 9.2).

5.5 Coded Block-DF: Measuring the Remark 4.2 Caveat

A note on constellation scope. The canonical and coded-study fixed-MCS relay-decoding comparisons of this thesis (block-DF vs. the learned relay at a fixed code rate and constellation) are conducted on QPSK throughout; the unknown-channel study’s relay comparisons mainly use BPSK (Chapter 6), with one exception (Section 6.2.2) that repeats the unknown-ISI relay comparison with QPSK in place of BPSK. 16-QAM is not studied as a relay-architecture question (Section 4.5): it does appear as one rung of the modulation-and-coding ladder that the link-adaptation study of Section 5.5.3 selects from, and that study does compare the block-DF and denoise-only relay strategies at 16-QAM rungs (e.g. Table 5.11), but the object of study there is the *rate-and-modulation choice* rather than the relay’s ability to process a denser constellation. Nothing in that study rests on how a learned relay handles four levels per axis, which is the question left to future work.

Remark 4.2 draws a distinction the rest of this thesis relies on: the DF baseline used everywhere here is *symbol-wise* (uncoded, per-symbol hard slicing), not the *block* DF of the information-theoretic relay-channel literature, and cautions that ”the reported DF results should not be read as bounds on coded block-DF performance” – asserted there, measured here: a practical finite-length coded block-DF relay, built from a rate-1/2 convolutional code and a soft-decision Viterbi decoder, added as a supplementary study on the canonical channel rather than a revision of the core comparison above.

Setup. A standard rate-1/2 convolutional code (constraint length K, 2^{K-1} trellis states, zero-tail terminated per 200-information-bit frame) is applied before QPSK modulation on the canonical Rayleigh channel; `CodedDecodeAndForwardRelay` decodes the full frame at the relay via soft-decision Viterbi using an unweighted squared-distance metric on the coherently compensated I/Q samples, then re-encodes and re-modulates a fresh codeword before forwarding. Because post-equalization noise is heteroskedastic, a matched metric would weight each symbol by its instantaneous channel reliability. The implemented decoder is therefore a practical mismatched block-DF baseline, not a matched-ML decoder for the fading observations. Two coded-aware learned relays are trained on the same task for comparison: a windowed 756-parameter MLP classifier (window 21, $\approx 5\times K$ for $K = 3$) and the Mamba-S6 architecture already used in Table 5.2 (24,084 parameters at this configuration), both predicting the transmitted QPSK symbol from a window of noisy coded observations with no explicit knowledge of the code. All results use the thesis-standard budget of 100 trials × 100,000 information bits per SNR point.

Table 5.4: Coded block-DF vs. uncoded symbol-wise DF and coded-aware learned relays, canonical QPSK/Rayleigh, rate-1/2 $K=3$ code. "coded-AF" forwards the noisy codeword unmodified (the relay does not decode), isolating the code's own gain from any relay intelligence

SNR (dB)	uncoded DF	coded AF	coded DF	MLP-coded	Mamba-coded
0	**0.3337**	0.4902	0.4204	0.4439	0.4505
4	**0.2216**	0.4468	0.2229	0.2974	0.3214
8	0.1205	0.2722	**0.0638**	0.1010	0.1174
12	0.0560	0.0539	**0.0154**	0.0214	0.0226
16	0.0239	0.0099	0.0042	0.0045	**0.0041**
20	0.0098	0.0020	0.0014	0.0011	**0.0010**

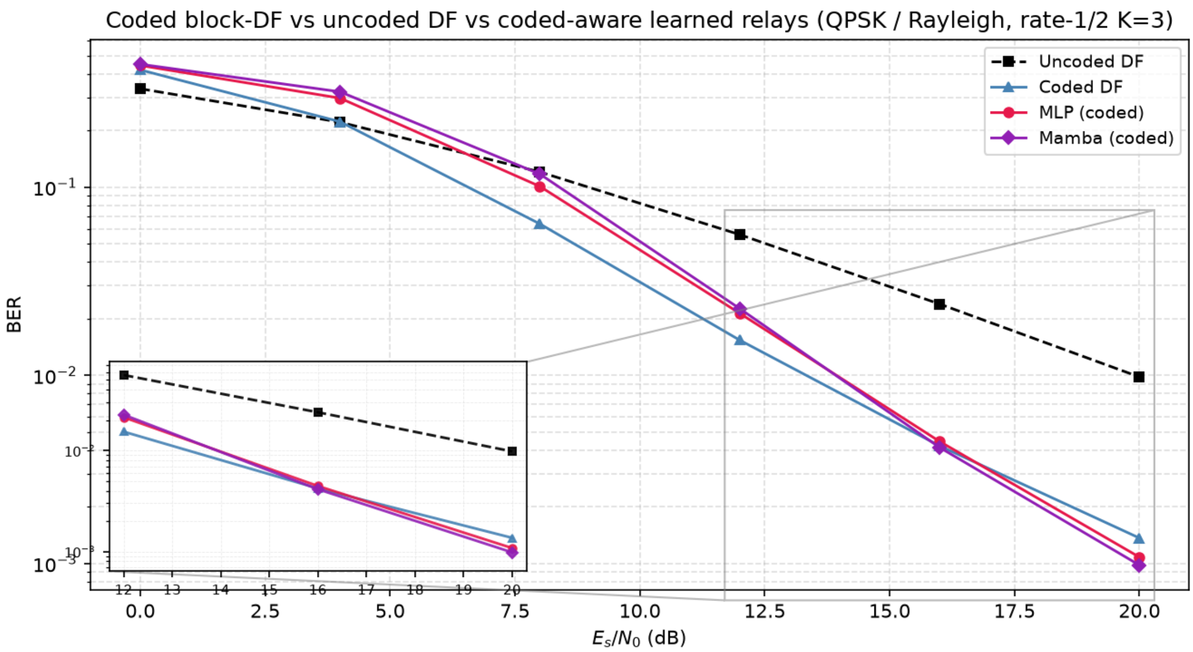


Figure 5.3: BER vs E_s/N_0 for coded block-DF, uncoded symbol-wise DF, and the two coded-aware learned relays (MLP and Mamba), canonical QPSK/Rayleigh, rate-1/2 $K=3$. Coding hurts below the measured crossover (between 4 and 8 dB for this code, decoder, frame and SNR convention) and helps decisively above it; the learned relays close on coded-DF only at the top of the sweep

The caveat holds precisely where it should, and no further. At equal E_s/N_0, coded block-DF beats uncoded symbol-wise DF decisively from 8 dB upward – a 5.7× reduction in BER at 16 dB, 7.2× at 20 dB – closing the gap the remark left open. Those factors are the right comparison for a detection question but the wrong one for a link budget, because the rate-1/2 code halves the information carried per channel use; Section 5.5.2 redoes the comparison at equal throughput and the advantage largely disappears. But below roughly 6 dB, coding *hurts* even on the favourable axis: both coded AF and coded DF are worse than plain uncoded DF at 0 and 4 dB. This is the measured crossover for this configuration, not an artifact: for this finite-length code, decoder, frame, modulation and SNR convention the coded and uncoded curves cross between 4 and 8 dB, and below that crossover a fixed-rate code produces decoding errors that cluster and propagate across the trellis, which a memoryless per-symbol slicer cannot do. The caveat is real, but it cuts both ways – coded block-DF is a strictly stronger baseline only above its own threshold, which itself has to be measured, not assumed.

At low-to-mid SNR the classical decoder is far ahead, and the learned relay's extra capacity buys nothing. Unlike the canonical memoryless channel of Table 5.2, where Section 5.3 found ”the memoryless channel simply offers no temporal structure for [the sequence models'] inductive bias to exploit,” a convolutional code *is* genuine temporal structure. It does not rescue the learned relays where the code is doing the most work: both are far behind coded-DF at 4–8 dB, and the 32×-larger Mamba-S6 relay shows no clear advantage over the 756-parameter MLP there, being worse at 4–8 dB (Table 5.4). Giving the sequence model a task with real memory to exploit therefore does not overturn the thesis's minimal-capacity finding (H3): extra capacity still buys no reliable advantage over the minimal architecture.

At high SNR the ordering reverses, and it is not noise. The differences at 16–20 dB in Table 5.4 are small in absolute terms, and they are not trial-to-trial variation. Re-measuring those two points with *paired* trials – every relay driven by identical information bits and identical hop-1/hop-2 fading realizations within each trial, so the channel-draw variance that dominates an unpaired comparison cancels in the per-trial difference – and testing with the Wilcoxon signed-rank test used elsewhere in this thesis gives Table 5.5. At 20 dB both learned relays beat coded-DF in essentially every one of 100 paired trials ($p < 10^{-4}$). At 16 dB the result splits by architecture: Mamba-coded wins (68 of 100 trials) while the MLP significantly *loses* (23 of 100), both at $p < 10^{-4}$. The effects are real in both directions, and no single sentence of the form “the learned relay beats/does not beat Viterbi” describes them.

Table 5.5: Paired re-measurement of the two high-SNR points of Table 5.4, 100 paired trials × 100,000 information bits. “Diff” is the mean per-trial BER difference against coded-DF (negative favours the learned relay); “W/L” counts trials won/lost; p is Wilcoxon signed-rank

SNR	Relay	BER	Diff vs. coded-DF	W/L	p
16 dB	coded-DF	0.004245	—	—	—
	MLP-coded	0.004437	+0.000192	23/77	$< 10^{-4}$
	Mamba-coded	**0.004124**	−0.000121	68/32	$< 10^{-4}$
20 dB	coded-DF	0.001362	—	—	—
	MLP-coded	0.001066	−0.000295	99/1	$< 10^{-4}$
	Mamba-coded	**0.000967**	−0.000395	100/0	$< 10^{-4}$

This is not evidence that a learned relay out-decodes matched Viterbi, and the mechanism

says why. Viterbi is the maximum-likelihood sequence decoder only when its branch metrics match the channel. The implementation here uses unweighted post-equalization distances and is therefore mismatched to the symbol-dependent Rayleigh reliability. Even against this implemented baseline, the learned relay is not the more accurate relay-stage decoder: instrumenting the relay output directly (Table 5.6) shows coded-DF is better by a wide margin, putting roughly $2.1\times$ fewer symbol errors on the air than the learned relay at both SNRs, consistent with its use of the code constraints. What separates them is what happens to those errors afterwards. Coded-DF decodes and then *re-encodes*, so a frame it gets wrong leaves the relay as a different but perfectly *valid* codeword: the code's redundancy has been spent and then regenerated around the error, and the destination's decoder has no way to detect or repair it. The learned relay never re-encodes, so its (more numerous) mistakes remain isolated bad symbols inside an otherwise-valid codeword, and the destination still holds the redundancy needed to fix them. Only 16–25% of coded-DF's relay errors are repaired downstream against 58–73% of the learned relay's. That also explains the SNR-dependent split above: at 16 dB the $2.1\times$ raw-error penalty still outweighs the repair advantage for the MLP, and by 20 dB it no longer does.

Table 5.6: Where the errors go, 100 trials $\times$ 500 frames. "Relay symbol ER" counts symbols the relay put on the air that differ from the transmitted ones; "repaired" is the fraction of those that do not survive the destination's decoder. The oracle relay forwards the clean codeword and so isolates the hop-2-only floor

SNR	Relay	Relay symbol ER	Repaired	Final BER
16 dB	coded-DF	**0.00505**	15.7%	0.004257
	MLP-coded	0.01065	**57.9%**	0.004487
	oracle	—	—	0.002140
20 dB	coded-DF	**0.00182**	24.7%	0.001369
	MLP-coded	0.00385	**73.0%**	0.001039
	oracle	—	—	0.000668

Does a stronger code help? The $K{=}3$ code used above is deliberately small. Sweeping the constraint length to $K \in \{5, 7\}$ on the same QPSK/Rayleigh link does *not* improve matters: the larger-K codes are worse than $K{=}3$ across 0–8 dB, a sharper error-propagation cliff for the bigger trellis, and by 20 dB all three are within trial-to-trial noise of one another (values 0.0014, 0.0015, 0.0012). Within this frame length and trial budget, the relay's re-encoding liability is therefore not something a stronger code repairs; the full sweep is available in `results/coded_df_experiment.json`.

Does reliable decoding remove it? Remark 4.2 defines block-DF with "a strong (ideally capacity-achieving) channel code", which the measurements above do not approach: at 20 dB the destination fails on 20.23% of frames under coded-DF and on 10.58% even under an *oracle* relay forwarding the clean codeword, so one frame in ten is lost for reasons unrelated to the relay.

Since the liability tracks the relay's frame-error rate rather than the code family behind it, the link is extended up the SNR axis, code fixed, until decoding is reliable (Table 5.7); two learned relays are run, one on the recipe above and one retrained on 5–30 dB, which brackets three of the four evaluated points and leaves only the 32 dB reading as a mild, 2 dB extrapolation, so no high-SNR reading rests on a large train/test mismatch.

Table 5.7: Block-DF into the reliable-decoding regime: the rate-1/2 K=3 code and canonical QPSK/Rayleigh link of Table 5.4, extended up the SNR axis, 100 trials $\times$ 1,000 frames (20,000,000 information bits) per point. "MLP (5/10/15)" uses that table's training recipe; "MLP (5–30)" is the same architecture retrained on 5–30 dB, bracketing all but the 32 dB point, hence a different instance whose 20 dB entry need not reproduce it. Bold marks the best *relay* in each row (both marked where they tie); the oracle forwards the clean codeword and is the single-decode floor they are measured against, not a competitor

SNR (dB)	coded-DF	MLP (5/10/15)	MLP (5–30)	oracle	coded-DF FER
20	0.001378	**0.001064**	0.001092	0.000703	0.1998
24	0.000485	**0.000308**	0.000318	0.000253	0.0804
28	0.000187	**0.000105**	0.000105	0.000092	0.0324
32	0.000069	**0.000037**	**0.000037**	0.000034	0.0122

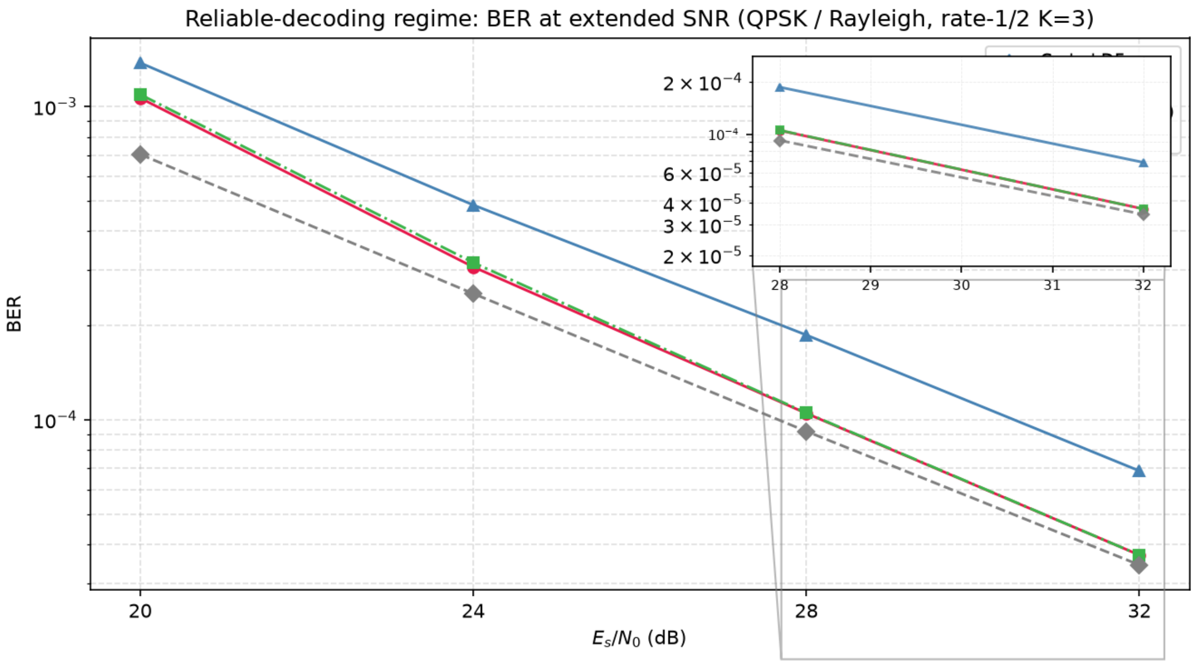


Figure 5.4: BER vs E_s/N_0 at extended SNR (20–32 dB), rate-1/2 K=3 QPSK/Rayleigh. Coded DF's BER remains near twice the oracle floor at every point; the retrained learned relay (MLP 5–30 dB) closes on that floor monotonically and reaches statistical parity with it at 32 dB, demonstrating that the high-SNR reversal survives into the reliable-decoding regime

The liability does not dissolve. Across 20–32 dB coded-DF's frame-error rate falls 16.3$\times$ (0.1998 to 0.0122), yet its BER stays near twice the oracle floor with no trend at all (1.96, 1.92, 2.03, 2.00; the scatter is trial noise, the absence of a downward slope is the point), while the retrained learned relay (MLP 5–30) closes on that floor (1.55, 1.26, 1.15, 1.08) and reaches statistical parity with it by 32 dB (636 against 607 frame errors in 100,000, $z = +0.8$) while still beating coded-DF there ($z = +13.7$). The cause is structural, not code strength: block-DF decodes once per hop, so two independent frame-error events accumulate and neither decoder repairs the other's, whereas denoising leaves a single decoding operation at the destination. The reversal is thus no artefact of a weak code or short frame; it survives, and relatively widens, as decoding becomes reliable. One limit stays out of reach: driving per-hop failure to zero makes the sum of two vanishing terms vanish, so a capacity-achieving code at long blocklength would recover the floor. No operating point here does: even at 32 dB the oracle's own decode – one hop's worth, by construction, and equal to the other hop's by the link's symmetry – still fails on 0.6% of frames.

5.5.1 Soft-Decision Relaying, and What the Liability Actually Is

The mechanism of Table 5.6 suggests an obvious repair. If the trouble with block-DF is that it commits to a codeword, replace the commitment with soft information: run BCJR (forward-backward MAP) at the relay instead of Viterbi and forward the posterior-mean symbol per trellis step, so a symbol the relay is unsure of is shrunk toward the origin rather than asserted. The same idea applies to the learned relay, whose softmax already carries a posterior it was throwing away: forwarding $\sum_k p_k s_k$ instead of $\arg\max_k p_k$ costs nothing, needs no retraining, and reuses the identical weights, so any difference isolates the read-out rule alone. Table 5.8 evaluates both against their hard counterparts and against an oracle relay that forwards the clean codeword, on paired trials.

Table 5.8: Soft- versus hard-decision relaying, canonical QPSK/Rayleigh, rate-1/2 K=3, 1
trials × 100,000 information bits. The MLP hard/soft pair shares one set of trained we
differs only in the read-out. The oracle relay forwards the clean codeword, giving the h
floor. Both block-DF variants are marked at 8 dB, where they are a statistical tie

SNR (dB)	hard block-DF	soft block-DF	MLP hard	MLP soft	orac
0	0.4208	**0.4150**	0.4443	0.4343	0.30
4	0.2231	**0.2218**	0.2969	0.2761	0.12
8	**0.0636**	**0.0636**	0.1008	0.0878	0.03
12	**0.01537**	0.01537	0.02142	0.01768	0.007
16	0.004222	0.004222	0.004403	**0.003633**	0.002
20	0.001354	0.002348†	0.001056	**0.000912**	0.000

†Nominal-noise-variance run; miscalibrated at high SNR (see text below). The variance-corrected figure is 0.001376, a statisti

Soft read-out helps the learned relay everywhere (Figure 5.5). At every SNR the posterior mean beats the argmax, on identical weights (columns 3–4 of Table 5.8). The gain is largest exactly where the mechanism predicts, since a relay that reports its uncertainty gives the destination's decoder more to work with than one that reports a confident guess. It is enough to

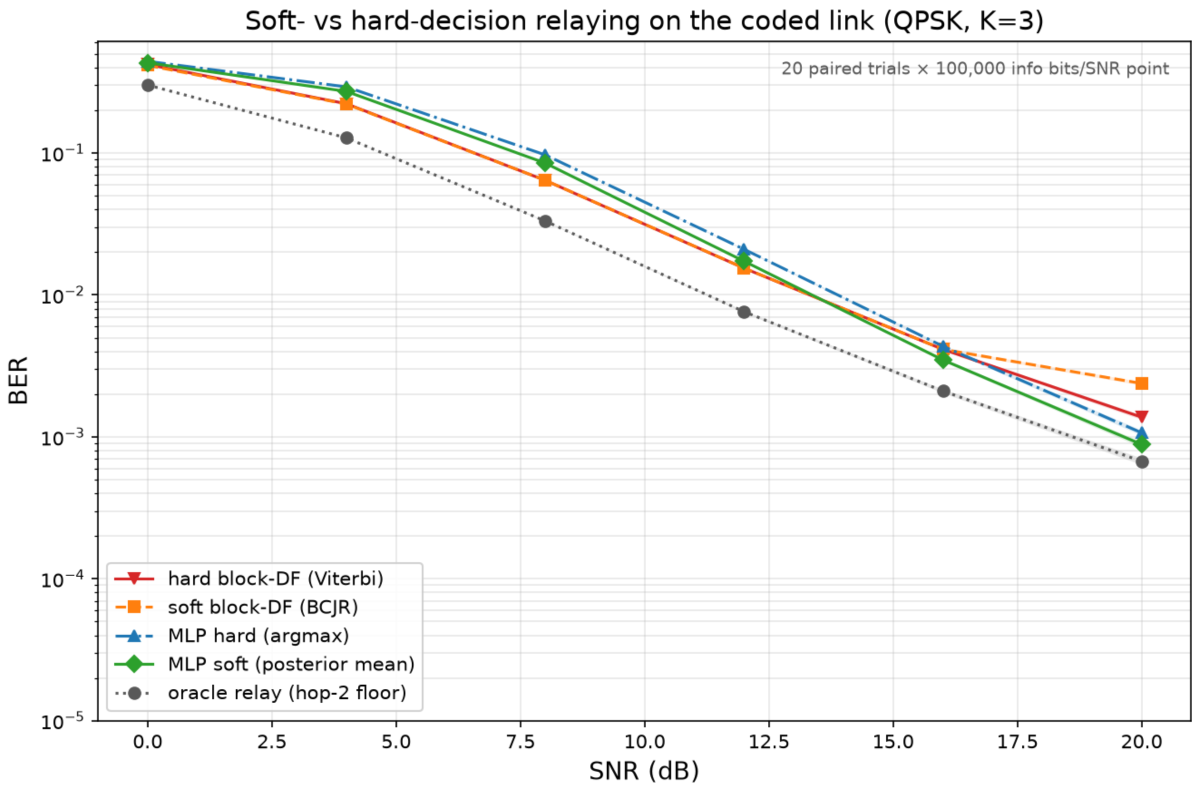


Figure 5.5: Soft- versus hard-decision relaying on the coded link (canonical QPSK/Rayleigh, rate-1/2 K=3, 100 paired trials). The soft MLP (posterior mean) tracks below the hard MLP at every SNR, with the largest gain visible at mid-to-high SNR where the relay's residual uncertainty is most useful to the destination decoder. The two block-DF variants (hard Viterbi, soft BCJR) converge at high SNR and are never separated by a reliable margin, confirming that the liability is consuming the code's redundancy at the relay rather than hard quantization per se. The oracle relay (dashed) is the hop-2-only floor

change a verdict: at 16 dB the hard MLP significantly *loses* to block-DF (19 wins, 80 losses and one tie) while the soft MLP significantly *beats* it (100 wins, no losses), and at 20 dB the soft relay comes within 0.00023 of the oracle floor that no relay in this fixed-rate, fixed-power chain can pass.

Soft block-DF does not rescue the implemented block-DF baseline. The relay is given a nominal noise variance, although post-equalization Rayleigh noise has variance $1/(2\gamma|h|^2)$ per real axis. Its BCJR metrics are therefore mismatched and overconfident on deep-fade symbols. Uniformly inflating the assumed variance removes the 20-dB gap, but this sensitivity sweep does not supply the required per-symbol reliability weights and should not be described as matched calibration. The same limitation applies to unweighted Viterbi distances: multiplying every branch metric by one constant leaves its decisions unchanged, whereas omitted symbol-dependent weights can change the selected path.

Within that metric family, soft block-DF ties hard block-DF at 16 and 20 dB to the reported precision. This suggests that hard quantization alone does not explain the measured block-DF gap. In the evaluated decode–re-encode chain, an erroneous relay decision is embedded in a newly valid codeword, whereas denoise-only forwarding leaves the original code constraints for the destination decoder. That mechanism explains these measurements, but it is not a theorem about every coded relay protocol, decoder metric, or iterative relay–destination design.

This sharpens the thesis's recurring claim rather than overturning it. A learned relay still does not out-decode a matched classical decoder: coded-DF remains far ahead through 12 dB and, per Table 5.6, is the more accurate decoder at every SNR measured. What the high-SNR reversal shows is an architectural point about *where* in a two-hop chain the decoding should happen, and the answer that emerges – leave the code alone at the relay, denoise only, and let the destination decode once – is a system-design conclusion, not a statement about the relative power of Viterbi and neural networks.

5.5.2 Throughput and Latency: What the BER Tables Leave Out

Every table above compares relays at equal E_s/N_0. That is the correct axis for asking which relay detects better, and the wrong one for asking which relay a link should use, because it silently ignores two costs that differ by orders of magnitude across these designs.

Throughput: the coded comparison was not like-for-like. A rate-1/2 code carries half the information per channel use. Concretely, the coded QPSK link spends 202 symbols to deliver 200 information bits (0.99 bits/symbol) where uncoded QPSK spends 100 (2.00 bits/symbol). The "$5.7\times$ better at 16 dB" quoted above is therefore bought with half the data rate, and a link designer given that trade would not read it as a $5.7\times$ improvement at all. The honest comparison holds spectral efficiency fixed, and the measurements already collected supply it: rate-1/2 *16-QAM* delivers $4 \times \frac{1}{2} = 1.98$ information bits per symbol, matching uncoded QPSK's 2.00. Table 5.9 places those two side by side.

Table 5.9: Coded versus uncoded at equal spectral efficiency (≈2 information bits per channel symbol): uncoded QPSK against rate-1/2 K=3 16-QAM, both on the canonical Rayleigh channel. Recomputed from the data of Table 5.4 and `results/coded_df_experiment.json`; no additional simulation

SNR (dB)	uncoded QPSK	rate-1/2 16-QAM	Coding gain
0	**0.3337**	0.4680	$0.71\times$ (worse)
4	**0.2216**	0.4203	$0.53\times$ (worse)
8	**0.1205**	0.2726	$0.44\times$ (worse)
12	**0.0560**	0.1001	$0.56\times$ (worse)
16	**0.0239**	0.0273	$0.88\times$ (worse)
20	0.0098	**0.0076**	$1.29\times$ (better)

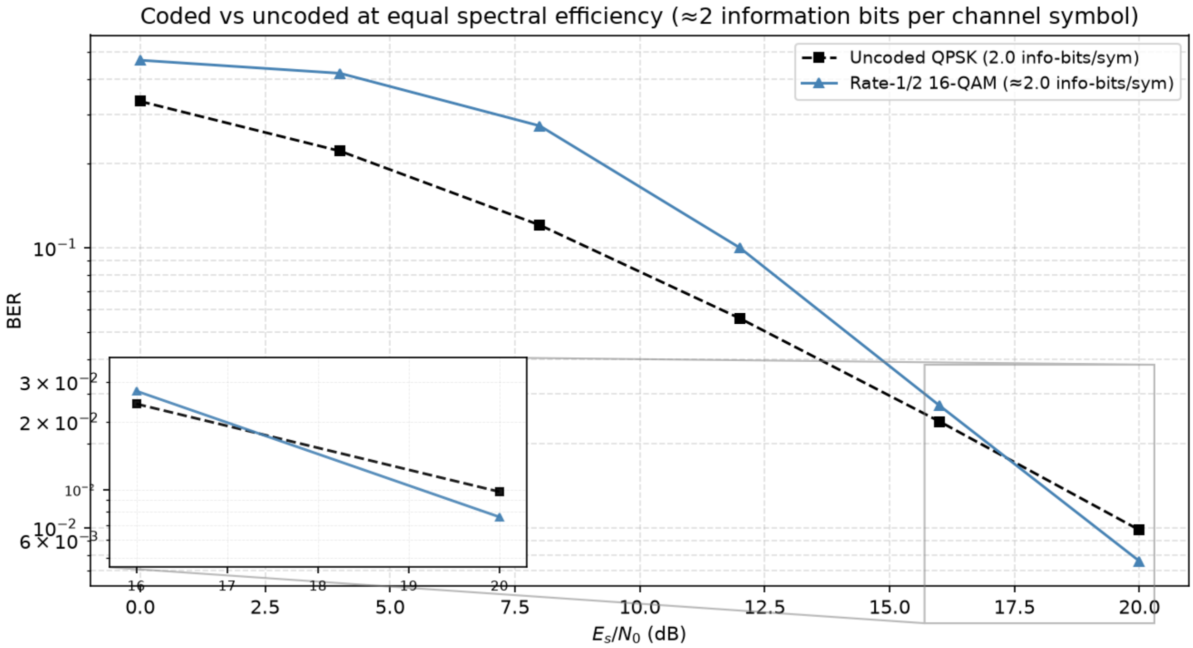


Figure 5.6: BER vs E_s/N_0 for uncoded QPSK and rate-1/2 16-QAM, both delivering ≈2 information bits per channel symbol on the canonical Rayleigh channel. Held to equal throughput, the code does not pay for itself below 20 dB; the $1.29\times$ gain at 20 dB is the entirety of the coding advantage at this rate and frame length

The result is a substantial qualification of Section 5.5. Held to equal throughput, the rate-1/2 code does not pay for itself anywhere below 20 dB, and where it finally does the gain is $1.29\times$, not the 5.7–$7.2\times$ the equal-E_s/N_0 reading suggested. Coding is not free reliability: it is reliability bought with bandwidth, and on this channel, at this rate and frame length, the purchase is a poor one until the link is already operating well. This does not overturn the earlier section – coded block-DF genuinely is the stronger *detector*, and Remark 4.2's caveat genuinely is real – but it does bound how much the caveat matters in practice.

Relay readiness: block relays buffer substantially more. The relays differ in how much they must accumulate before producing their first output. Symbol-wise AF and DF can process a current symbol; the symmetric 21-sample learned relay needs $w = 10$ future symbols; and block-DF must receive a 202-symbol frame before decoding. These are relay-processing readiness delays, not complete end-to-end latency: in the half-duplex protocol every strategy must still spend channel uses on both source–relay and relay–destination transmissions.

Table 5.10: Relay readiness and measured compute cost. "Buffer" is the number of symbols the relay must accumulate before it can produce its first output; it excludes the two half-duplex transmission phases and is not end-to-end latency. Compute is the median of 7 runs over 49,894 symbols after a discarded warm-up, single-threaded CPU. The identical benchmark was run on two different machines, both reported here: neither is more correct than the other, and the pair shows that absolute figures and ratios involving sub-microsecond MLP timings are machine-dependent (`results/coded_latency_compute_machines.json`)

Relay	Buffer (symbols)	μs/symbol Machine A	μs/symbol Machine B	Relative to Viterbi
AF / symbol-wise DF	**0**	—	—	—
MLP hard (756p)	10	0.30	0.34	50–63$\times$ faster
MLP soft (756p)	10	0.22	0.35	61–68$\times$ faster
hard block-DF (Viterbi)	202	15.07	21.23	1$\times$
soft block-DF (BCJR)	202	29.30	41.21	1.94$\times$ slower

The compute column points the same way as the latency column, and the two machines make clear which parts of it can be relied on. On the two block decoders, which are the large and reliably measurable costs here, Machine B is about 40% slower than Machine A on both (Table 5.10). No absolute figure in this table should therefore be read as a property of the algorithms. What survives the change of machine is the structure. The learned relay is faster than Viterbi by 50$\times$ on Machine A and 63$\times$ on Machine B: the factor moves, the order of magnitude does not. BCJR costs 1.944$\times$ Viterbi on Machine A and 1.941$\times$ on Machine B, a ratio reproduced to three decimal places across hardware differing by 40% in absolute speed, exactly as its forward, backward and marginalization passes imply.

The MLP rows do not track that 40% and should not be expected to. At a third of a microsecond per symbol they sit close to the resolution of the measurement, so their run-to-run scatter is comparable to the difference being measured: hard moves +14% between the machines and soft +57%, bracketing the block decoders' 41% from both sides rather than agreeing with it. The same noise floor is why the two read-outs are *not* separable from each other – soft is cheaper on Machine A, hard on Machine B, and repeat runs on a single machine flip the ordering too. The defensible claim is that replacing an `argmax` with a posterior-mean inner product costs nothing measurable, and that both read-outs are cheaper than Viterbi by something in the region of 50–70$\times$; the precise factor is not resolved by this benchmark. Taken together with the buffering column, which is exact and architectural rather than measured, the ordering is not in question even though the absolute timings clearly are.

Taken together with Table 5.9, this reframes the coded study's practical conclusion. Where the code pays for itself at all, it does so at 20 dB, for a $1.29\times$ gain at equal throughput, while asking for roughly $20\times$ the relay readiness buffer and at least $50\times$ the measured reference arithmetic of the learned relay that comes within a few ten-thousandths of it there. The learned relay's case in this chapter, as in Chapter 6, rests on cost and constancy rather than on a lower error floor.

5.5.3 Adaptive Modulation and Coding: Choosing the Rate, Not Just the Relay

Every comparison above, including the equal-throughput one, holds the modulation-and-coding scheme (MCS) fixed and asks which relay achieves the lower BER. A real link does not hold the MCS fixed; it adapts. The question a link actually faces is not "which relay wins at rate R" but "what is the highest *useful* rate this link can sustain at this SNR", and answering it requires more than one code rate to choose from. The mother code of Section 5.5 is extended here with the standard punctured rates $2/3$ and $3/4$ (deleting a periodic subset of the rate-1/2 output, re-inserted as soft zeros at the decoder, so the same Viterbi decoder runs unmodified at every rate), and evaluated via bit-interleaved coded modulation across the full grid $\{\text{QPSK}, \text{16-QAM}\} \times \{\text{uncoded}, 1/2, 2/3, 3/4\}$.

The objective is an idealized one-shot *goodput*, $G = R\,(1 - \text{FER})$, assuming perfect frame-error detection and discarding failed frames without accounting for feedback or retransmission overhead. G and R are both measured per channel symbol transmitted on the active hop, matching the ergodic-capacity reference C introduced below; because the relay is half-duplex, expressing goodput instead as a fraction of the total two-slot cycle (source-to-relay plus relay-to-destination) would halve every figure in this section without changing which MCS or relay strategy is preferred, since that factor of $\frac{1}{2}$ applies identically to both. Maximizing G over the MCS grid at each SNR traces the link-adaptation envelope, compared here for the two relay strategies the mechanism of Table 5.6 contrasts directly: *block-DF*, which decodes the code at the relay and so spends its redundancy before transmitting, against *denoise-only*, which hard-decides each bit and re-modulates without ever invoking the code, leaving the redundancy intact for the destination and requiring no training and no code knowledge at the relay.

Table 5.11: Link-adaptation envelope: the MCS maximizing goodput at each SNR, block-DF versus denoise-only relaying, canonical Rayleigh channel, 100 trials $\times$ 200 frames $\times$ 200 information bits per point. Columns are SNR in dB, the goodput-maximizing MCS and its goodput G for each relay, and C, the ergodic Rayleigh (Shannon) capacity at that SNR, $C = \log_2(e)\, e^{1/\gamma} E_1(1/\gamma)$ – an unconstrained-input upper bound no achievable goodput may exceed, included as a sanity ceiling rather than a target: the gap to it here is dominated by the finite constellation and coding loss, not by relay strategy. The capacity column is a *single-link* reference evaluated per active-hop channel symbol; it is not a capacity bound for the half-duplex two-hop relay system studied here, whose own bound would additionally account for the two-slot cycle and the relay constraint

SNR	MCS (block-DF)	G	MCS (denoise)	G	C
8	QPSK $\frac{1}{2}$	**0.0039**	QPSK $\frac{1}{2}$	0.0003	2.40
12	QPSK $\frac{1}{2}$	**0.1708**	QPSK $\frac{1}{2}$	0.1342	3.45
16	QPSK $\frac{1}{2}$	0.5406	QPSK $\frac{1}{2}$	**0.6046**	4.63
20	QPSK $\frac{3}{4}$	0.9545	QPSK $\frac{3}{4}$	**0.9635**	5.88
24	16-QAM $\frac{2}{3}$	**1.5790**	16-QAM $\frac{2}{3}$	1.4739	7.17
28	16-QAM $\frac{3}{4}$	**2.3003**	16-QAM $\frac{3}{4}$	2.2625	8.48
32	16-QAM $\frac{3}{4}$	**2.6807**	16-QAM $\frac{3}{4}$	2.6726	9.80
36	16-QAM uncoded	3.3920	16-QAM uncoded	3.3920	11.13
40	16-QAM uncoded	3.7336	16-QAM uncoded	3.7336	12.46

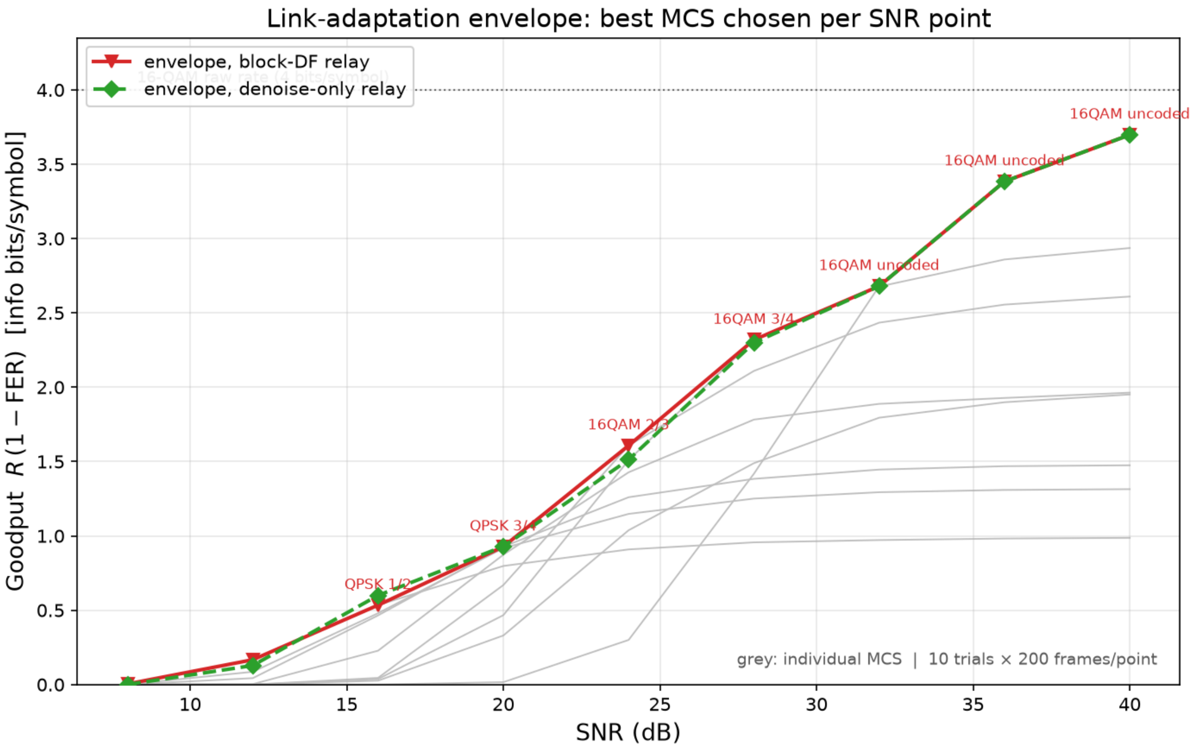


Figure 5.7: The envelope of Table 5.11 (colored) against every individual fixed MCS on the block-DF relay (grey), which is the visual case for adaptation: the envelope tracks the upper hull of the grey curves rather than any single one of them, switching MCS as SNR crosses each rate's operating threshold. The two relay strategies' envelopes are visually almost the same curve

Two results follow, and the second qualifies the mechanism finding of Table 5.6 substantially.

The envelope is a real staircase, and its top step is "turn the code off". Both relay strategies climb QPSK 1/2 $\rightarrow$ QPSK 3/4 $\rightarrow$ 16-QAM 2/3 $\rightarrow$ 16-QAM 3/4 $\rightarrow$ *16-QAM uncoded* as SNR rises from 8 to 40 dB, each switch occurring once the next rate's threshold has cleared. The optimum at high SNR is no code at all: once the channel is good enough, coding redundancy is pure overhead, and the highest-goodput choice is the raw 4 bits/symbol of 16-QAM with zero coding loss. This is the same lesson as the equal-throughput correction above, now reached by direct optimization over the grid rather than by hand-picking one comparison. Table 5.11's C column confirms the envelope never approaches the information-theoretic ceiling: at 40 dB the achieved 3.73 sits at 30% of the 12.46-bit Shannon capacity there, a gap set by the fixed 16-QAM constellation (itself capped at 4 bits/symbol) and residual FER, not by anything a smarter relay strategy could close.

Once the rate adapts, which relay decodes the code barely matters. The two envelopes in Table 5.11 are close at every SNR and identical for SNR ≥ 36 dB, where both select 16-QAM uncoded and there is no code for the strategies to differ over. Where they do differ, the gap is small and does not consistently favor one side: block-DF leads at 8, 12, 24, 28 and 32 dB, denoise-only leads at 16 and 20 dB, and the largest gap over the whole sweep is 0.105 information bits/symbol at 24 dB, under 7% of the envelope value there. Compare this to how much the *MCS* choice moves goodput at fixed relay strategy: at 24 dB the spread across the eight MCS options the block-DF envelope searches over is 1.25, roughly $12\times$ the 0.105 gap between relay strategies; at 32 dB and above the MCS spread exceeds 1.7 while the relay-strategy gap is under 0.01. Rate adaptation, in other words, dominates the relay-decoding decision by more than an order of magnitude across most of this sweep.

This bounds the mechanism finding of Section 5.5.1 rather than reversing it. That the destination can repair a denoising relay's errors but not a re-encoding relay's remains true and remains the reason the two strategies differ at all – but once the system is allowed to pick its own operating rate, the room that difference has to matter shrinks to a few percent of the achievable goodput, because the adaptive link mostly avoids operating in the regime (a fixed rate pushed past its comfortable threshold) where the mechanism produces a large gap in the first place. The practical recommendation this chapter's coded study converges on is therefore rate adaptation first: get the MCS right for the channel, and only then does it become worth asking which relay strategy to run within it.

5.5.4 Goodput Under an Idealized Buffer-Readiness Budget

The coded relay of Section 5.5 decodes a whole frame before it forwards, and every comparison made of it so far has left the frame length free. A link with a deadline cannot: the deadline bounds the blocklength directly, and with it the redundancy the relay has to spend. The question that follows – the achievable rate once blocklength (equivalently, latency) is fixed rather than left free – is a standard one in the coding-theory literature: the finite-blocklength capacity of Polyanskiy, Poor and Verdú [35], and the delay-limited capacity of Hanly and Tse [36] for fading channels specifically. Both frame it the same way this section does: constraining the number of channel uses (Polyanskiy et al.) or imposing a hard per-block deadline (Hanly–Tse) costs achievable rate relative to the unconstrained (Shannon, or ergodic) capacity, and by how much is exactly the object of study.

The following calculation uses an idealized *buffer-readiness budget*, not physical round-trip latency. It counts one destination frame buffer for both strategies and an additional relay frame buffer for block-DF, while assigning only the learned relay's look-ahead to denoise-only relaying. It omits the channel-use duration of the two half-duplex transmission phases, which both strategies incur. Re-deriving the goodput envelope of Table 5.11 under this accounting illustrates the effect of relay buffering, but does not establish performance under a complete end-to-end deadline.

Table 5.12: Link-adaptation envelope re-derived under the idealized 150-symbol buffer-readiness budget, same layout and data source as Table 5.11

SNR (dB)	Best MCS (block-DF)	G (block-DF)	Best MCS (denoise)	G (denoise)
8	QPSK uncoded	0.0000	QPSK uncoded	0.0000
12	QPSK uncoded	0.0000	QPSK 3/4	**0.0105**
16	QPSK uncoded	0.0251	QPSK 3/4	**0.3582**
20	16-QAM 3/4	0.4966	QPSK 3/4	**0.9635**
24	16-QAM 3/4	**1.5357**	16-QAM 2/3	1.4739
28	16-QAM 3/4	**2.3003**	16-QAM 3/4	2.2625
32	16-QAM 3/4	**2.6807**	16-QAM 3/4	2.6726
36	16-QAM uncoded	3.3920	16-QAM uncoded	3.3920
40	16-QAM uncoded	3.7336	16-QAM uncoded	3.7336

Under this idealized readiness budget, block-DF can be locked out of coding. At $L_{\text{max}} = 150$ symbols (Table 5.12), block-DF's cheapest coded option barely clears the budget while its next-cheapest does not. Under the stated accounting, the 16-dB entries differ by approximately $14.3\times$ (0.3582 versus 0.0251 information bits/symbol), and the 20-dB entries by about $1.9\times$. These figures illustrate buffering sensitivity only; they are not end-to-end latency results.

Tighten the budget further and the reversal spreads to SNRs where block-DF led unconstrained. At $L_{\text{max}} = 100$ symbols, block-DF has *no* coded option left at any of these four SNRs – every one of its coded MCS choices exceeds the budget – so it is pinned to uncoded transmission throughout, while the denoise-only relay still reaches 16-QAM 2/3 or 3/4 at 20, 24 and 28 dB. At 24 and 28 dB, this is a direct reversal: Table 5.11's unconstrained envelope has block-DF *ahead* at both points, but under the 100-symbol budget denoise-only leads by $1.46\times$ and $1.54\times$ respectively, because the MCS that gave block-DF its unconstrained edge is no longer reachable in time.

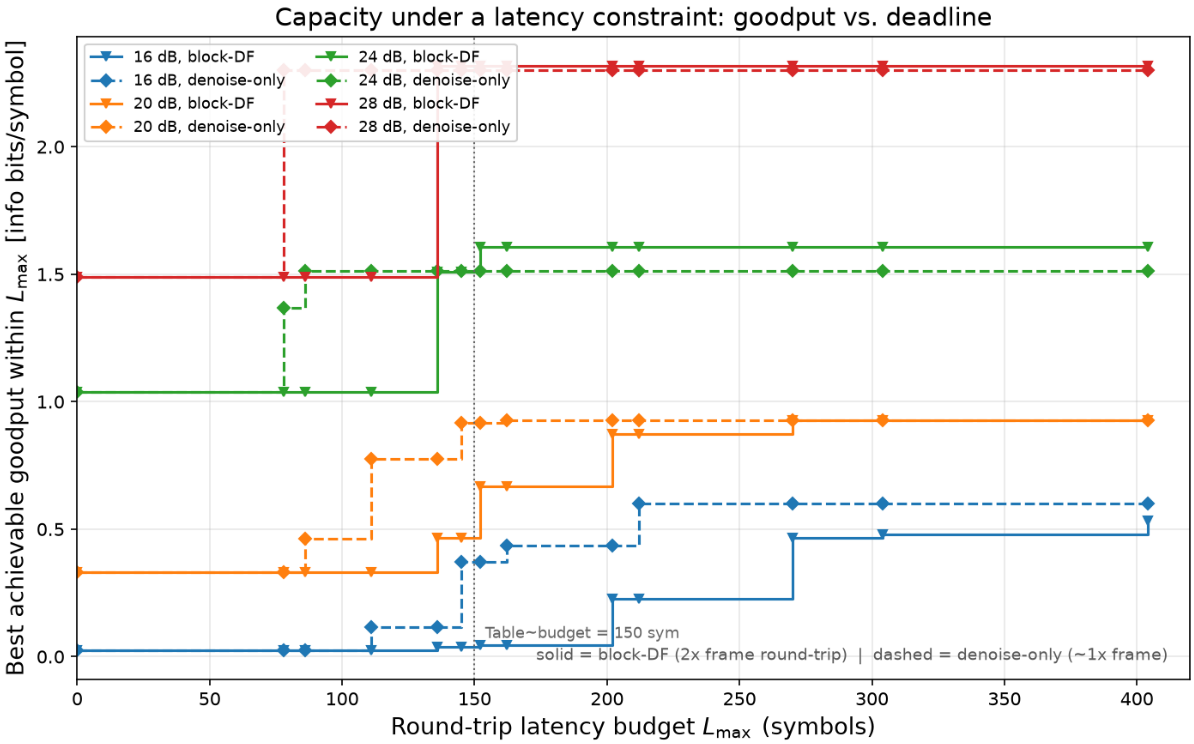


Figure 5.8: Best achievable goodput as a function of the idealized buffer-readiness budget $L_{\max}$, at the four SNRs where the two relay strategies differ (16, 20, 24, 28 dB). Each curve is a step function under this accounting. The omitted two-hop transmission time prevents interpreting this axis as end-to-end latency

The mechanism finding of Section 5.5.1 concerned decoding location; Section 5.5.3 showed that rate selection dominates it in the unconstrained grid. The present calculation indicates that relay buffering can alter that result, but a deployment recommendation under a hard deadline requires a slot-level latency model including both half-duplex transmissions, processing time, and the reliability target. That analysis is not performed here.

It is important to emphasize that the core relay-comparison conclusions of this chapter (E2, E3) are specific to the canonical memoryless channel model; whether those conclusions remain valid when the underlying channel assumptions are violated is the subject of Chapter 6. The supplementary coded study's conclusions are scoped separately, to coding, rate adaptation, and latency constraints on that same canonical Rayleigh channel, and are not revisited under channel mismatch.

Chapter 6

Learned Relaying under Unknown and Mismatched Channels

This is the central chapter of the thesis. Everything before it is either calibration or a confirmatory result: on a matched, known channel a minimal learned relay does *not* improve on classical decode-and-forward (H2), and raising the constellation order does not change that. Here the learned relay provides its clearest advantage over the memoryless classical relay baselines, while remaining below the performance of correctly informed sequence-optimal or BER-optimal classical detectors.

The condition under which it wins is precise, and stating it is as much a part of the claim as the win itself. **When the channel's structure is unknown to the receiver chain, or lies outside the model class the classical relay assumes, a fixed 170-parameter learned relay restores reliable relaying where the evaluated memoryless classical relays, AF and zero-threshold symbol-wise DF, fail outright.** Under three-tap intersymbol interference (Table 6.3) the classical relays do not degrade gracefully: they saturate near the analytic 0.25 floor and then get *worse* as the SNR rises, symbol-wise DF bottoming out near 8 dB and climbing thereafter as hard decisions lock the interference in. No increase in transmit power removes this error floor for the memoryless relay architecture considered here. The learned relay, whose length-11 window spans the channel memory, restores a monotonically falling BER, reaching 4.79×10^{-8} by 16 dB. What separates the two is the function class and not the information available: this filter is fixed, and the relay is trained on it, so it is as well informed about the channel as the genie-CSI Viterbi detector it is measured against. The evaluated zero-threshold symbol-wise slicer cannot compensate for the selected ISI channel at any transmit power. A detector whose observation spans the channel memory can exploit the additional temporal information; the windowed MLP evaluated here does so successfully, though spanning the memory is a necessary condition rather than a sufficient one. Whether a learned relay also copes with a channel it has *not* seen is a separate question, answered at layers 3 and 4 below, where the channel is redrawn every block. That is not a marginal gain over a classical baseline; it is the difference between a working link and a broken one, and it is the thesis's answer to whether learning can defeat classical processing in relaying.

The boundary of the claim is equally important. No statistically resolved advantage over matched MLSE is observed: given a reliable channel estimate, pilot-aided Viterbi remains ahead by approximately 1–1.5 dB over the reported operating range. This is empirical, not an impossibility result, because MLSE minimizes sequence error rather than BER; the BER-optimal matched comparator is a bit-MAP/BCJR receiver. That comparator is reported at QPSK (Section 6.2.3), where it is never worse than MLSE but does not overturn the margin reported here; it is unmeasured at BPSK. The learned relay's advantage is therefore not demonstrated superior detection but the absence of a per-block channel-estimation precondition. Where the channel is fixed and known to both, it is simply the cheaper detector, constant and parallelizable per symbol against the trellis's M^L branch metrics over M^{L-1} states; where the channel must be identified and cannot be, it is the one that still functions, needing no per-block estimate — though it is trained on the impairment family, and so is realization-agnostic rather than family-agnostic. Where the classical pipelines evaluated here can identify the channel they should be used; where they cannot, the learned relay is what still works. That is a statement about these comparators, not about classical receiver design in general.

Rather than a single comparison, the chapter is a systematic map, along every axis relevant to practical deployment, of *when* the learned relay adds value over classical processing, and, equally, when it does not.

The argument as a ladder of assumptions. The thesis as a whole is organised as a monotone sequence in which each layer relaxes a further assumption about the channel, and the learned relay is compared at every layer against the strongest classical baseline evaluated here. Table 6.1 is that sequence with its outcomes. Reading it downward is the thesis's argument in one page: classical processing generally is not beaten where it is well posed, and is beaten precisely where its preconditions fail – except once, where sequence-optimal is not BER-optimal (Section 6.2.2).

Table 6.1: The four layers of the argument. Each layer relaxes a further channel assumption; the classical baseline is strengthened at each layer to remain the toughest available opponent (a relay function at layer 1, a receiver from layer 2 onward). BER figures are the representative operating points cited in the sections named

Layer	Assumptions	Strongest classical	Outcome
1	Memoryless, known channel, perfect CSI (Chapter 5)	Symbol-wise DF, zero trainable parameters	**Classical wins.** On the canonical QPSK/Rayleigh setup itself (Table 5.2), DF 0.1218 against the MLP's 0.1229 at 8 dB: the learned relay only matches DF, at 169 parameters against none (H2).

Layer	Assumptions	Strongest classical	Outcome
2	+ channel memory (3-tap ISI), CSI still perfect (Section 6.2)	Viterbi MLSE with exact taps	**Split.** The memoryless relays break outright, DF rising from 0.1802 at 8 dB to 0.2457 at 20 dB. The MLP restores the link (0.0065 at 8 dB), but genie-CSI Viterbi is still ahead by 1–1.5 dB (0.0072). Both are equally informed here, the filter being fixed and known to each, so this layer separates function classes rather than channel knowledge.
3	+ CSI uncertainty: a finite pilot budget (Section 6.3)	Pilot-aided LS estimate then MLSE	**Crossover.** At 10 dB, Viterbi-est holds 0.0283 on 200 pilots and 0.0335 on 20, both beating the pilot-free MLP's 0.0486; by 10 pilots it has degraded to 0.0545 and the MLP leads. The classical advantage is real but purchased with pilots.
4	+ unseen realization from the trained family, redrawn per block (Sections 6.2.4, 6.4)	CMA blind equalization	**Learned relay wins.** At 20 dB the MLP reaches 0.00262 against CMA's 0.00329, while decision-directed blind MLSE is unstable, non-monotonic and worse at 20 dB (0.0498) than at 16 dB (0.0374). The margin in BER is modest; the margin in stability and in per-block cost is not.

Two features of the ladder are worth naming before the evidence is presented. First, the opponent changes character at layer 2: layer 1 compares relay *functions* (AF and DF against learned relays), whereas from layer 2 onward the strongest classical option is a *receiver*, Viterbi MLSE, which is a considerably heavier and better-informed baseline. This is deliberate, since comparing against a weaker classical option at the layers where the learned relay wins would make the result worthless. Second, what degrades down the ladder is not the classical algorithm itself but its model match. Viterbi is sequence-ML only for the channel represented by its branch metrics; as the estimated model departs from the true channel, that guarantee no longer applies. The learned relay remains trained on the impairment distribution and requires no per-block estimate.

The investigation uses progressively stronger classical references: symbol-wise DF for memoryless cases, differential detection for unknown phase, matched Viterbi MLSE where exact taps are supplied, LS-plus-MLSE where pilots are available, and CMA in the blind comparison. These are the evaluated baselines, not an exhaustive proof that no stronger classical method exists. The learned relay receives no per-block channel estimate and never adapts online, but it is trained and tested within the same parametric impairment family. At layer 2 the filter is fixed and represented directly during training; layers 3 and 4 redraw realizations from the trained family. The supported claim is therefore *realization-agnostic*, not family-agnostic: the relay amortizes the family distribution into its weights, and generalization to a structurally different family is not tested. Matched Viterbi provides a sequence-error reference rather than a BER-optimal bound. The bit-MAP/BCJR benchmark that closes that gap is reported at QPSK in Section 6.2.3 and is still open at BPSK.

Relationship to the canonical model. The canonical setup of this thesis (Chapter 5) assumes a memoryless two-hop relay channel with white noise, perfect receiver-side CSI, and uncoded transmission. The present chapter deliberately departs from those assumptions by introducing impairments that are not part of the canonical model, including channel memory (3-tap intersymbol interference), amplifier nonlinearity, unknown phase, and channel-model mismatch. Consequently, sequence-detection and equalization techniques such as Viterbi MLSE, pilot-aided channel estimation, and blind equalization become relevant only in this chapter. The goal is not to replace the canonical study, but to evaluate whether the conclusions established under matched memoryless conditions remain valid when the underlying channel model no longer represents the true channel.

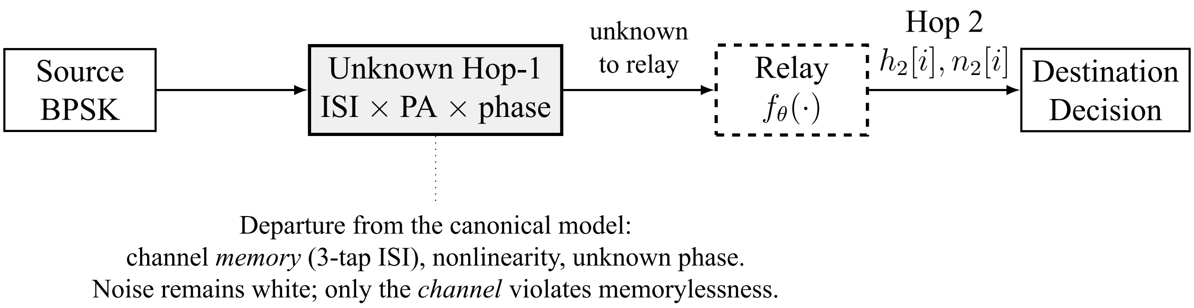


Figure 6.1: Unknown/mismatched-channel study (Chapter 6), a deliberate departure from the canonical benchmark of Figure 1.1. The Hop-1 channel is replaced by an unknown impairment (3-tap ISI, amplifier nonlinearity, and unknown phase) that is not disclosed to any relay. The noise remains white; only the channel violates the memoryless assumption, which is why sequence detection (Viterbi MLSE) becomes the relevant classical benchmark here and nowhere else in the thesis

6.1 Layer 1 — Matched Memoryless Channel: The Control

The first rung is not run in this chapter. It is the canonical comparison of Chapter 5, restated here so the ladder has all four of its rungs in one place: a memoryless channel, known to the receiver chain, with perfect CSI, on which the strongest classical opponent is a relay function rather than a receiver. It is the control the rest of the ladder is read against, and its role is to establish that the learned relay's later advantages come from the assumptions being relaxed rather than from the comparison being unfair. Sections 5.2 and 5.4 report it; the row for layer 1 in Table 6.1 gives the operating point.

6.2 Layer 2 — Channel Memory with Perfect CSI: Classical Failure and Learned Mitigation

Goal: Test whether the learned relay's value changes qualitatively when the channel violates the assumptions under which AF and DF are (near-)optimal: do the *memoryless* classical relays fail on an unknown channel while the minimal 170-parameter MLP mitigates by learning it from data (H5)?

Unlike the canonical experiments, where each transmitted symbol can be processed independently, the 3-tap ISI channel couples adjacent symbols in time. The sufficient statistic for detection is therefore no longer a single received sample, making sequence detection the relevant classical benchmark. To make the classical side as strong as possible, the study benchmarks against the optimal sequence detector: Viterbi MLSE applied to the 3-tap FIR/ISI channel [28] with genie CSI, and a practical variant with a 200-pilot least-squares estimate.

MLSE is introduced here solely because the channel now contains memory. In the canonical memoryless setting of Chapter 5, per-symbol detection is sufficient and sequence detection provides no additional benefit. The need for MLSE therefore arises from the deliberate introduction of ISI in this chapter rather than from the system model studied elsewhere in the thesis.

Relation to data-driven detection, and what is claimed here. That a learned detector can approach model-aware performance without channel state information is not a new result and is not claimed as one. ViterbiNet [37] establishes it directly: by replacing only the channel-dependent computation inside the Viterbi recursion with a neural network, it attains near-CSI-based Viterbi performance while remaining ignorant of the channel, and it does so with a considerably more sophisticated, model-based architecture than the minimal feedforward relay used here. Learned detection has likewise been applied within relay channels, including DNN detectors for MIMO decode-and-forward relaying. The present chapter therefore does not ask *whether* learning can substitute for channel knowledge; that question is settled.

What it asks instead is *when* doing so is worth anything, which the prior work does not address, because a method demonstrated at one operating point does not delineate its own boundary. Three elements of the design serve that question and none of them is a capability demonstration. The flat-channel control (Section 6.2.1) is deliberately a *negative* result: against unknown phase, gain and per-branch asymmetry, all memoryless, the learned relay merely matches symbol-wise DF, so parameter uncertainty alone turns out not to be what learning mitigates. The pilot-budget sweep locates the crossover quantitatively rather than asserting it, showing classical estimate-then-MLSE ahead until its channel estimate degrades past a pilot-budget threshold, and the learned relay indifferent to it. The boundary is one of estimation conditioning, not of strict identifiability: three taps remain identifiable from five pilots, and what fails is the conditioning of the least-squares system (Appendix 9.6). And where the channel family *is* known and adequately pilots, classical detection stays ahead by 1–1.5 dB, which is reported here rather than omitted. The contribution is the boundary these three locate together, not the fact that a network can be trained on an impairment.

Trials.

- **Topology:** SISO (both hops)
- **Modulation:** BPSK (DBPSK for the unknown-phase control and the composite cascade)
- **Hop-1 channel (unknown to all relays); Hop-2 (equal SNR per hop):**
  - 3-tap ISI, $h = [1.0, 0.7, 0.5]/\|h\|$ (Hop 2: AWGN)
  - Flat unknown channels (phase, gain, asymmetry; Section 6.2.1) (Hop 2: Rayleigh)
  - Composite cascade of ISI, amplifier nonlinearity, and unknown phase (Section 6.2.4) (Hop 2: Rayleigh)
- **Relays:**
  - AF
  - Symbol-wise DF
  - MLP (window 11, one hidden layer of 13 tanh units, 170 parameters; trained at SNR $\in \{5, 10, 15\}$ dB using the canonical training protocol)
  - For ISI setups only: Viterbi MLSE applied to the 3-tap FIR/ISI channel, evaluated with either genie CSI or a 3-tap least-squares channel estimate obtained from 200 pilot symbols
- **Evaluation:** 10 trials $\times$ 100,000 bits per SNR point, SNR range 0–20 dB. To rule out the influence of the particular random initialization and training data used to fit the MLP, the network is retrained independently with three different random seeds, giving 30 BER samples per cell (3 training instances $\times$ 10 inference trials). The mean is their average. The confidence interval is *not*: the ten trials inside one seed share a trained network, so they resample Monte Carlo noise but not training variance and are not 30 independent

draws from the population the claim concerns. Intervals are therefore computed hierarchically — average within each seed, then a Student-t interval on the three seed means, $t_{2,0.975} = 4.303$ rather than 1.96, deliberately wide because three points say little about a spread. Pooling all 30 as i.i.d. divides by $\sqrt{30}$ where the effective sample size is nearer 3; on the MLP rows here that understates the interval by a factor of about seven to eight (at 8 dB on S1, $\pm 1.3 \times 10^{-3}$ against a pooled $\pm 1.8 \times 10^{-4}$). AF and DF are barely affected, as expected, since neither has a trained network to vary across seeds.

The ISI channel admits a sharp analytic prediction, and it is not confined to the asymptote. With BPSK symbols i.i.d. and equiprobable, the relay observes $y[n] = h_0 x[n] + h_1 x[n-1] + h_2 x[n-2] + v[n]$ and a memoryless slicer returns sign$(y[n])$. Conditioned on $x[n] = +1$ the two interfering symbols take one of $2^2 = 4$ equiprobable sign patterns, each fixing an effective amplitude $A(s_1, s_2) = h_0 + s_1 h_1 + s_2 h_2$, so

$$P_e(\gamma) \;=\; \tfrac{1}{4} \sum_{s_1, s_2 \in \{\pm 1\}} \Phi\!\left(\frac{-A(s_1, s_2)}{\sigma}\right), \qquad \sigma^2 = \frac{1}{2\gamma}, \quad \gamma = 10^{\mathrm{SNR_{dB}}/10}. \tag{6.1}$$

For $h = [1.0, 0.7, 0.5]$ normalized the four amplitudes are 1.6678, 0.9097, 0.6065 and -0.1516. As $\sigma \to 0$ each term tends to 0 where $A > 0$ and to 1 where $A < 0$, so the limit is the fraction of interference patterns that *flip the symbol sign* — here exactly one of four, giving the floor of 0.25, approached from below.

AF reaches the same floor, but this needs its own argument rather than an appeal to inheritance, since AF performs no hard decision. The relay forwards $g(A + n_1)$ with g the power-normalization gain and the destination slices $g(A + n_1) + n_2$; a positive scale cannot change a sign, so the effective statistic is $A + n_1 + n_2/g$. That is the same four-amplitude structure with an inflated variance $\sigma^2(1 + g^{-2})$, $g^2 = 1/(\mathbb{E}[A^2] + \sigma^2)$, so the $\sigma \to 0$ limit is again the fraction of sign-flipping patterns. The floor belongs to the four amplitudes, not to the decision rule. The two relays approach it differently, and AF is the better of the two at high SNR *because* of its extra noise: on the one pattern whose amplitude is negative, DF errs with probability tending to one, while AF's additional noise still occasionally rescues the decision.

Both closed forms predict the measured curves across the sweep, not the asymptote alone. The measured rows below are the S1 columns of the same three-seed run that produces Table 6.3, read at six of its eleven SNR points rather than the four that table tabulates (`isi_slicer_floor.py` for the closed forms, `e6_sim_ported.py` for the measurements):

SNR (dB)	6	8	12	16	18	20
DF closed form, Eq. (6.1)	0.1786	0.1803	0.2009	0.2280	0.2389	0.2460
DF measured	0.1802	0.1802	0.2009	0.2296	0.2391	0.2457
AF closed form	0.1939	0.1814	0.1835	0.2075	0.2213	0.2338
AF measured	0.1943	0.1813	0.1834	0.2067	0.2213	0.2349

Three to four decimals throughout, including DF's minimum near 6 dB, AF's near 10 dB, and the rise of both thereafter. Two columns of Table 6.3 are therefore predicted in closed form by an argument that shares no code with the simulator, which is an independent check on those columns and on the rare-event estimator behind their 16–20 dB entries. The DF row alone sits low below 6 dB, because that model treats the relay slicer in isolation and omits the second hop; the AF model includes both hops and tracks the measurement from 0 dB up.

Two caveats bound the claim. The value 0.25 belongs to this tap vector, not to ISI in general: the floor is $k/4$ with k the number of *negative amplitudes* $A(s_1, s_2)$ among the four listed above — here one, giving $1/4$; equivalently, two of the eight equiprobable triples $(x[n], x[n{-}1], x[n{-}2])$ are decided wrongly, $2/8 = 1/4$. The count is of amplitudes, not of taps: this channel has no negative tap. so another 3-tap channel gives $0, 0.25, 0.5$ or 0.75, and one with $|h_1|+|h_2| < h_0$ has no floor at all. What is general for a memoryless relay, and what the argument of this chapter rests on, is that the limit is a fixed fraction independent of SNR, so transmit power cannot reach it. The floor is likewise a property of the detector, not of the channel: Viterbi MLSE over the 4-state trellis has no floor when the channel is known. It is more precisely a property of the *zero-threshold* symbol-wise rule than of memoryless detection as a class. The eight noiseless observations separate by symbol: conditioned on $x[n] = +1$ they are $\{1.6678, 0.9097, 0.6065, -0.1516\}$ and on $x[n] = -1$ their negatives, two disjoint sets whose closest members differ by 0.3032. A non-monotonic memoryless rule that exploited the interference distribution could therefore separate them as the noise vanishes, so symbol-by-symbol MAP detection under ISI is not excluded by this result. What the experiment isolates is the benefit of temporal context relative to the AF and zero-threshold DF baselines evaluated here.

6.2.1 Control: Flat (Memoryless) Unknown Channels

Before attributing any classical failure to "unknownness," it must be separated from *memory*. This subsection isolates unknown channels that are *flat* (memoryless): the receiver chain does not know the channel parameters, but the channel introduces no intersymbol coupling. Three physically motivated cases are tested, each with the parameter redrawn per block and unknown to all relays: (F1) an unknown carrier phase $\theta \sim \mathcal{U}[0, 2\pi)$ (oscillator/synchronization offset), with a DBPSK source and conventional differential detection as the classical relay; (F2) an unknown positive gain $g \sim \mathcal{U}[0.3, 2.0]$ (path loss / AGC error), coherent BPSK; and (F3) an unknown per-branch amplitude asymmetry $a_+ \neq a_-$ (per-symbol saturating-PA gain), coherent BPSK. The learned relay is trained identically to before. Its size differs slightly by setup, because the phase case consumes complex I/Q pairs (22 inputs into 7 hidden units, 169 parameters) while the two real-valued cases take an 11-sample window into 13 (170); the two are the same architecture class at the same budget, and are labelled by their exact counts in the table.

These channels remain memoryless and therefore stay within the scope of the canonical model despite the parameter uncertainty. Consequently, they serve as a control that separates the effect of parameter uncertainty from the effect of channel memory: any performance change here arises from model uncertainty, not from memory.

Conclusion (memorylessness $\Rightarrow$ no classical failure, and nothing to mitigate): On all three flat channels the classical relay does *not* fail, and the MLP matches it rather than beating it (largest BER difference 0.0012 across the tabulated SNRs, and 0.0064 over the full sweep, at the 0 dB floor). The reasons are structural: for F1, unknown phase is exactly what differential detection absorbs; for F2, sign(y) is *scale-invariant*, so unknown positive gain is irrelevant by construction; for F3, the fixed threshold is only mildly suboptimal because the asymmetry preserves the noiseless symbol's sign. The classical assumption, memorylessness, still holds in each case, so there is nothing for a learned relay to remove. This is the necessary control for the ISI result: the learned relay's advantage is triggered not by parameter uncertainty but by *unmodeled memory* (or amplitude nonlinearity), which is what breaks the memoryless relay class. This conclusion is stable across all three independent MLP training seeds: the ordering and the size of the DF–MLP difference are reproduced in every training instance, confirming that the result is not an artefact of a lucky or unlucky parameter initialisation.

Table 6.2: Flat unknown channels: classical relay vs. the learned relay (MLP-169 on the complex phase setup, MLP-170 on the two real-valued ones) at 8/12/16/20 dB (mean over 3 training seeds $\times$ 10 trials = 30 samples). The learned relay matches the classical one rather than beating it — the point of the control: unknown parameters alone do not defeat classical relays; memory does (gap sizes in the accompanying conclusion)

Flat channel	Relay	8 dB	12 dB	16/20 dB
Unknown phase (DBPSK)	DF (diff.)	0.0654	0.0286	0.0121 / 0.0050
	MLP-169	0.0666	0.0286	0.0121 / 0.0050
Unknown gain	DF	0.0739	0.0298	0.0122 / 0.0050
	MLP-170	0.0739	0.0300	0.0122 / 0.0050
Branch asymmetry	DF	0.0669	0.0286	0.0121 / 0.0050
	MLP-170	0.0671	0.0287	0.0121 / 0.0050

Conclusion (H5: confirmed as stated, with one boundary): Against the flat-channel control of Section 6.2.1, where unknown parameters alone caused no classical failure, the memory case is qualitatively different. On the unknown ISI channel, both evaluated memoryless classical relays — AF and zero-threshold symbol-wise DF — fail: DF's BER is *non-monotonic*, bottoming out around 8 dB and then rising toward the analytic 0.25 floor (Table 6.3; more transmit power makes DF strictly worse), and AF behaves identically. The 170-parameter MLP, whose window spans the 3-tap memory, learns a joint equalizer–detector and restores reliable relaying: its BER falls monotonically across the same sweep (Table 6.3). The Viterbi benchmarks set that boundary honestly: classical theory *does* solve this channel once its family is known, and genie-CSI MLSE leads the MLP by 1–1.5 dB across the clearly-separated mid-range (roughly 2–10 dB); a 200-pilot LS estimate recovers genie performance almost exactly there too. At the low-SNR extreme the ordering is not clean: near the 0 dB floor the two are within noise of each other (both near the analytic 0.25 ceiling). This does not contradict Viterbi's sequence-ML optimality; it means the *margin* by which it holds is SNR-dependent and vanishes at that end of the sweep, not that the MLP overtakes it. Two things follow, and neither is about channel knowledge, since the filter here is fixed and the relay is trained on it exactly as Viterbi is given it. The first is that the *function class* decides the outcome: the evaluated zero-threshold slicer cannot compensate for this channel at any transmit power, which is why DF worsens with SNR, while a relay whose window spans the memory recovers the link. The second is a complexity result: with the same channel information, 170 parameters land within 1–1.5 dB of the optimal sequence detector at a per-symbol cost that grows only linearly in the window spanning that memory, against the trellis's exponential M^L branch metrics per symbol. The stronger property, mitigating channels whose *family* is unknown or outside linear MLSE's model class, is not demonstrated by this experiment and is established later, by the composite cascade of Section 6.2.4 and the posterior-free study of Section 6.4. The learned relay's distinctive value is near-MLSE reliability on an unseen realization from the impairment family it was trained on, with no per-block identification stage, and arithmetic that grows with the observation window rather than exponentially in the channel state. All of the above qualitative findings replicate across all three independent MLP training seeds: the non-monotonic DF failure, the MLP's monotonically decreasing BER, and the 1–1.5 dB gap to genie-CSI Viterbi are each present in every training instance. The pooled mean (Table 6.3) therefore reflects a structural property of the relay class rather than a fortunate parameter initialisation.

Note on simulation validity. The AF/DF rise above their 8 dB minimum as SNR increases is *not* a simulation artefact or consequence of insufficient data. It is a structural property of any memoryless relay acting on an ISI-distorted signal: at high SNR the hard decision amplifies a louder but equally wrong version of the corrupted waveform, so more transmit power strictly worsens the output. The simulation itself rules out any noise floor: in Table 6.3, genie-CSI Viterbi's BER descends normally through the same SNR range and reaches $< 5 \times 10^{-5}$ by 12 dB, demonstrating that the simulation framework is capable of producing low BER when the channel is correctly modelled. The MLP BER reduction is evidenced by a three-tier measurement programme. (i) At 12 dB, the adaptive-budget evaluation uses 3 seeds $\times$ 10 trials $\times$ $10{,}000{,}000$ bits, yielding approximately $3{,}000$ observed MLP errors across 300M payload bits and a mean BER of 0.0001 with 95% CI $\pm 1.3 \times 10^{-5}$. (ii) At 14 dB, a dedicated $10 \times 1{,}000{,}000$-bit run found 59 errors (BER 5.9×10^{-6}, 95% CI $\pm 2.0 \times 10^{-6}$), confirming a smooth monotone descent rather than a sampling artefact. (iii) At 16 dB and above, the rare-event rule of Table 6.3 was used: transmit until the first error at N_1 bits, then continue to $10N_1$ bits and divide accumulated errors by total exposure. At 16 dB this accumulated 16 errors over 3.34×10^8 bits, giving BER 4.79×10^{-8}; at 18 dB and 20 dB no error occurred within the 10G-bit cap, so the entries are the rule-of-three 95% upper bound $3/N = 3.0 \times 10^{-10}$ rather than an exact zero. Extending past the first error matters: an estimator that stopped at the first event and reported its reciprocal waiting time would rest on a single sample per cell and returned values as implausible as 0.5 for DF at 16 dB, where error counting over the same channel gives the entry in Table 6.3.

Table 6.3: Unknown-channel BER, all entries from one three-seed run. Columns at 8 dB use 3 seeds $\times$ 10 trials $\times$ 100k bits; at 12 dB, 3 seeds $\times$ 10 trials $\times$ 10M bits; at 16–20 dB every relay uses the rare-event error-counting rule described in the accompanying note on simulation validity, and where no error is seen at all the entry is the rule-of-three 95% upper bound $3/N$, not $1/N$. The ISI setup exhibits a memoryless-relay error floor; the MLP restores reliable relaying within ≈1–1.5 dB of genie-CSI Viterbi MLSE

Setup	Relay	8 dB	12 dB	16 dB	20 dB
Unknown ISI $\to$ AWGN	AF	0.1813	0.1834	0.2067	0.2349
	DF	0.1802	0.2009	0.2296	0.2457
	MLP (170p)	**0.0065**	9.60×10^{-5}	$4.79 \times 10^{-8\dagger}$	$< 3.0 \times 10^{-10\ddagger}$
	Viterbi (genie CSI)	0.0072	<5e-5	<5e-5	<5e-5
	Viterbi (200-pilot LS)	0.0074	<5e-5	<5e-5	<5e-5

† 16 dB: 16 accumulated errors over 334,002,040 bits (first error at 33.4M, extended tenfold).

‡ 20 dB: no error in 10G bits, giving the rule-of-three 95% bound $3/N = 3.0 \times 10^{-10}$.

AF and DF at 16–20 dB now come from the same error-counting run as every other cell, not from a separate pooled measurem

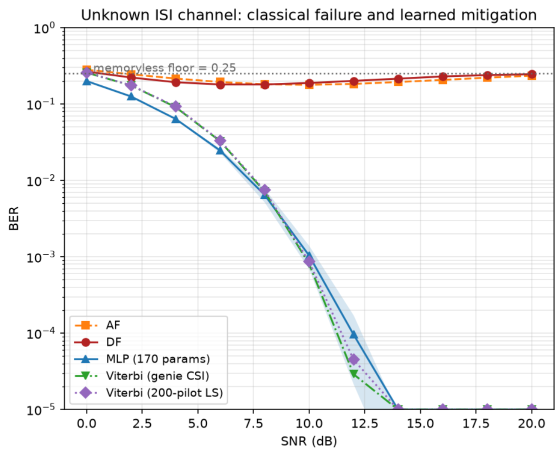


Figure 6.2: Unknown-channel relaying, BER vs. SNR with 95% CI bands. On the unknown ISI channel, the memoryless AF/DF relays are pinned near the analytic 0.25 floor (DF non-monotonic in SNR); the 170-parameter MLP restores monotonically decreasing BER within ≈1–1.5 dB of genie-CSI Viterbi MLSE, which a 200-pilot LS estimate matches almost exactly

6.2.2 Does the Result Generalize to QPSK?

The unknown-ISI study above uses BPSK throughout. The canonical setup of Chapter 5 is QPSK, so this subsection checks that the failure mode is a property of the channel rather than of the constellation it was first observed on. The 0.25 floor derived in Section 6.2 is a property of the *binary* slicer: it counts the $2^2 = 4$ equiprobable interfering-symbol patterns that flip a ± 1 decision. That argument does not extend to QPSK, whose four constellation points carry two levels per axis, so it is not obvious a priori whether either the elevated error floor or the ordering between the learned relay and genie-CSI Viterbi MLSE survives a move to a higher-order modulation. This subsection repeats the unknown-ISI study with Gray-coded QPSK in place of BPSK, keeping every other element fixed: the same 3-tap ISI channel $h = [1.0, 0.7, 0.5]$ (normalized) on Hop 1, AWGN on Hop 2 at the same nominal per-hop SNR (identical convention, $\gamma = 10^{\mathrm{SNR_{dB}}/10}$, memory-bank/techContext.md), and the same relay set generalized to the complex alphabet: AF, symbol-wise DF (hard per-axis decision), a 193-parameter 4-class MLP-QPSK classifier (window 11, softmax cross-entropy, trained at SNR $\in \{5, 10, 15\}$ dB), and Viterbi MLSE generalized to the complex QPSK trellis ($4^2 = 16$ states, complex branch metrics).

One difference from the BPSK study matters more than it first appears. Hop 1 here is not the plain ISI channel of Section 6.2 but

$$y[n] \;=\; g[n]\,(h * x)[n] \;+\; v[n], \qquad g[n] = |\,\mathcal{CN}(0,1)\,| \text{ i.i.d.}, \tag{6.2}$$

an independent Rayleigh magnitude on every symbol on top of the fixed filter. A trellis built from h alone is therefore *not* a genie-CSI detector on this channel: its branch metrics compare $y[n]$ against unfaded expected values, leaving a residual $A^2(g[n] - 1)^2$ that does not shrink as the noise does. Both detectors are reported below — the taps-only trellis, and a genie-CSI trellis given $g[n]$ as well as h, which scales each branch's expected observation accordingly. Evaluation uses 10 trials $\times$ $100{,}000$ bits per SNR point over the same 0–20 dB range. No closed-form floor is derived for QPSK, so the curves below are reported as measured, not fit to a predicted asymptote. Because the QPSK relays are applied jointly to the complex signal rather than decomposed into two independent BPSK problems, and the per-relay power normalization is computed on total (I+Q) power, the resulting AF/DF operating points are not required to, and empirically do not, numerically match the BPSK figures of Table 6.3 at the same SNR label; no such match is claimed.

Table 6.4: QPSK unknown-channel BER (mean over 10 trials) on the per-symbol faded ISI channel of Eq. (6.2). AF and DF plateau at an elevated, non-monotonic BER across the SNR range tabulated, and the 193-parameter MLP-QPSK restores monotonically decreasing BER. The last two rows are the same trellis given different information: with the taps alone it is model-mismatched here and the MLP passes it above about 2 dB; given the per-symbol fading gains as well, it is the genie-CSI detector for this channel and leads the MLP everywhere.

Setup	Relay	8 dB	12 dB	16 dB	20 dB
QPSK: ISI $\rightarrow$ AWGN	AF	0.2447	0.2116	0.2019	0.2100
	DF	0.2197	0.2024	0.2074	0.2211
	MLP-QPSK (193p)	**0.1292**	**0.0827**	**0.0601**	**0.0508**
	Viterbi (taps only)	0.1307	0.0852	0.0670	0.0618
	Viterbi (genie CSI)	**0.0739**	**0.0155**	**0.0018**	**0.0001**

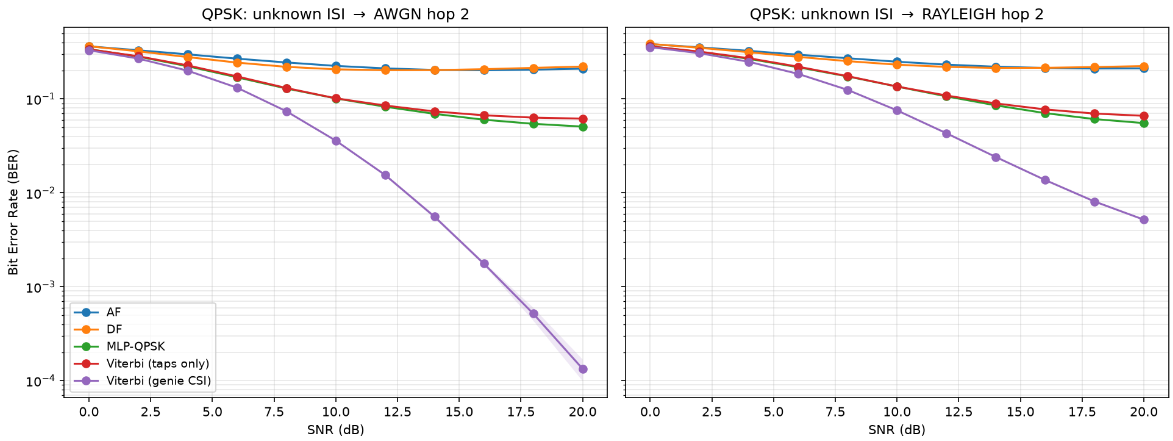


Figure 6.3: QPSK unknown-channel relaying, BER vs. SNR (10 trials, 100,000 bits/point). As with BPSK, AF and DF plateau far above the MLP and Viterbi curves. The MLP-QPSK curve separates below the *taps-only* trellis beyond about 2 dB, but the correctly informed genie-CSI trellis (given the per-symbol fading gains as well as the taps) leads the MLP at every SNR, by a gap that widens with SNR

Conclusion (the ISI failure mode generalizes to QPSK): The qualitative pattern of Section 6.2 survives the move to QPSK: AF and DF plateau at an elevated, non-monotonic BER broadly similar in magnitude to BPSK's memoryless-relay floor of 0.25 (Table 6.4; no equivalent closed form is derived here), while the MLP restores monotonically decreasing BER, confirming that unmodeled channel memory, not modulation order, is what breaks the memoryless relay class (H5).

Two trellis detectors have to be distinguished on this channel, and Eq. (6.2) is why. A trellis built from h alone, on a channel that also fades per symbol, is model-mismatched rather than genie-informed, and its error rate flattens for that reason. Three controls on the same trellis (`qpsk_trellis_controls.py`, run under the exact configuration of Table 6.4) separate the two explanations at 20 dB: given the taps on the faded channel it reaches a symbol error rate of 0.113; on the same channel with the fading removed, 0.000; and given the fading gains as well, 0.0003. The trellis is not at fault (it is verified never to return a path less likely than the transmitted sequence, and to reduce exactly to the nearest-symbol slicer when the ISI is removed) only its channel model was.

With the genie-CSI detector in place the classical benchmark leads MLP-QPSK at every SNR measured, by a margin that grows from 0.3300 against 0.3389 at 0 dB to 0.0001 against 0.0508 at 20 dB. The BPSK ordering therefore generalizes to QPSK, and the comparator rather than the modulation is what decides it. What separates the two studies is not the constellation but the channel: BPSK's hop 1 is the plain ISI filter, on which a taps-only trellis *is* genie CSI, whereas the QPSK study's fades on top of it.

The ordering is not an artefact of the error criterion either. A decomposition of the QPSK error events (`qpsk_error_decomposition.py`) separates the detectors on symbols as well as on bits: measured against the taps-only detector, the MLP is ahead on *symbol* error rate as well from about 8 dB up (0.0939 against 0.1126 at 20 dB), and the two lose almost the same number of bits per symbol error (1.073 against 1.090), so the Gray map is not the route either. It cannot be, for a structural reason stronger than that measurement. The taps are real and the fading gain is a real magnitude, so $y[n] = g[n](h * x)[n] + v[n]$ separates into two independent real ISI channels carrying the in-phase and quadrature bits; the symbol posterior therefore factorises as $P(b_0)P(b_1)$, and bit-wise MAP and symbol-MAP are the same detector on this channel, not merely close ones (`tests/test_bcjr_qpsk.py`, agreement to 5.6×10^{-16}).

Two conclusions follow, and the second is the more important. H5 stands and is unweakened: a 193-parameter windowed relay restores a link that both memoryless classical relays lose, on a channel with memory it was trained on but whose realization it does not see, at a per-symbol cost that does not grow with the trellis state. What does not stand is the stronger reading the earlier text drew from it — that the learned relay was the best-performing option among either family here. It is not. A correctly informed sequence detector beats it comfortably, and by more at QPSK than at BPSK. The learned relay's claim on this channel is the same one the rest of the chapter makes: near-optimal reliability at fixed arithmetic and with no per-block identification stage, not superiority over a model-aware receiver that has been given the model. A BER-optimal BCJR/APP benchmark is reported at QPSK (Section 6.2.3) and is unmeasured at BPSK; where it is measured it confirms the ordering rather than disturbing it.

6.2.3 A BER-Optimal Benchmark: BCJR/APP Against Viterbi MLSE

Every classical comparator used so far is Viterbi MLSE, which minimizes the probability that the whole *sequence* is wrong. This thesis reports BER, and on a channel with memory the two criteria differ: the detector that minimizes bit error probability is the bit-wise MAP rule, read off the posteriors a BCJR forward–backward recursion delivers [29]. The benchmark was therefore never BER-optimal, and the margin it reports over the learned relay had not been measured against one. It has now been (`qpsk_bcjr_benchmark.py`), under the configuration of Table 6.4 and with both detectors given the taps *and* the per-symbol fading gains.

Two controls run first and gate the comparison, because a benchmark can look plausible and still be model-mismatched. With the fading removed the two detectors agree (0.021424 against 0.021764 at 8 dB, a paired difference of $+0.00034 \pm 0.00016$; neither detector produced a bit error at 20 dB, 0 of 10^6 bits each, which bounds both BERs above by 3.0×10^{-6} at one-sided 95% rather than establishing that either is zero), and with the ISI removed the BER-optimal rule reduces to the nearest-symbol slicer on every one of $10{,}000$ symbols.

Conclusion (the margin does not change): BCJR is never worse, as optimality requires, but the advantage it buys shrinks with SNR and is no longer resolvable at 16 dB and above, where the confidence interval on the paired difference includes zero. At 20 dB the classical benchmark moves from 0.000127 to 0.000126 against MLP-QPSK's 0.0508. No ordering reported in this chapter depends on the distinction, and the reading of Section 6.2.2 stands unchanged: the

Table 6.5: Genie-CSI Viterbi MLSE against the BER-optimal BCJR/APP detector on the same trellis, relay output, 10 trials of 100,000 bits. The interval is on the paired per-trial difference, $t(9, 0.975) = 2.262$

SNR (dB)	Viterbi BER	BCJR BER	Relative gain	Paired 95% CI
0	0.250177	0.239113	4.42%	resolved
4	0.162702	0.155861	4.20%	resolved
8	0.068918	0.067245	2.43%	resolved
12	0.015075	0.014857	1.45%	resolved
16	0.001726	0.001725	0.06%	includes zero
20	0.000127	0.000126	0.79%	includes zero

learned relay's claim is near-optimal reliability at fixed arithmetic without a per-block identification stage, not superiority over a model-aware receiver given the model. The value of the measurement is that the claim no longer rests on a comparator that was optimal for a different criterion than the one reported.

6.2.4 A Composite (Mixed) Channel: Mismatch with a Full Posterior

The preceding studies vary one impairment at a time. Real links stack several. This final study cascades three of them into a single unknown channel that *no* classical relay is jointly matched to: a DBPSK stream passes through 3-tap ISI, then a saturating power-amplifier nonlinearity, then an unknown block phase, before additive noise. Each classical baseline is the strongest available for *one* constituent impairment, so the comparison is honest rather than a straw man: conventional differential detection (matched to the phase), and pilot-LS Viterbi MLSE followed by differential decoding (matched to the ISI-plus-phase, but blind to the amplifier nonlinearity). The 169-parameter MLP and a $1{,}153$-parameter MLP receive raw I/Q windows and no knowledge of any constituent. This composite channel departs from the canonical model in two ways: it introduces explicit channel memory through ISI and nonlinear distortion through the amplifier. The resulting benchmark therefore tests robustness to both memory and model mismatch simultaneously. **Evaluation:** 10 trials $\times$ 100,000 bits per SNR point, the same budget as the main unknown-ISI study (Section 6.2); 95% CIs use the $1.96\,\sigma/\sqrt{n}$ convention throughout. The hop-1 impairment here is fixed rather than redrawn per block, so bits per trial and trials both reduce the reported uncertainty.

Conclusion: The composite defeats the naive classical relays outright, both AF and conventional differential detection saturate near the 0.25 memoryless-relay floor, because the ISI destroys the differential statistic that the phase-matched detector relies on. The 169-parameter MLP again restores reliable relaying, reaching 2.5×10^{-3} at 20 dB from raw I/Q with no impairment model. This reproduces the H5 pattern on a realistically mixed channel: memoryless-and-matched classical processing fails, and a minimal learned relay recovers. Two familiar boundaries also reappear. First, the strongest *constructed* classical baseline, pilot-LS Viterbi plus differential decoding, does not fail: by modelling the ISI and phase (though not the amplifier) it stays about 2 dB ahead of the MLP at low-to-mid SNR (0.0674 vs. 0.1073 at 8 dB) and converges with it by 16–20 dB, the two becoming indistinguishable at 0.0025 each at 20 dB. The learned relay's value here is that it needs *no analytic model* of the impairment, rather than superiority over a matched classical receiver: it reaches near-Viterbi reliability with no analytic model of any of the three cascaded effects, whereas estimate-then-MLSE must assume a linear FIR channel and therefore cannot represent the amplifier at all. This cascade is itself a fixed channel that the relay is trained on, so it speaks to the model class the two methods can express, not to generalization across channels; that is the subject of Section 6.4, where a fresh channel is drawn for every block. Second, the $1{,}153$-parameter network is close to the 169-parameter one across the whole range, ending at the identical 0.0025 at 20 dB. The mid-SNR comparison is the one place in this chapter where extra capacity may buy something measurable, and the three-seed re-run sharpens it rather than settling it: the larger network leads at both mid-SNR points, by 0.0057 at 8 dB (0.1016 against 0.1073), which sits inside the combined confidence intervals of 0.0099, and by 0.0059 at 10 dB (0.0482 against 0.0542), which is marginally outside the corresponding 0.0054. Nearly seven times the parameters therefore buys nothing at 8 or 20 dB and possibly a little at 10 dB, at the very edge of what this trial budget resolves. That is weaker than a flat reconfirmation of the minimal-capacity finding (H3), and it is recorded here as the exception rather than folded into it. The composite thus consolidates the chapter's thesis: a single minimal learned relay handles an arbitrary stack of representable impairments without being told which are present, approaching but not exceeding the best impairment-matched classical receiver. These conclusions are consistent across all three independent MLP training seeds: the H3 finding (MLP-large $\approx$ MLP-169), the convergence of both with Viterbi-diff at 20 dB, and the 2 dB mid-SNR margin of Viterbi over the learned relay are each reproduced in every training instance.

6.3 Layer 3 — CSI Uncertainty: The Pilot-Budget Crossover

This is layer 3 of the ladder in Table 6.1: the perfect-CSI assumption is dropped, and nothing else changes. Every classical result so far has been handed either exact channel taps or a 200-pilot estimate good enough to match them (Section 6.2.4), which is what a full posterior buys. Here the posterior is made *partial*: the pilot budget is finite, and the classical receiver must work from a correspondingly noisy estimate. Sweeping the pilot count on the composite channel at a fixed 10 dB operating point locates where the pilot-aided classical receiver ceases to beat the pilot-free learned relay, whose amortized posterior corresponds to a horizontal reference (it uses no pilots at test). **Evaluation:** 50 trials $\times$ 20,000 bits per operating point ($1{,}000{,}000$ bits), matching Section 6.4 and for the same reason: the channel is redrawn per block, so trials are draws from the impairment family. In panel (b) each trial is further subdivided into independent blocks of the swept length, so every block length is measured over the same total bit budget and

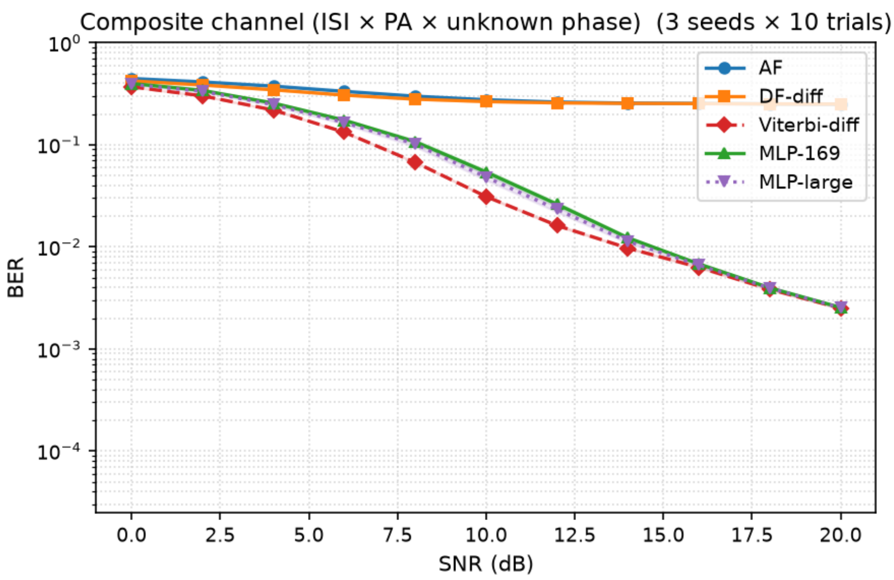


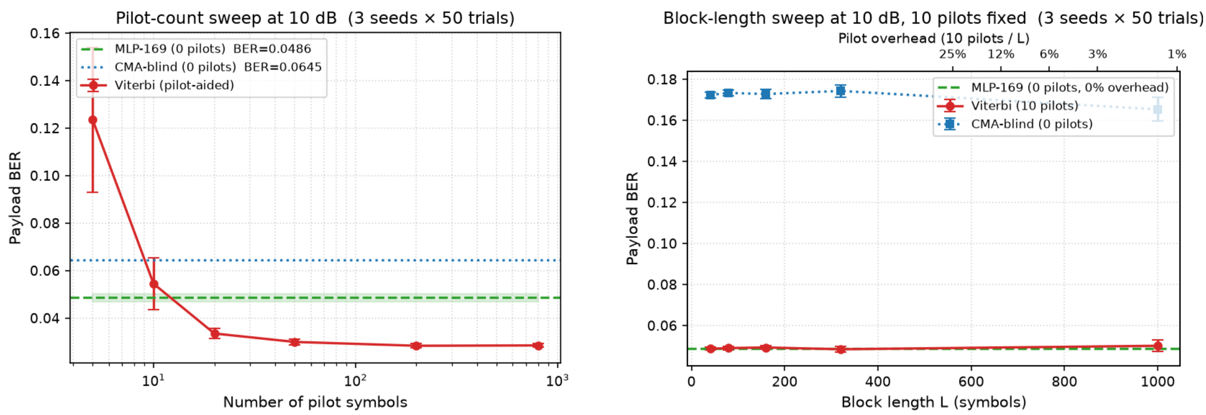
Figure 6.4: Composite channel (ISI $\times$ PA-nonlinearity $\times$ unknown phase, DBPSK; mean over 10 trials, 100,000 bits/point, 95% CI bands). Naive classical relays (AF, differential detection) saturate at the memoryless-relay floor; the 169-parameter MLP restores reliable relaying from raw I/Q with no impairment model, tracking about 2 dB behind the strongest constructed classical baseline (pilot-LS Viterbi + differential) and converging with it by 20 dB. The $1{,}153$-parameter MLP adds nothing (H3)

over many channel draws rather than a single one.

Conclusion (an estimation-variance cliff at 10 dB, and a rate–reliability tradeoff): With a generous budget the classical partial posterior wins, as expected: pilot-aided Viterbi reaches payload BER 0.0285 at 800 pilots and degrades only gently down to 0.0335 at 20 pilots, comfortably below the MLP's pilot-free 0.0486. At 10 pilots Viterbi has lost its edge, 0.0545 against the MLP's 0.0486, and at 5 pilots it reaches 0.1235, because MLSE is then running on a badly estimated channel. The mechanism is estimation variance rather than rank deficiency: with 5 pilot observations and 3 complex taps the least-squares problem is formally overdetermined, but there are too few observations to average down the noise. The 95% interval at that point is 0.1235 ± 0.0304, against ± 0.0011 at 50 pilots. The MLP remains at 0.0486 because it requires no per-block pilots after training on the channel family. Thus, at the measured 10-dB operating point, the crossover lies between 20 and 10 pilots. Its location is necessarily SNR-dependent because least-squares estimation error depends on noise power; no pilot-count sweep was run at another SNR, so a universal crossover is not claimed. The ordering is stable across the three trained MLP instances.

The practical force of this depends on block length, because a fixed pilot *count* is a growing pilot *fraction* as blocks shorten. Panel (b) sweeps block length at the same operating point with the classical receiver spending its minimum viable ten pilots per block, and adds blind CMA as the second pilot-free reference. The payload BERs of Viterbi and the MLP are now statistically indistinguishable at every block length, both flat at approximately 0.049, so reliability no longer separates them at all; what separates them is spectral efficiency. At a 1000-symbol block the ten pilots cost 1% of the block, whereas at a 40-symbol block, representative of low-latency or fast-varying links, they cost 25%, so the classical receiver carries data on only 75% of the block against the learned relay's 100% for identical error performance.

The blind reference settles a further question. Blind CMA is also pilot-free, so the learned relay is not unique in avoiding pilots; the distinction claimed for it was that it needs *zero per-block adaptation*, a single fixed forward pass, whereas CMA must converge its adaptation loop inside each block. Measuring CMA per block confirms this directly: its payload BER is 0.1723 at $L = 40$ and only improves to 0.1653 at $L = 1000$, against the 0.0645 it achieves when given a $20{,}000$-symbol block to adapt over. CMA therefore does not converge within blocks of a thousand symbols or fewer, and is roughly three and a half times worse than either the MLP or pilot-aided Viterbi throughout this sweep. The partial-posterior picture therefore refines H5's boundary once more, and more sharply than before: against a classical receiver that can afford accurate channel estimation the learned relay achieves no better BER, but at short block lengths it matches that receiver's BER while removing its pilot overhead entirely, and at these block lengths it is the only pilot-free option that stays reliable, because the classical pilot-free alternative has no time to converge: CMA sits at 0.1723 on 40-symbol blocks and is still at 0.1653 at 1,000. This is a statement about short blocks specifically, and it does not carry over to Section 6.4, where blocks are long enough for CMA to converge and it comes within a factor of 1.3 of the learned relay.



(a) pilot-budget sweep at 10 dB (b) block-length sweep at 10 pilots

Figure 6.5: Partial-posterior regime; both panels are means over 50 channel draws at 20,000 bits/-point. **(a)** Pilot-aided Viterbi outperforms the pilot-free MLP down to approximately 20 pilots per block, loses its edge by 10 pilots, and collapses at 5 pilots when the least-squares estimate becomes too noisy to construct a reliable trellis (0.1235 ± 0.0304); the MLP is flat across the sweep, operating with an amortized posterior and requiring no pilots. **(b)** With 10 pilots per block, pilot-aided Viterbi and the MLP are indistinguishable in payload BER at every block length (both ≈ 0.049), but the classical receiver's pilot overhead grows from approximately 1% at $L = 1000$ to 25% at $L = 40$. Blind CMA, the other pilot-free option, cannot converge within blocks this short and sits roughly three and a half times worse throughout. The MLP therefore matches the pilot-aided receiver's reliability at no overhead, and is the only pilot-free method that stays reliable for short packets

6.4 Layer 4 — Unknown Per-Block Realization: The Blind Regime

This is layer 4, the bottom of the ladder. Per-block CSI and pilots are removed, and a fresh realization is drawn from the same impairment family used during MLP training. The comparison is against CMA, which assumes constant modulus and adapts an equalizer from the received block, and decision-directed MLSE, which bootstraps a channel estimate from its own decisions. Thus no method receives per-block CSI, but neither is assumption-free: the MLP knows the training distribution and CMA knows signal/equalizer structure. **Evaluation:** 50 independent channel draws $\times$ 20,000 bits per SNR point. Classical-method intervals use those 50 draws; neural uncertainty is assessed across trained instances rather than by treating repeated inference under one network as independent training evidence.

A correction to the blind equalizer. The constant-modulus algorithm used here was originally implemented with a fixed (unnormalized) step size. Because the CMA error term is cubic in the equalizer output, a fixed step is a positive-feedback loop: the implementation happened to remain bounded over the short blocks first used, but diverges to numerical overflow on longer ones, after which the equalizer output is noise rather than an estimate. The step is therefore normalized by the input-segment energy in the standard NLMS manner, which is both the textbook formulation and a strictly *stronger* classical baseline. All CMA results below use the corrected equalizer; the uncorrected one is not reported, since its outputs past the divergence point measure an implementation fault rather than the algorithm.

Conclusion (learning narrowly leads CMA and avoids per-block adaptation): The corrected CMA reaches BER 3.3×10^{-3} at 20 dB, and the family-trained MLP reaches 2.6×10^{-3}. The MLP holds a small measured margin over the evaluated range, the MLP at 0.0955 against CMA's 0.1182 at 8 dB, while decision-directed blind MLSE is non-monotonic (0.0374 at 16 dB and 0.0498 at 20 dB). Whether that margin is resolved is a question the intervals have to answer rather than the point estimates: the decision-directed MLSE interval at 8 dB is ± 0.0201, against the MLP's ± 0.0026 and CMA's ± 0.0078. This does not make CMA unstable: the observed instability belongs to the decision-directed MLSE implementation. CMA instead carries an online adaptation cost and requires enough samples to converge, whereas the MLP uses one fixed forward pass after offline training. For uncertainty reporting, the three trained MLP instances must be treated hierarchically rather than pooling their inference trials; classical methods have no training seed, so their effective sample size is the 50 independent channel draws. The supported conclusion is therefore narrow: on unseen per-block realizations from its training family, the MLP slightly outperforms the evaluated CMA implementation while avoiding per-block adaptation, and it avoids the non-monotonic behavior of the evaluated decision-directed MLSE.

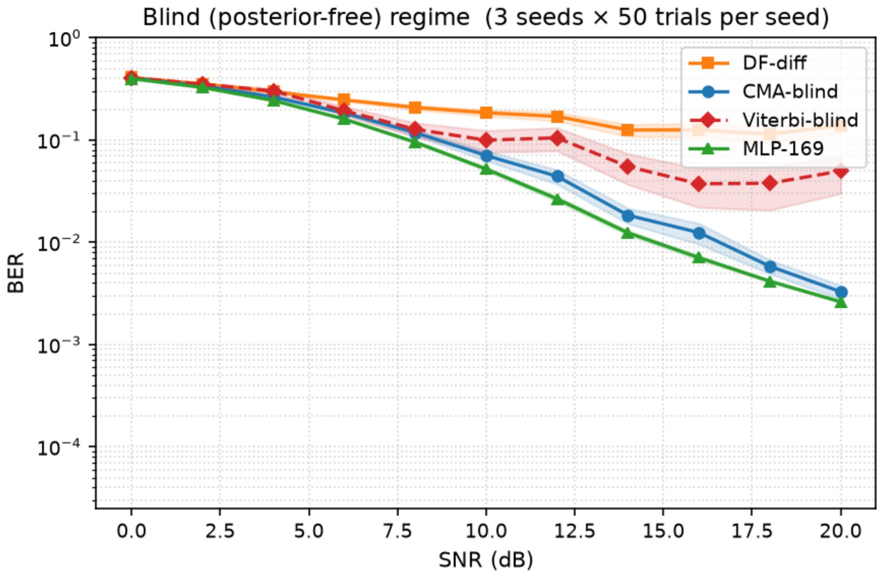


Figure 6.6: Blind composite channel (no pilots or per-block CSI; fresh realization from the trained family per block; mean over 50 channel draws, 20,000 bits/point, 95% CI bands). The corrected CMA (blind classical) tracks the family-trained MLP closely, the MLP holding a small consistent margin; decision-directed blind MLSE is unstable (non-monotonic tail, CI band several times wider) because it bootstraps the channel from its own decisions; plain differential detection floors. Without a posterior, learning only narrowly outperforms the best blind classical equalizer, but does so with one-shot, stable inference and no online adaptation

6.4.1 A Low-Complexity Linear Baseline: MMSE Equalization

The comparisons above set the learned relay against two classical relays, and neither is matched to it in cost. Symbol-wise DF is memoryless and cannot undo intersymbol interference at any transmit power, so outperforming it establishes only that the relay uses its window. Viterbi MLSE is the optimal sequence detector but costs $O(M^L)$ per symbol in the channel memory, which is precisely the expense Section 6.5 invokes on the learned relay's behalf. The case between them was not tested: an N-tap MMSE linear equalizer is a natural low-complexity remedy an engineer might reach for before a trellis. If it recovered most of what the learned relay recovers, the contribution of Chapter 6 would be a narrower claim about a nonlinearity rather than about learned processing.

It does not. Table 6.6 gives the equalizer's SNR penalty against MLSE for equalizer lengths from 3 to 11 taps, measured on the same two-hop chains, with the exact channel taps supplied and the weights redesigned at every SNR point — the same genie CSI granted to MLSE, and a longer window than the learned relay is given.

Two readings follow. The learned relay's advantage over the evaluated memoryless relays is not solely an artifact of comparing against detectors without temporal context: it also leads this low-complexity linear equalizer by 1.8 to 2.4 dB. This does not establish superiority to every cost-matched classical method; decision-feedback and reduced-state sequence equalizers were not evaluated, and comparable operation counts are not reported here. And on the composite channel

Table 6.6: SNR penalty against Viterbi MLSE for an MMSE linear equalizer of N taps, with the best learned relay on the same channel for comparison. Negative values beat MLSE. The reader should observe that the learned relay leads the equalizer by 1.8 to 2.4 dB on every channel, despite the equalizer being given exact channel knowledge, per-SNR weight redesign, and up to 11 taps against the relay's window of 7

Channel	3 taps	5 taps	7 taps	11 taps	Learned relay
ISI (BPSK)	$+6.58$	$+4.40$	$+4.23$	$+4.04$	$+\mathbf{1.67}$
ISI (QPSK)	$+7.81$	$+9.11$	$+5.02$	$+4.84$	$+\mathbf{3.03}$
Composite	-0.33	-0.37	-0.37	-0.44	$-\mathbf{2.26}$

both the equalizer and the relay beat MLSE, which is expected rather than anomalous, since a fixed-tap trellis is mismatched to that channel's amplifier nonlinearity and phase term while neither of the other two detectors assumes a purely linear channel. The one non-monotonicity — five taps ($+9.11$ dB) worse than three ($+7.81$) on the complex ISI channel — is an artefact of the metric, not the equalizer. A longer filter nests a shorter one, so the attained MMSE is non-increasing in tap count by construction, and is verified so here ($0.0666, 0.0374, 0.0354, 0.0333$ at 16 dB for $N = 3, 5, 7, 11$). The tabulated figure is the *worst* penalty over targets 10^{-1}, 10^{-2}, 10^{-3}; at the first two it is monotone as that argument requires ($+3.24, +2.98, +2.79, +2.65$ and $+6.34, +4.27, +4.13, +4.02$ dB). Only 10^{-3} misbehaves, and being the maximum it sets every entry in the row — its crossing lies at the top of an SNR grid sampled every 4 dB, where interpolating a steep curve is coarse. The conclusion is unaffected: the learned relay leads at every target and tap count.

6.4.2 Does a Sequence Architecture Help Where a Window Does?

Chapter 5 finds that Transformer, Mamba-S6 and Mamba-2 buy nothing over a feedforward relay on the canonical channel, and attributes this to the channel rather than to the architectures: a memoryless channel offers no temporal structure for their inductive bias to exploit. That is a prediction about what would happen given such structure, and until now it was untested here, no sequence model having been evaluated anywhere in this chapter. The minimum-size study sharpens the prediction, finding the relay's *window* to be the binding resource on channels with memory, worth 8 to 11 dB between window 1 and window 7 while width and depth buy almost nothing. A window is a blunt way of granting a feedforward network temporal context; sequence models are built for it.

The prediction fails. Table 6.7 compares the four architectures at an equal budget of approximately $3{,}000$ parameters, each trained on the channel under test rather than on a memoryless surrogate, with identical hyperparameters and three initializations apiece.

The feedforward relay leads on all three channels, by 1.55 and 2.11 dB on the two ISI channels and by 0.32 dB on the composite. The conclusion is therefore the opposite of the natural one, and it is a positive statement rather than a null result: channel memory raises the size floor without

Table 6.7: SNR penalty against Viterbi MLSE at an equal budget of ≈3,000 parameters, best of three initializations, with the across-initialization spread in brackets. Negative values beat MLSE. The reader should observe that the feedforward relay leads every sequence architecture on every channel, and that the two Mamba variants (the most reproducible architectures measured on the canonical channel, at 0.019 and 0.041 dB) become the least reproducible here

Channel	MLP-3K	Transformer-3K	Mamba-S6-3K	Mamba2-3K
ISI (BPSK)	$+$**1.97** (0.39)	$+3.52$ (0.33)	$+4.30$ (1.15)	$+4.55$ (0.14)
ISI (QPSK)	$+$**2.58** (0.07)	$+5.96$ (0.45)	$+5.72$ (1.33)	$+4.69$ (2.12)
Composite	$-$**2.08** (0.03)	-1.76 (0.03)	-1.56 (0.08)	-1.60 (0.06)

calling for a sequence architecture. What a relay needs on such a channel is sufficient temporal context, and a fixed window supplies it more cheaply than a learned sequence mechanism, both in parameters at equal accuracy and in training time, which ran 35 to 66 seconds per initialization for the feedforward relay against 217 to 609 seconds for the sequence models.

A second observation was not anticipated. The sequence models' reproducibility inverts. On the canonical channel Mamba-2 and Mamba-S6 are the two most stable architectures measured across initializations; on the complex ISI channel the same architectures spread by more than a decibel (bracketed spreads in Table 6.7), while the feedforward relay stays inside 0.07 dB throughout. They are worse on channels with memory, and substantially less predictable there, which compounds the case against them for this task.

One limitation is carried. All four architectures receive identical hyperparameters, matching the equal-budget protocol of Section 5.3; no per-architecture tuning was attempted, and a tuned sequence model might close part of the gap.

6.5 The Cost Side: Decision Delay and Arithmetic

Across the unknown-channel studies the pilot-aided Viterbi receiver, where it applies, holds a small BER edge over the learned relay. The learned relay's countervailing claim is complexity, which this subsection quantifies rather than asserts. Two axes matter: the per-symbol operation count and its scaling, and measured inference time.

Per-symbol cost and scaling. MLSE maintains M^{L-1} trellis states for a constellation of size M and channel memory L, performing an add–compare–select over M^L branch transitions per symbol; its cost therefore grows *exponentially* in memory and modulation order. The 170-parameter MLP performs one fixed forward pass, about 330 floating-point operations per symbol. That figure is a constant of the *deployed* architecture, not a quantity independent of the channel: the input window has to span the memory, so its length grows with L, and the output dimension grows with the constellation, from a single real output at $M = 2$ to the 193-parameter QPSK relay used earlier in this chapter, and further still for denser constellations. Once those are fixed, the per-symbol cost is the same for every symbol and every channel realization, which is the property the comparison below relies on. The contrast with MLSE is therefore one of *growth rate*, not of a constant against a variable: the network's cost rises roughly linearly in window length and output dimension, the trellis's exponentially in M^L.

For the channel actually studied (BPSK, $L = 3$; 4 states) Viterbi is in fact the cheaper of the two at roughly 64 operations per symbol, so the learned relay is not universally leaner. But the scaling diverges immediately: at $M = 4$, $L = 5$ the trellis has 256 states ($\sim 8{,}000$ operations/symbol), and at $M = 16$, $L = 5$ it has $65{,}536$ states (~ 8.4 million operations/symbol), whereas a network resized for that same channel, an 11-sample window and a 16-way output head, is on the order of 10^3 operations by direct count of its layer dimensions, leaving a ratio in the thousands. The measured 330 applies to the BPSK, $L = 3$ case; the larger figure is arithmetic from the architecture, not a timing measurement. The learned relay's advantage is thus not lower cost on a small channel but *constant* cost as the channel scales, exactly where classical MLSE becomes intractable and practical receivers must fall back on suboptimal reduced-state or decision-feedback equalizers [38]. Those methods are not benchmarked here: full Viterbi is taken as the steepest classical comparator, which makes it the conservative choice for the comparison but leaves the reduced-state middle ground unmeasured. This constant-cost advantage is a property of the present relay's size, not of learned equalizers as a class: neural equalizers are often parameterized heavily enough that their complexity becomes comparable to, or larger than, the trellis they replace [39]. The relay studied here, at 170–193 parameters, sits at the favourable extreme of that range, and the claim should be read as scoped to that regime.

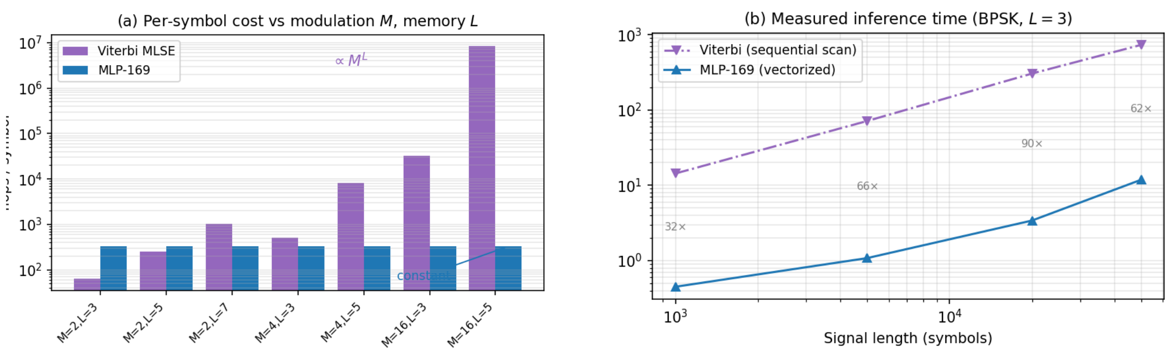


Figure 6.7: Complexity of Viterbi MLSE against the 170-parameter MLP. **(a)** Per-symbol computational cost versus modulation order M and channel memory L: Viterbi grows as M^L (intractable by M=16, L=5), while the network grows only linearly in window length and output dimension ($\sim$330 flops for the deployed BPSK, L=3 relay). On the small studied channel (BPSK, L=3) Viterbi is actually cheaper; the MLP wins on *scaling*, not on the small case. **(b)** Measured inference time for BPSK with $L = 3$: the MLP is 30–90$\times$ faster as a vectorized matmul versus Viterbi's interpreted sequential trellis scan

Measured inference time. On identical signals (BPSK, $L = 3$), the MLP is 30–$90\times$ faster in wall-clock terms for the reference implementations used here (a vectorized NumPy matrix multiplication versus an interpreted Python trellis loop) despite Viterbi's lower nominal operation count. Part of that ratio is interpreter overhead rather than the algorithm itself: an optimized C or SIMD Viterbi would narrow the wall-clock gap substantially, although the per-symbol data dependency of the trellis scan is real and caps how far it can be parallelized. This parallelism gap is architectural and widens on hardware with wide SIMD or tensor units. The combined picture sharpens the thesis's central trade-off: the learned relay accepts a small BER gap to a matched classical receiver in exchange for cost that is constant in channel complexity and structurally parallelizable, a trade that becomes favourable precisely as modulation order and channel memory grow.

6.5.1 A Channel with Memory Under a Latency Constraint

The accounting above and the ladder that precedes it answer different questions and never meet. The ladder asks which detector is more accurate on a channel with memory, imposing no budget; this section's opening asks what each detector costs, without reference to a channel. What a deployment actually faces is both at once, and the two constraints do not have to bind in the same place. This subsection measures the intersection directly, and includes block DF, which the ladder omits: of the classical options it is the one whose cost is structural rather than arithmetic, so it is exactly the case the ladder's accuracy-only comparison cannot price.

Every scheme is placed inside one identical coded system so that the comparison is between relays and nothing else: a rate-$1/2$, $K = 3$ convolutional code on QPSK, 200 information bits per frame (202 symbols after tail), with soft Viterbi decoding of the code at the *destination*. All schemes therefore deliver an information BER at the same information rate. Hop 1 carries the unknown L-tap ISI response ($h_k = 0.7^k$, unit-energy normalized) plus complex AWGN and hop 2 is AWGN; 150 frames per trial and three seeds per point.

Each relay is then labelled with the structural decision delay it forces, in symbols: zero for AF and symbol-wise DF; $\lfloor W/2 \rfloor$ for a relay of window W, which must wait for the later half of its own window; D for an MLSE with traceback depth D; and the full frame, 202 symbols, for block DF, which cannot begin until the last symbol of a codeword has arrived.

No latency separation was resolved on the tested short-memory channel. Table 6.8 gives the result at $L = 3$. The BER at 12 dB is unchanged at the reported precision from traceback depths $D = 2$ through $D = 15$. This finite-budget observation does not prove that all survivor paths merge by three symbols or that full-asymptotic accuracy is reached. Among the tested settings, MLSE at $D = 2$ has less delay than the window-11 relay's five symbols and lower measured BER. Block DF spends 202 symbols without a resolved accuracy gain in this experiment; with an equalizer it reproduces the reported MLSE value but adds frame buffering.

The symbol-wise relays confirm the chapter's Layer 2 finding inside the coded chain: AF and hard DF do not merely lose, they fail, DF's BER *rising* with SNR from 0.2913 at 8 dB to 0.4253 at 20 dB as the interference rather than the noise comes to dominate its decisions.

Table 6.8: Coded information BER at 12 dB on the three-tap channel, against the structural decision delay each relay forces and its arithmetic per symbol. Mean of three seeds. No BER difference among the tested MLSE traceback depths is resolved at the reported precision; this is not an equivalence or survivor-merger test

Relay	Delay (symbols)	MACs/symbol	BER at 12 dB
AF	0	2	0.0609
DF-hard (symbol-wise)	0	0	0.3027
MLP $W = 5$	2	112	0.0010
MLP $W = 11$	5	208	0.000189
MLP $W = 21$	10	368	0.000278
MLSE $D = 2$	2	128	**0.000033**
MLSE $D = 3$	3	128	**0.000033**
MLSE $D = 15$	15	128	**0.000033**
block DF	202	16	0.0411
block DF + MLSE	205	144	0.000033

The cost convention. Counts below are multiply–accumulates per transmitted symbol, stated here so they can be checked. Additions and comparisons carrying no multiply are excluded; they dominate the trellis search, so the comparison is conservative against this thesis's own conclusion. MLSE has M^{L-1} states and M branches from each, so M^L branch metrics per symbol; with the expected observation precomputed each is $|y - c|^2$ on a complex difference, two real multiplies, giving $2M^L$ (its further M^L additions and $M^{L-1}(M-1)$ comparisons are not counted). The relay's $2W$-value input window, the I and Q streams, into H hidden units is $2WH$, and the four-class head $4H$, giving $2WH + 4H$; biases are additions and the activations transcendental, so neither is counted.

What separates them is arithmetic, and only as memory grows. MLSE costs $2M^L$ MACs per symbol. The cleanest statement of the contrast is asymptotic: covering a channel of memory L requires a window $W = O(L)$, so the relay costs $O(LH)$ where the trellis costs $O(M^L)$ — linear against exponential in the memory. Holding the architecture fixed is the special case of that: a relay with W pinned costs $2WH + 4H$ regardless of L — with the important caveat that holding W fixed while L grows is a choice, not a law: preserving observation coverage as the memory lengthens generally requires widening the window, and the relay's cost then grows roughly linearly in W rather than not at all. Setting the two equal locates the crossover for one stated architecture, with no appeal to a processor, clock rate or symbol rate:

$$L^\star = \log_M\left(\frac{2WH + 4H}{2}\right) = 3.35 \text{ taps (MAC-count crossover)} \tag{6.3}$$

for the window-11, eight-unit relay used here (2.90 taps at $W = 5$, 3.76 at $W = 21$). This figure is an arithmetic crossover for a fixed (W, H, M) and the MAC convention stated above; it is not a universal boundary for learned relaying, and it moves with the window — which is precisely the quantity a longer memory may force upward. Below $L^\star$ the learned relay has no case: at $L = 3$ MLSE needs 128 MACs per symbol against the relay's 208, decides in three symbols against its five, and is the more accurate of the two. Above $L^\star$ the gap compounds, because one cost grows as M^L while the other grows at worst linearly in the window. The window-11 relay is measured here to hold its accuracy out to $L = 7$ (Table 6.9), so over the span swept the comparison is like-for-like; extrapolating it to longer memories would require re-checking that a fixed window still suffices. Table 6.9 sweeps it.

Table 6.9: Coded information BER at 12 dB and arithmetic per symbol as channel memory grows, with MLSE given genie CSI and a traceback of $5L$. The relay is the same window-11 network throughout. BER is measured at $L \in \{3, 5, 7\}$ over 1,500,000 information bits per point (five seeds), with 95% Wilson intervals; the $L = 4$ and $L = 6$ MAC counts are arithmetic from the architecture rather than measured points, and are given only to show the growth rate. The reader should observe that the accuracy ratio does not grow with L: the relay trails by a factor between five and eight at every measured memory length, while MLSE's arithmetic rises by 256

L	MLSE states	MLSE MACs	Relay MACs	MLSE BER (95% CI)	Relay BER (95% CI)
3	16	128	208	3.20×10^{-5} [2.41, 4.24]	2.36×10^{-4} [2.13, 2.62]
4	64	512	208	—	—
5	256	2,048	208	4.20×10^{-5} [3.28, 5.37]	1.96×10^{-4} [1.75, 2.20]
6	1,024	8,192	208	—	—
7	4,096	32,768	208	2.47×10^{-5} [1.79, 3.40]	1.91×10^{-4} [1.70, 2.14]

Across the measured span the relay's cost and error are both flat while MLSE's cost rises by a factor of 256, for an accuracy advantage that does not grow with the channel either: the ratio is 7.4 at three taps, 4.7 at five and 7.7 at seven, and the three confidence intervals overlap one another. What grows with channel memory is the price of the classical detector, not the penalty for declining to pay it. This is the trade the chapter's opening paragraphs assert qualitatively, now stated as a measured crossover: under the MAC-counting convention of this section, and for the fixed architecture deployed here, the learned relay crosses full-state MLSE in multiplication count between the evaluated three- and five-tap settings, after which the margin grows as $M^L/(2WH+4H)$. The crossover is a property of that architecture and that counting convention, not a universal threshold in channel memory.

Two limits are worth recording. The threshold is a statement about *arithmetic*, not about accuracy: above $L^\star$ MLSE remains the more accurate detector wherever it can be afforded, and the learned relay's case rests on it becoming unaffordable rather than on it becoming wrong. And the three-tap channel used throughout this chapter sits *below* $L^\star$, so this chapter's earlier results should be read as establishing the failure of the memoryless relays on channels with memory, which they do, and not as a cost argument against MLSE, which at $L = 3$ they cannot support.

The experiments of this chapter deliberately extend the thesis beyond the canonical memoryless model. The resulting conclusions should therefore be interpreted as robustness results under model mismatch rather than as replacements for the matched-model conclusions of Chapter 5.

Together, these two principal contributions of the thesis delineate where learning provides value: not under matched memoryless conditions, where classical methods remain strongest, but when the assumptions underlying those methods are violated.

Chapter 7

Discussion and Conclusions

This chapter draws on all three studies; which claim belongs to which is governed by the reading rule of Section 1.

The canonical core (Chapter 5) settles hypotheses H1–H4 and establishes the baseline finding that, on a matched known channel, a minimal learned relay does not improve on classical decode-and-forward. The thesis's principal contribution (Chapter 6) settles H5 and is where the substantive new claim lies: a systematic delineation of when a learned relay adds value under unknown and mismatched channels. The interpretation below treats these two as the thesis's two principal contributions, with the unknown-channel results carrying the main argument; the supplementary coded study is the smaller, third part, discussed separately below where it bears on the relay-decoding and rate-adaptation questions it was designed to measure.

7.1 Interpretation of Results

The experimental results reveal several consistent patterns across the canonical setup, its supplementary coded-link study, and the unknown-channel contribution. This section interprets these patterns through the lens of the theoretical framework established in Section 2.5 and evaluates each research hypothesis.

7.1.1 Low-SNR Advantage of Neural Relays over AF (H1: Confirmed)

Low SNR (0-4 dB): AI advantage over AF. At low SNR, the feedforward learned relays outperform AF on the canonical channel, confirming H1 for the tested configurations. The scalar BPSK-AWGN posterior mean below is qualitative motivation, not the exact canonical estimator. The canonical link uses Gray-QPSK and complex Rayleigh fading; after zero forcing, the conditional real-axis noise variance is $\sigma^2/(2|h|^2)$. If $|h|$ is not supplied to the relay, the relevant estimator marginalizes over that reliability and is not a single fixed-variance tanh.

At low SNR (e.g., 0 dB, $\sigma^2 = 1$), the fixed-variance form is a gentle sigmoid, $f^*(y) = \tanh(y)$;

the fading case rescales its slope per symbol and, unconditioned, softens it into a mixture, but the qualitative point is unchanged: the optimal relay map is a smooth saturating non-linearity rather than a linear gain. The neural network can approximate such a mapping accurately, producing a **soft estimate** that preserves information about the confidence of the decision. By contrast:

- **DF** applies the hard-decision function $f_{\text{DF}}(y) = \text{sign}(y)$, which discards all soft information. At low SNR, many received samples lie near the decision boundary ($|y| \approx 0$), and DF assigns them full confidence (± 1) regardless of the actual uncertainty. This information loss leads to excess errors on the second hop.
- **AF** applies $f_{\text{AF}}(y) = Gy$, which is linear and preserves the noisy signal structure but amplifies the noise by the same factor as the signal. The effective SNR degradation is given by $\text{SNR}_{\text{eff}} = \gamma^2/(2\gamma + 1) \approx \gamma/2$ at high SNR: a 3 dB penalty.

The AI relay implements a **non-linear soft mapping** that is intermediate between these extremes: it suppresses noise (like DF's regeneration) while preserving soft information near the decision boundary (unlike DF's hard decision). This explains why learned non-linear relay processing can outperform AF and, under selected channel conditions, also outperform DF at low SNR.

Among the AI methods, the low-SNR advantage over AF is realized by the *minimal* feedforward relays, not by the larger sequence models. On the canonical Rayleigh channel (Table 5.2, 0 dB) the Hybrid, MLP and VAE relays reach 0.3329, 0.3349 and 0.3359 respectively, essentially matching symbol-wise DF's 0.3337 and improving substantially on AF's 0.4076, while the higher-capacity Transformer, Mamba-S6 and Mamba-2 (SSD) relays sit at 0.4014, 0.4071 and 0.4032 – that is, at essentially AF's level, capturing almost none of the available advantage. The advantage over AF is therefore *not* driven by parameter count: the smallest network already captures it, consistent with the complexity ordering discussed under H3 and with the convergence observed in the equal-parameter-budget comparison (H4, Section 4.4).

7.1.2 The Best Evaluated Symbol-Wise Classical Relay at High SNR (H2: Confirmed)

Medium-to-high SNR (≥6 dB): the best evaluated symbol-wise classical relay. At medium and high SNR, DF matches or exceeds all AI methods with exactly zero trainable parameters and zero training time. The theoretical explanation is straightforward: at high SNR, $\sigma^2 \to 0$ and the posterior-mean estimate on each real decision axis tends to the corresponding hard slice. The symbol-wise DF relay therefore implements the limiting *hard symbol-regenerating* rule in this regime, while any neural network approximation introduces a small but non-zero reconstruction error due to the finite precision of the learned mapping and the tanh output saturation (which approaches but never reaches ± 1).

Under the thesis's E_s/N_0 convention, the AWGN limit at 10 dB gives a two-hop per-bit DF BER of $2Q(\sqrt{10})[1 - Q(\sqrt{10})] \approx 1.6 \times 10^{-3}$. Empirically on the canonical Rayleigh channel, DF attains the lowest or tied-lowest BER at every reported point from 6 dB upward. Componentwise slicing is the MAP local symbol decision after coherent equalization, but this does not prove end-to-end BER optimality among all power-constrained relay mappings. The data show only that the learned models match or trail DF here; they do not isolate optimization, architecture, or run-to-run variation as the cause of the sequence-model gap.

7.1.3 Robustness Across the Fading Ensemble

Robustness across fades. The relay ranking on the canonical channel is stable across the evaluated SNR grid, even though each transmitted symbol experiences an independent fading realization. After zero forcing ($\tilde{x} = y/h = yh^*/|h|^2$), the relay processes Gaussian noise at a random conditional SNR. The network is trained across multiple nominal SNRs and evaluated without per-realization retraining or explicit CSI input; the measurements establish performance over the sampled Rayleigh ensemble, not adaptation to arbitrary fading laws.

7.2 Capacity and Overfitting (H3: Not Supported)

One of the most practically useful findings is that relay BER does not improve with model size over the range evaluated: the Minimal MLP architecture (169 parameters) matches the performance of models with 100–140$\times$ more parameters (3K–24K), and no larger evaluated model improved on it under the training protocol used. The stronger reading of H3, that the curve turns upward because excess capacity causes *overfitting*, is not established by the evidence collected here. Each model was trained once, with no seed replication, no train/test-gap measurement and no held-out validation curve, so a size effect cannot be separated from run-to-run variance or from an optimization difficulty that more capacity happens to introduce. The claim carried forward is therefore the weaker, directly measured one: larger evaluated models did not improve BER. What follows in this section is offered as a candidate explanation for that observation, not as a demonstrated mechanism.

This result has important theoretical and practical implications:

7.2.1 Information-Theoretic Perspective

The canonical memoryless relay-denoising task maps a window of $2w + 1$ real-valued noisy observations to a single real-valued clean estimate, applied per axis to the canonical QPSK symbol (equivalently, the real BPSK-over-AWGN form of the same problem). Each real axis carries exactly 1 bit of information, and under the canonical i.i.d. model the sufficient statistic for the current bit is the current observation alone, with Bayes-optimal posterior mean $f^*(y_i) = \tanh(y_i/\sigma^2)$. The neighboring window samples used by the learned relays are therefore redundant architectural inputs in this setting rather than part of the sufficient statistic, so the **effective dimensionality** of the target mapping is still 1: far below the $2w+1 = 5$ or 11 input dimensions.

As a heuristic, the minimum description length (MDL) principle suggests matching model complexity to the effective dimensionality of the target mapping. Since the target here is a one-dimensional monotone function of the current sufficient statistic, even the 169-parameter network has capacity far in excess of what the mapping requires; this is offered as an intuition for why the larger models did not improve on the smallest one, not as a formal MDL bound.

7.2.2 Bias-Variance Analysis

Applying the bias-variance decomposition of Equation 2.7 to the three evaluated sizes:

- **169 parameters:** This model is sufficient to attain the reported BER, but its bias and variance were not estimated separately.
- **3,000 parameters:** Performance is nearly identical to 169 parameters; the measurements do not establish higher predictive variance.
- **24,001 parameters (Mamba-S6):** Despite having 140$\times$ more parameters than the MLP, it does not improve on it; on the canonical channel it is worse at low SNR (Table 5.2). The marginal utility of the additional parameters is at best vanishingly small.

This decomposition is a framework for interpreting the observation, not a measurement of it: bias and variance were not estimated separately, which would require training each configuration repeatedly across seeds and comparing training against held-out error.

7.2.3 Does the Complexity Curve Turn Upward?

The preceding subsections offer bias-variance and MDL readings of H3 but note that the stronger, overfitting reading, BER rising again once capacity exceeds what the mapping requires, was not established, because each model had been trained once and a size effect could not be separated from run-to-run variance. A dedicated size sweep was therefore run to supply the missing evidence. It covers the nine channel families used across Chapters 5 and 6 plus the coded scenario, varying relay window, depth and width independently, with three independent initializations at every configuration and each configuration held to its *worst* initialization, so that a size counts only if it works whichever way it initializes — a necessary precaution given the spread reported below. Degradation is measured as the additional SNR required to reach a given BER, which unlike a relative BER penalty does not divide by a vanishing denominator at high SNR.

Because seed replication is the specific evidence this section previously lacked, the across-initialization spread is treated as a measurement in its own right rather than as a caveat. Table 9.5, in Appendix 9.5, gives per channel, the spread in SNR penalty between the best and worst of the three initializations at the same configuration, summarized over all configurations swept.

Two things follow. First, initialization variance is not a small correction: across all 152 configurations the median spread is 0.180 dB and 57% exceed 0.15 dB, with a worst case of 3.18 dB on the complex ISI channel at eleven parameters. On the nonlinear-bias channel the spread is

1.23 dB at essentially every configuration, which is the two-attractor behaviour noted below rather than a gradual variance. Second, the variance is strongly channel-dependent: the canonical Rayleigh channel is highly reproducible at 0.008 dB median, and the composite and AWGN channels nearly so, while four channels sit above 0.3 dB. Any single-run size comparison on the latter group measures which initialization was drawn, not which architecture is better — which is precisely the confound this section identified in the original H3 evidence, now quantified.

Against that yardstick, **the upward arm is not observed**, and the sweep had the resolution to see it had it been there. Holding the window fixed so that capacity is the only variable, the worst-case penalty against the classical baseline is flat on the memoryless channels and monotonically decreasing on the channels with memory. On the canonical Rayleigh channel at window 5 the penalty moves from $+0.03$ dB at one hidden unit to $+0.06$ dB at twenty-four; on the composite channel at window 7 it improves monotonically from -1.97 dB to -2.23 dB as width grows from four to forty-eight. Where a rising tail appears at all it is at most 0.15 dB, which is below the 0.180 dB median across-initialization spread of Table 9.5 and therefore not separable from the draw. The one clearly non-monotone case, the nonlinear-bias channel, oscillates between two distinct outcomes (-1.63 and -2.87 dB) rather than tracing a curve, which is the signature of training landing in one of two attractors rather than of capacity being harmful.

Two corrections to the reading of H3 follow. First, the weaker claim is now confirmed under seed replication rather than merely unrefuted: additional parameters neither help nor hurt on a memoryless channel over roughly two orders of magnitude of model size. Second, an effect that appears substantial under a relative BER criterion largely dissolves under the SNR-penalty criterion. On AWGN the penalty grows from $+11\%$ to $+28\%$ in relative BER as the window widens, which invites the conclusion that a window is actively harmful; the same measurements correspond to $+0.06$ dB and $+0.11$ dB. The effect is real in sign but negligible in magnitude, and the relative criterion overstates it because the absolute BER gap it divides by falls to 4×10^{-4}.

What the sweep does establish is that the binding resource is not parameter count but **window matched to channel memory**. On the memoryless channels the smallest configuration examined (four parameters, a single hidden unit with no window) is within 0.15 dB of the classical baseline on three of five channels and within 0.03 dB on the canonical channel, some $42\times$ smaller than the 169-parameter relay of Table 5.2. On the three-tap channels the same four-parameter configuration is between 11 and 13 dB adrift, and closing that gap requires widening the window rather than the layer: on the real ISI channel the penalty falls from $+12.95$ dB at window 1 to $+1.67$ dB at window 7 (Section 5.4 gives the depth-sweep result behind this).

The coded scenario behaves differently again and is the one case where size is demonstrably not the operative variable. Against block DF, every configuration from 18 to 1,076 parameters lies between $+1.86$ and $+1.92$ dB, so a sixty-fold range in model size buys 0.06 dB. The relay is not capacity-limited there but interface-limited: it commits to a hard constellation decision per symbol, at 90–94% symbol accuracy, before the outer code is decoded, so the destination decoder receives no reliability information that block DF would have exploited. This identifies a design change rather than a scaling law, and it is taken up in Section 7.5.

Taken together, these measurements support the weaker claim and leave the overfitting hypothesis unsupported — now on the basis of a sweep designed to detect it, rather than for want of evidence.

The framing that replaces it is not that smaller models are better, but that **architectural complexity is only exercised by channel memory**. Capacity is neither good nor bad in itself; whether it can be used at all depends on the channel. On a memoryless channel the sufficient statistic for the current symbol is the current observation, so there is nothing for additional window taps to represent and nothing for additional units to fit: every configuration from four parameters to several hundred lands within a fifth of a decibel of the same value, and the extra capacity is not harmful so much as idle. Introduce three taps of intersymbol interference and the same capacity becomes load-bearing — the relay must now represent a function of several consecutive samples, a window-1 network cannot express it at any width, and widening the window is worth between 8 and 11 dB. The size axis looks flat in the first case and decisive in the second because the channel, not the architecture, determines whether the parameters have work to do.

This reading accounts for both halves of the sweep, which "less is more" does not: a principle that smaller is better would predict the memoryless behaviour and mispredict the ISI channels, where the minimum rises from four parameters to between 73 and 145 and the wider network is unambiguously the better one. It also explains the shape of the failure at the two extremes. Where the window is too narrow for the channel's memory the relay fails structurally rather than gradually, as on the unknown-phase channel of Section 6.2.1, whose information symbol is a product of two consecutive samples that a one-tap network cannot represent at any width. Where the window is far wider than the memory, as at window 31 on the coded channel, the additional inputs carry no information about the current symbol and the optimization becomes harder rather than the model more expressive. The design rule is therefore to match the window to the channel's impulse response first and to treat width as a secondary knob, which is a statement about the channel as much as about the network.

7.2.4 Practical Implications

1. **Occam's Razor for relay AI.** On the canonical memoryless task, adding width or depth beyond an adequate small model bought no resolved accuracy over the evaluated range. Channels with memory are different: they require a wider observation window, so the statement does not extend below the context required by the channel.

2. **Deployment footprint.** A 169-parameter model occupies approximately 0.7 KB as float32 and trains in seconds on a CPU. A footprint this small removes storage as an obstacle on constrained relay nodes, and is small enough to sit in the on-chip SRAM of a typical microcontroller, so an embedded deployment is a plausible implication of these results. It

is not demonstrated by them. Establishing it would require what this thesis does not measure: fixed-point or quantized accuracy, sustained throughput, memory traffic, energy per symbol, and an actual hardware implementation. Parameter count alone bounds none of these. The footprint is of the same order for the window-11 relay that Chapter 6 holds fixed out to $L = 7$ (170 parameters, Table 6.9), so the claim does not depend on staying at the canonical channel's window width.

3. **Generalization.** The 169-parameter MLP handles the full Rayleigh fading ensemble, every effective SNR the fades produce, with a single set of weights and no per-condition retraining, an instance of generalization that would be harder to achieve with a larger, more specialized model.

7.3 Practical Deployment Recommendations

Based on the comprehensive evaluation, we propose the following deployment strategy:

Table 7.1: Practical deployment recommendations: recommended relay strategy by operating regime

Operating Regime	Recommended Relay	Rationale
Low SNR (0-4 dB)	MLP Minimal / Hybrid / channel-specific selection	Neural relays often help; exact best choice is channel-dependent, with DF remaining strong on some fading scenarios
Medium SNR (4-8 dB)	Hybrid (MLP + DF)	Automatic switching at empirically chosen crossover
High SNR (>8 dB)	DF	Zero parameters; lowest BER among the evaluated per-symbol relays
Resource-constrained	MLP Minimal (169 params)	0.7 KB memory, <5 s training
Best overall (BPSK/QPSK)	Hybrid	Combines AI advantage with DF reliability

Within the simulated canonical benchmark, the Hybrid relay is a candidate when operating SNR can be estimated reliably, because it switches between MLP processing and DF. This is not a general deployment recommendation: threshold robustness, quantization, energy, synchronization, and hardware behavior were not evaluated.

7.3.1 Latency and Adaptivity Considerations

Two practical questions bear on whether a learned relay can be deployed with minimal impact: the latency it adds relative to AF/DF, and its ability to adapt to varying channel conditions without retraining.

Latency. Two distinct latency components must be separated. The first is *buffering* latency, incurred by the sliding window $y_R[i-w:i+w]$. In the half-duplex protocol studied here this component is absorbed by the protocol itself: the relay buffers the entire listening-phase block before the transmission phase begins (Remark 4.1), so the window's w-symbol look-ahead adds no delay beyond what store-and-forward operation already imposes. Only in a streaming or ultra-low-latency protocol, outside the scope of this thesis, would the symmetric window cost w symbols of look-ahead that the memoryless AF/DF baselines ($w = 0$) do not pay. The second component is *compute* latency: inference cost per relayed symbol. Here the minimal-capacity result (H3) is decisive: since the 169-parameter MLP matches the performance of models $100-140\times$ larger, the deployed network reduces to two small matrix–vector products per symbol, a per-symbol cost of a few hundred multiply-accumulates, small in absolute terms even if it cannot literally match the single comparison of DF's $\mathrm{sign}(y)$. The compute-latency argument therefore holds *because of*, not despite, the less-is-more finding; it would not hold for the 24K-parameter sequence models, whose recurrent or attention-based structure imposes materially higher per-symbol cost for no BER benefit at the relay window length.

The coded study of Section 5.5.2 adds a third component that the canonical comparison never had to consider: *buffering*. A block relay cannot emit until it has decoded a whole frame (202 symbols there), and the soft BCJR variant cannot even in principle, its backward recursion running from the end of the frame. A windowed learned relay with an 11-sample symmetric window needs five future symbols, and that figure is constant in frame length where the block relays' grows with it.

Section 6.5.1 measures these components together on a channel with memory, and two of the arguments above do not survive it. Treating buffering as the learned relay's advantage presumes that sequence detection is latency-expensive. In the finite-budget experiment, no BER difference was resolved among the tested MLSE traceback depths, including three symbols, which is fewer than the window-11 relay's five; so a latency budget tight enough to exclude MLSE excludes the learned relay first. Latency separates the block relays from everything else, and nothing else from anything. The appeal to wall-clock measurement fares no better — that the measured advantage over Viterbi spans a factor of 50 to 63 *depending on the machine* (Table 5.10) is itself the objection — and counting multiply-accumulates instead turns the argument into a threshold at $L^\star = 3.35$ taps (Equation 6.3), below which the learned relay is worse on cost, delay and accuracy at once. The measurement and its two tables are given in Section 6.5.1 and not repeated here; what matters for deployment is that the learned relay's cost advantage is real but conditional, and the condition is a property of the channel rather than of the processor it runs on.

Adaptivity. A single network, trained across the range of simulated SNRs, handles the entire canonical fading ensemble without retraining or explicit CSI input (Section 7.1.3): after zero-forcing equalization, each fading realization presents the relay with an AWGN-like observation whose conditional noise variance is $\sigma^2/|h|^2$, i.e. a random effective SNR, and multi-SNR training already covers this mixture. Two honest qualifications apply. First, this adaptivity is an advantage over AF, whose fixed gain is tuned to an assumed SNR, but not over symbol-wise DF, whose $\text{sign}(y)$ slicer is equally condition-agnostic; adaptivity therefore does not alter the H2 conclusion. Second, adaptation has been demonstrated only within the canonical Rayleigh ensemble; generalization to other channel families (Rician, bursty interference, phase noise, hardware non-linearities) remains untested and is identified as future work.

7.4 Limitations

Several limitations of this study should be acknowledged, as they define the boundary conditions under which the conclusions hold:

1. **Modulation scope.** The canonical relay comparisons use QPSK, and the unknown-channel study (Chapter 6) mainly uses BPSK, with a QPSK re-check of the unknown-ISI result (Section 6.2.2); 16-QAM enters only as a modulation-and-coding rung in the link-adaptation study, never as a relay-architecture question. In that sense higher-order constellations are not investigated: 16-QAM and denser grids break the per-axis relay formulation and require a joint N-class 2D relay whose output dimension scales with M, which is left to future work.
2. **Perfect CSI (canonical comparison).** The canonical comparison assumes perfect channel state information at each receiver; the finite-pilot and blind regimes of Chapter 6 relax this deliberately and are not limitations of that study. Within the canonical comparison, estimation error would affect the coherent compensation step and hence relay performance, introducing a model mismatch this thesis does not evaluate there; whether it would help or hurt the AI relays relative to AF/DF is untested.
3. **Static offline training.** AI relays are trained once on synthetic data and deployed without adaptation. While multi-SNR training covers the canonical fading ensemble (Section 7.1.3), and the unknown-channel study (H5) shows a fixed network can learn a *stationary* unknown impairment, this protocol does not exploit online information; adaptive or continual training would be required if the deployment channel drifts over time.
4. **Single relay.** The framework considers a single relay node; multi-relay cooperation introduces relay selection, power allocation and cooperative-diversity dimensions not captured here, where system-level optimization (which relay, how to combine observations) becomes as important as the per-relay function studied in this thesis.
5. **Two-hop only.** Extension to 3+-hop networks introduces noise accumulation (AF) and

error propagation (DF) that grow with hop count, a regime this thesis does not measure.

6. **Limited window and memory sweep.** Window sizes were swept for selected channel families and channel lengths through seven taps. The study does not establish the required window, accuracy, or cost for longer memories or structurally different temporal channels.

7. **No coding, in the core comparison.** The core comparison operates without error-correcting codes, so the DF baseline there is its symbol-level variant (Remark 4.2). Sections 5.5–5.5.4 measure a finite-length coded block-DF baseline, its soft-decision counterparts, a rate-adaptive envelope over a modulation-and-coding grid, and that envelope re-derived under an idealized buffer-readiness budget, directly rather than leaving any of this as an assertion, but that study is confined to one code family, one frame length, and one channel model, and the relay and destination each decide once with no iterative exchange between them.

8. **Seed replication is complete on both axes; two narrower qualifications remain.** The original cross-architecture canonical comparison used one trained instance per architecture, and the architectures did not all receive identical training budgets (e.g. the cGAN trains for 200 epochs against 100 for the other relays, Section 4.2), a confound the normalized-parameter study (Section 4.4) does not remove, since it equalizes parameter count and input window but not epoch count. Where a result rests on a single trained instance, the confidence intervals accompanying it are Monte Carlo intervals over the *evaluation* channel and bit realizations for that fixed model, not over the training procedure. Both axes have since been replicated. The MLP size axis was re-run across three independent initializations per configuration (Section 5.4), and the resulting across-initialization spread is reported in Table 9.5: median 0.180 dB over 152 configurations, but strongly channel-dependent, from 0.008 dB on the canonical channel to 1.231 dB on the nonlinear-bias channel. That spread is itself the reason the single-run caveat mattered, since on four of the nine channels it exceeds the entire effect of varying model size. The *cross-architecture* comparison, which carries the claim that architecture is secondary to capacity (Section 5.3), was replicated separately. Every architecture in the canonical comparison was retrained from three seeded initializations on an *equalized* budget of 100 epochs at $25{,}000$ samples — equalized because the published run gives the MLP and Hybrid $25{,}000$ samples and the VAE and sequence models $50{,}000$, confounding architecture with training budget — and evaluated on identical channel draws and payload bits (`seed_spread_native_architectures.py`). The across-seed spread in SNR penalty against DF is 0.004 dB for Mamba-S6, 0.007 for the Hybrid, 0.013 for Mamba-2, 0.043 for the MLP, 0.075 for the Transformer and 0.091 for the VAE. All six sit inside the ≈ 0.18 dB that the high-SNR architecture spread corresponds to, so initialization variance is not large enough to have produced the reported ordering by chance. Two qualifications survive the measurement. The Transformer instability of Table 9.6 did not recur, but a failure mode occurring in roughly one initialization in eight will not always appear in

three draws, and three draws cannot bound its rate. And the VAE's per-seed range overlaps the MLP's, so between those two a single draw could have reported either ordering; the published confidence intervals for that pair already overlap at every SNR point, so no claim in this thesis rests on separating them.

7.5 Future Work

Several directions warrant further investigation:

1. **Time-selective fading.** This study addresses channels static within a block. Extending to time-selective fading with continuously varying unknown phase (e.g. Jakes/Clarke Doppler) is a natural next step, likely requiring phase-aware architectures (complex-valued state, learned phase unwrapping, or a noncoherent-detection training objective) evaluated against prediction-based noncoherent receivers.
2. **Higher-order constellations (16-QAM and denser).** Constellations with more than two levels per axis break the per-axis relay formulation and need a joint N-class 2D relay whose output dimension grows with M; that reformulation, and whether the complexity ordering discussed under H3 shifts with constellation density, are open questions this thesis does not address.
3. **Imperfect CSI.** Introduce channel estimation errors to assess robustness of AI relay processing under realistic conditions.
4. **A BER-optimal (BCJR/APP) benchmark for the unknown-ISI channel.** Section 6.2 benchmarks against genie-CSI Viterbi MLSE, the optimal *sequence* detector. Sequence-ML and bit-BER-optimal detection are distinct criteria on any channel with memory, at both modulation orders studied here, so the benchmark is not BER-optimal in either case; for Gray-coded QPSK there is the additional effect that a wrong trellis path can differ from the truth by one or two bits with unequal probability. That second effect has since been measured (Section 6.2.2) and is in fact exactly zero here: the taps are real and the fading gain is a real magnitude, so the channel separates into independent in-phase and quadrature ISI channels, the symbol posterior factorises, and bit-wise MAP and symbol-MAP coincide (the measured 1.073 against 1.090 bits per symbol error is the same statement empirically). A BCJR/APP detector, which directly minimizes per-bit error probability rather than sequence error probability, is measured over the same 3-tap trellis (Section 6.2.3). It is never worse than MLSE, as optimality requires, but the advantage shrinks with SNR and is no longer resolvable at 16 dB and above; at 20 dB the benchmark moves from 0.000127 to 0.000126 against the learned relay's 0.0508. The classical margin therefore does not change when the comparator is made BER-optimal, at QPSK. It is unmeasured at BPSK, where the same I/Q factorisation argument is expected to hold but has not been separately measured.

5. **Online learning.** Develop relay strategies that adapt their parameters during operation, tracking time-varying channel conditions.

6. **Multi-relay networks.** Extend to cooperative relay selection and multi-hop routing with AI-optimized relays at each node.

7. **Other channels and MIMO.** Evaluate the canonical conclusions on additional channel families (Rician line-of-sight fading, and AWGN treated as a full operating point rather than the bounded calibration role it has here) and on a multi-antenna second hop, where destination equalization interacts with relay processing and the additivity of the two gains becomes a testable question.

8. **End-to-end learning.** Train the entire communication chain (modulation, relay, demodulation) jointly using autoencoder-based approaches, and compare against the modular classical-plus-learned-relay cascade studied here.

9. **Coded block-DF baseline (measured; open items remain).** Remark 4.2 scopes the core comparison's DF baseline to the uncoded, symbol-level variant. Section 5.5 measures the coded, practical finite-length block-DF baseline, together with the soft-information relays (BCJR at the relay, posterior-mean read-out of the learned relay) and an adaptive-rate envelope over a modulation-and-coding grid. The substantive finding is mechanistic: a relay that decodes the code has spent its redundancy before transmitting, so a wrong decode arrives as a valid-but-wrong codeword the destination cannot repair, whereas a relay that only denoises leaves the redundancy intact. Rate selection then dominates the relay-decoding question by an order of magnitude, unless a latency deadline is imposed, at which point block-DF's double buffering reopens the gap. What remains open: one code family, one frame length, one channel model; the soft block-DF relay was given the nominal noise variance rather than per-symbol CSI, which caused its 20 dB mis-calibration; iterative (turbo-style) relay–destination exchange was not evaluated; and the latency analysis uses structural buffer counts rather than a 10^{-5}-class reliability target, so it does not establish ultra-reliable low-latency performance.

10. **Longer relay windows with state-space architectures.** Future work should explore whether longer relay windows (e.g., 128–512 symbols) combined with a chunk-parallel state-space architecture can improve both BER performance and processing speed simultaneously; the training-time crossover between sequential and chunk-parallel formulations, removed from this thesis's scope, is the natural companion measurement.

7.6 Hypotheses Revisited

Chapter 8 states what this thesis found, and Table 8.1 there records the verdict on each hypothesis. This section addresses the different question of *how* each hypothesis fared under measurement, and why three of the five required qualification.

Do the other hypotheses survive the size sweep? The sweep of Section 5.4 re-measures four of the five hypotheses' subject matter with a different relay class, a corrected two-hop model, per-channel classical comparators and three initializations per configuration, so it is worth asking explicitly whether any of them moves. Only H3 does.

H1 is reconfirmed on the canonical channel: the smallest matching relay reaches 0.3294 at 0 dB and 0.2198 at 4 dB against AF's 0.4088 and 0.3140, so the low-SNR advantage over AF is not an artifact of the particular architectures originally compared. H2 is likewise reconfirmed and, for the first time, quantified in the units that matter: on the canonical channel the learned relay trails symbol-wise DF by 0.05 to 0.09 dB, which is a real margin in DF's favour but two orders of magnitude smaller than the "$+2.06\%$ relative BER" figure that the same measurements produce at 20 dB. H2's direction stands; its magnitude was previously overstated by the choice of metric.

H5 receives a qualitatively consistent check. Chapter 6 reports the learned relay 1–1.5 dB behind genie-CSI Viterbi on the three-tap ISI channel with a window of 11. A separate sweep, using a different relay class, training procedure, two-hop model, and Monte Carlo budget, reports penalties decreasing as its window widens through seven samples. Because several factors differ, that trend cannot attribute the residual gap to window width alone or support extrapolation beyond the measured windows.

H4 needs qualifying, and the qualification could not be inferred from the size sweep. That sweep measures initialization variance on MLPs, where the canonical channel is the most reproducible case anywhere in this thesis at 0.008 dB median. That reassures only for MLPs, the sequence models never having been measured; measuring them directly reverses the conclusion for one of the six.

Each equal-budget architecture was retrained from three seeded initializations and evaluated on identical channel draws and payload bits. The resulting spread in SNR penalty against DF is tabulated as Table 9.6 in Appendix 9.5.

Four of the five are stable to within 0.16 dB and the two Mamba variants are the most reproducible architectures measured, at 0.019 and 0.041 dB. The Transformer is not: its three initializations span 0.906 dB, five times the effect H4's high-SNR claim must resolve, with one run at $+2.039$ dB against two at approximately $+1.14$ dB. The instability is therefore specific to that architecture rather than generic to sequence models, which is the opposite of what a capacity or inductive-bias account would predict.

What the Transformer's instability is. Running eight initializations rather than three, and recording the training objective alongside the BER, identifies the cause as an outright optimization failure rather than a property of the architecture's solution. Figures 9.3 and 9.4 show the eight runs and the loss that separates them. Seven of the eight seeds converge to a final training loss between 0.0018 and 0.0023 and land within 0.033 dB of one another, which is tighter than the Mamba control's 0.041 dB. The eighth converges to 0.0256, eleven times the median, and is the $+2.039$ dB run. Final training loss and SNR penalty correlate at $+0.998$ across the eight runs, but that figure should not be read as a general relationship between loss and error rate: it is produced entirely by the single outlier's leverage. Among the seven runs that converge, loss varies by roughly 30% with no detectable effect on BER ($r = -0.38$, $p = 0.41$). The correct reading is a two-cluster one — the failed run is worse on both measures simultaneously, which is what identifies it as a failure of optimization rather than an alternative solution reached by a different route, while within the converged cluster the training loss carries no information about which relay will perform better. Training loss is therefore a reliable detector of catastrophic non-convergence and useless as a fine-grained quality signal, and nothing here suggests the objective is mismatched to the error rate the relay is judged on.

Three practical points follow. One failure was observed among eight Transformer initializations at this budget. That sample is too small to estimate a failure probability or the chance that a three-seed study would contain one. It is trivially detectable at training time, the bad run's loss being an order of magnitude off, so seed selection on final training loss would remove the instability entirely at no evaluation cost. And it is *not* detectable from validation accuracy, which reads 0.9971 for the failed seed and spans 0.9971 to 0.9992 across the healthy ones: the metric the training loop already monitors does not separate the failure, which is presumably why it was never noticed. Excluding the single failed run, the Transformer is as reproducible as any other architecture measured, so the correct statement is not that the Transformer is an unstable architecture but that it fails to converge occasionally and that the failure is silent under the diagnostics in use.

Two consequences follow for H4. H4's low-SNR finding stands: the feedforward and sequence groups separate by roughly a decibel here ($+0.15$ against $+1.1$ dB), far outside any measured spread, so the 17.4% split at 0 dB is not a seeding artifact. H4's high-SNR convergence claim does not stand as a seed-robust result, because the 4.1% spread it rests on is smaller than the Transformer's own run-to-run variation; it should be read as an observation from one run of each architecture. Notably, this was not detectable from the published experiments: none of the three sequence checkpoints seeds `torch` at all, so the reported comparison was drawn from an uncontrolled initialization and its variance was not merely unreported but unmeasurable after the fact.

The unknown-channel contribution (Chapter 6) settles the complementary question of *when the learned relay is not merely competitive but necessary*. Its flat-channel control isolates the cause: against unknown phase, gain and per-branch asymmetry, all memoryless, the classical relay does not fail and the learned relay only matches it (largest difference 0.0012 over the tabulated SNRs). Parameter uncertainty alone is therefore not what learning mitigates; failure appears only with unmodeled *memory*. The chapter's remaining findings are established there and enumerated in Chapter 8.

Taken together these results place neural relay processing as a complement to classical methods rather than a replacement for them, and Chapter 8 states the resulting deployment recommendation and the two boundaries that bound it. The interpretive point this chapter adds is the one the individual findings do not state on their own: **model complexity should be matched to task complexity, and the task's complexity is set by the channel rather than by the constellation or the budget**. Where the channel is memoryless the relay denoising task is small enough that minimal architectures suffice and the choice between feedforward and sequential paradigms matters less than sizing; where the channel has memory, the same reasoning raises the floor rather than lowering it, and does so through the relay's window rather than through its parameter count.

Chapter 8

Summary

This work examined classical and neural network–based relay strategies for *two-hop relay communication* using a unified simulation framework. The objective was to analyze performance–complexity tradeoffs and determine when learning-based relays provide practical advantages over analytically derived methods. The core comparison is carried out on one canonical setup, fixed in advance and never departed from within it: a two-hop SISO link over i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER, with the relay function as the only varied axis. The coded study and the unknown-channel contribution depart from it deliberately. The study showed the following:

1. At the evaluated low-SNR points, the best learned relays outperform amplify-and-forward (AF), where their learned soft denoising avoids AF's noise amplification (H1).
2. Classical *decode-and-forward (DF)* matched or outperformed all neural approaches at **medium-to-high SNR** while requiring no trainable parameters, confirming that DF is the best-performing evaluated symbol-wise classical relay in this regime (H2).
3. Increasing model size did not produce a **systematic BER improvement**, and the replicated size sweep found no statistically meaningful upward BER trend attributable to model capacity (H3).
4. A **minimal neural relay** (169 parameters) achieved performance comparable to substantially larger models; no larger evaluated model improved on it. The replicated size sweep provides no evidence that overfitting explains this plateau, and the single-run protocol used elsewhere (one training run per model, no train/test-gap measurement) could not have established it either way.
5. A **size sweep across every channel studied**, replicated over three initializations, located the floor and showed it is set by *channel memory* rather than parameter count: four parameters suffice on a memoryless channel, $42\times$ below the 169-parameter relay, while three-tap channels need 73 to 145 and a window spanning the interference. No upturn in error with capacity was observed, so the overfitting hypothesis H3 is unsupported on

evidence rather than untested (Sections 5.4 and 7.2.3).

6. At an equal parameter budget, feedforward and sequence relays remained separated at low SNR. The apparent high-SNR convergence was smaller than the Transformer run-to-run spread and is not seed-robust; equalizing parameters also left depth, width, state dimension, window, and optimization confounded (H4).
7. Within the simulated benchmark, a `hybrid relay` combining neural processing at low SNR with DF at high SNR tracked the best evaluated relay when operating SNR was known. Its switching-threshold robustness, quantization, energy use, synchronization, and hardware behavior were not evaluated.

The first seven findings concern the canonical memoryless relay model; the remaining findings summarize the deliberate model-mismatch extension of Chapter 6. The supplementary coded study's findings, being a smaller third part of the thesis, are summarized in prose rather than as numbered items, at the end of this chapter.

8. Beyond the canonical memoryless model, AF and symbol-wise DF exhibited the derived 0.25 floor on the evaluated three-tap channel, while a 170-parameter relay trained on the impairment family restored a decreasing BER curve. It remained approximately 1–1.5 dB behind matched MLSE on pure ISI. With no per-block CSI, it narrowly led the evaluated CMA implementation without online adaptation and avoided the non-monotonic behavior observed for decision-directed blind MLSE; CMA itself was stable (H5).
9. Sweeping the intermediate *partial*-posterior case at 10 dB located a crossover between 20 and 10 pilots; its SNR dependence was not measured. Pilot-aided LS-plus-MLSE wins when its estimate is sufficiently accurate, while the family-trained relay avoids pilot overhead and per-block estimation. The result is specific to the tested estimator, channel family, block lengths, and SNR.
10. A **cost accounting on processor-independent axes** completed the trade-off and bounded it. Measured in multiply-accumulates per symbol rather than wall-clock, and under the MAC-counting convention adopted here, the implemented full-state Viterbi MLSE requires $2M^L$ MAC-equivalents against a learned relay's $O(LH)$, linear in the memory against the trellis's exponential $O(M^L)$, and $2WH + 4H$ exactly when the window is pinned rather than widened with the channel — preserving coverage at longer memory generally means widening it, and the cost then grows linearly in the window rather than not at all. For the architecture deployed here the two are equal at $L^\star = 3.35$ taps, an arithmetic crossover at fixed (W, H, M) rather than a universal threshold. Below that threshold the learned relay has no cost case — at three taps MLSE is the cheaper detector (128 MACs against 208), decides sooner, and is more accurate — so the three-tap channel used throughout the unknown-channel study establishes the failure of the *memoryless* relays and not a cost argument against sequence detection. Above it the gap compounds as M^L, reaching $158\times$ at seven taps, and there the learned relay trades a BER gap of roughly

five to eight (flat in L, not widening with it) for arithmetic that does not grow with the channel (Section 6.5.1).

11. **Latency did not separate the learned relay from sequence detection at the available resolution** on the tested channel. No BER difference was resolved among the tested traceback depths; a three-symbol MLSE setting had fewer symbols of decision delay than the window-11 relay's five. The block relay required a full frame before producing output and showed no resolved accuracy gain in this experiment. The two research objectives of Chapter 3 therefore have boundaries set by different resources in the comparisons evaluated here — channel-estimate reliability for the unconstrained question, arithmetic for the constrained one — and the distance between them is a result rather than a restatement.

Table 8.1: Research hypotheses, their statements, and experimental outcomes

Hypothesis	Statement	Result
H1	Some AI relays outperform AF at low SNR	**Confirmed**: the best neural relays beat AF at 0–4 dB on the canonical channel
H2	DF lowest-BER among per-symbol relays at high SNR	**Confirmed**: DF matches or beats all AI methods at ≥ 6 dB with 0 parameters
H3	Overfitting at excess capacity	**Not supported; weaker size claim holds**: 169 params matches 24K, and larger evaluated models did not improve BER. The size sweep of Section 5.4, replicated over three initializations, finds no upturn above the 0.180 dB across-initialization spread, so the overfitting hypothesis is unsupported on evidence rather than untested. The overfitting *mechanism* remains unestablished, no train/test-gap analysis having been performed

Hypothesis	Statement	Result
H4	Equal parameter count eliminates architecture-dependent differences	**Rejected at low SNR; high-SNR convergence not seed-robust** (Table 9.6): at 3K params the six architectures span 4.1% BER at 20 dB but 17.4% at 0 dB, splitting into feedforward and sequence groups. Equalizing the budget required changing width, depth, state dimension and input window jointly (Table 4.2), so this equalizes capacity, not the other architectural degrees of freedom
H5	Memoryless classical relays fail on unknown channels; a minimal MLP mitigates by learning, near the model-aware optimum	**Confirmed**: on unknown ISI, AF/DF hit the analytic 0.25 floor (DF non-monotonic in SNR) while the 170-parameter MLP reaches BER 4.79×10^{-8} by 16 dB, within $\approx$1–1.5 dB of genie-CSI Viterbi MLSE; the same architecture mitigates a composite ISI$\times$PA$\times$phase cascade with no impairment model, tracking $\approx$2 dB behind pilot-aided Viterbi and matching blind CMA when no posterior exists; the advantage is bounded to structural model-class mismatch representable at minimal capacity

Overall the evidence supports something narrower than a learned-versus-classical verdict. A compact learned relay supplies useful amortized processing where the modelling or estimation requirements of the particular classical receiver it is measured against are not met, and on channels with memory it offers a different complexity–accuracy tradeoff rather than a uniformly better one. Nothing measured here shows it superior to classical receiver design as a whole; the comparators are the specific pipelines implemented in this thesis. Within that scope, learned relays are best used as targeted enhancements rather than universal replacements, with simplicity, parameter efficiency, and hybrid design as the practical principles that follow. The canonical conclusions hold for the *uncoded*, symbol-level setting studied here (Remark 4.2); a supplementary coded study (Section 5.5) measures where coding changes that standing: a rate-1/2 convolutional code with a genuine block-DF relay beats uncoded DF from 8 dB upward, but by 20 dB both learned relays overtake block-DF (one architecture already does at 16 dB, the other still trails there), since a relay that decodes the code spends its redundancy before forwarding while a denoising relay leaves that redundancy intact for the destination to use. That advantage is bounded, not erased, once the code rate is itself allowed to adapt: maximizing goodput over a modulation-and-coding grid shows rate selection dominates the relay-decoding question by an order of magnitude, unless a latency deadline reopens the gap by forcing the block-DF relay's double buffering out of budget first. Extending the study to other channel families, multi-antenna configurations, and higher-order constellations as a relay-architecture question (16-QAM and denser, which need a joint N-class 2D relay) is identified in Chapter 7 as the principal direction for future work.

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Chapter 9

Appendices

9.1 Appendix A: Mathematical Notation

Symbol	Description
b	Binary bit ($\in \{0, 1\}$)
x	Modulated BPSK symbol ($\in \{-1, +1\}$)
y	Received signal
n	Noise sample
h	Fading coefficient
σ^2	Noise variance
SNR	Signal-to-Noise Ratio (linear)
G	AF amplification gain
$\mathbf{W}$	Neural network weight matrix
$\mathbf{b}$	Bias vector
P_e	Bit error rate
$Q(\cdot)$	Gaussian Q-function
β	VAE KL weight
λ	Regularization parameter
η	Learning rate
Δ	State space discretization step
$\mathbf{A}, \mathbf{B}, \mathbf{C}, D$	State space model matrices

9.2 Appendix B: Model Architectures and Hyperparameters

Relay	Network Topology	Train Loss (LR)	Total Params	Implementation
MLP (Minimal)	Feedforward MLP, 5-symbol sliding window	MSE, lr=0.01, 100 epochs	169	NumPy (CPU)
Hybrid	MLP + DF switching via learned SNR threshold ($\sim$5 dB)	MSE (MLP only), lr=0.01	169	NumPy (CPU)
VAE	Encoder–decoder latent model (7$\rightarrow$8$\rightarrow$1)	β-VAE ($\beta = 0.1$), Adam lr=10^{-3}	1,777	PyTorch (CUDA)
cGAN (WGAN-GP)	Conditional GAN (generator + critic)	WGAN-GP, Adam lr=10^{-4}	2,946	PyTorch (CUDA)
Transformer	2-layer encoder, 4 heads, $d_{\text{model}} = 32$	MSE, Adam lr=10^{-3}	17,697	PyTorch (CUDA/CPU)
Mamba-S6	Selective SSM, 2 S6 layers	MSE, Adam lr=10^{-3}	24,001	PyTorch (CUDA/CPU)
Mamba-2 SSD	Selective SSM with SSD kernel (2 layers)	MSE, Adam lr=10^{-3}, clip=1.0	26,179	PyTorch (CUDA)

Table 9.2: Comparison of relay models, training objectives, parameter counts, and implementations

9.3 Appendix C: Reproducibility

9.3.1 Testing

The simulation framework carries 187 automated pytest tests covering channels, modulation, relay strategies, simulation, and statistics. In the recorded environment, the full suite passed; with PyTorch unavailable, 183 passed and four checkpoint-dependent tests were skipped.

9.3.2 Reproducibility

Evaluation seeds are controlled for source-bit and noise generation. Training initialization was not consistently seeded for every original sequence-model checkpoint, so exact retraining of those models is not guaranteed; the later replicated sweeps report training-seed variation explicitly.

Provenance of the reported results. Each reported result is associated with an experiment artifact and a verification script in the source-code repository, using the experiment-specific budgets documented in the relevant section and ledger. Independent audit requires that repository at the archived revision identifier; the ledger below documents the claimed provenance but cannot by itself reproduce every table cell.

9.4 Appendix D: Master Experiment Ledger

In the source-code repository, each numerical table and figure is associated with a result file and checked by `verify_thesis_tables.py` (Section 4.3.4). This ledger lists the claimed modulation, channel, and result source for each major study, grouped in the order they appear. Result-source filenames matching `e6_*.npy` are relative to `e6_unknown_channel_results/`.

Table 9.3: Master experiment ledger: modulation, channel, and result-file provenance for every major study

Study	Modulation	Channel	Tbl/Fig	Result source
Canonical benchmark	QPSK	Rayleigh	tbl:2, fig:10	`results/modulation/qpsk_rayleigh.json`
Normalized 3K-param. comparison	QPSK	Rayleigh	tbl:8	`results/normalized_3k/3k_rayleigh.json`
Coded block-DF, K-sweep	QPSK	Rayleigh	tbl:34, 35	`results/coded_df_experiment.json`
Coded high-SNR paired re-test	QPSK	Rayleigh	tbl:37	`results/coded_high_budget_test.json`
Coded relay-error mechanism	QPSK	Rayleigh	tbl:38	`results/coded_error_mechanism.json`
Coded soft-decision relaying	QPSK	Rayleigh	tbl:39	`results/coded_soft_decision.json`
Equal-throughput & latency	QPSK, 16-QAM	Rayleigh	tbl:40, 41	`results/coded_latency_throughput.json`
Adaptive modulation/coding (AMC)	QPSK, 16-QAM	Rayleigh	tbl:42	`results/coded_rate_adaptation.json`
Capacity under a latency budget	QPSK, 16-QAM	Rayleigh	tbl:43	`results/coded_latency_capacity.json`
Unknown ISI (layer 2)	BPSK	AWGN (2nd hop)	tbl:E6	`e6_sim_ported_results.npy` + `e6_viterbi_*.npy`
Flat-channel control	BPSK	Rayleigh	tbl:E6flat	`e6_flat_ported_results.npy`
Unknown ISI, QPSK generalization	QPSK	AWGN	tbl:E6qpsk	`e6_qpsk_unknown_channel_results.npy`
Composite impairment cascade	DBPSK	Rayleigh	E6composite	`e6_composite_ported_results.npy`
Partial posterior (pilot sweep)	BPSK	Rayleigh	E6partial	`e6_partial_ported_results.npy`
Blind channel (layer 4)	BPSK	Rayleigh	E6blind	`e6_blind_ported_results.npy`
Four-layer summary	mixed (above)	mixed (above)	tbl:layers	combination of the above

The result files above are themselves generated by named, committed scripts (e.g. `coded_rate_adaptation.py` produces `coded_rate_adaptation.json`); the correspondence between script and output file is one-to-one and is not separately tabulated here since the filenames make it explicit.

9.5 Appendix E: Minimum Relay Size and Initialization Variance

Supporting material for the minimum-size study of Section 5.4 and the initialization-variance measurements discussed in Section 7.2. The findings themselves are stated in those sections; what follows is the underlying tabulation and the figures, collected here to keep the experiment and discussion chapters to their conclusions.

Table 9.4: Smallest relay meeting a given degradation budget against each scenario's own published relay (MLP-169, or MLP-756 for the coded row), written as parameters and window. Most memoryless rows settle at window 1, but flat gain and nonlinear bias require wider windows at the strictest budget; every evaluated three-tap row requires window 5 or 7 until a full decibel is conceded. A dash indicates that no smaller configuration meets that budget

Channel	Taps	$\leq$0 dB	$\leq$0.1 dB	$\leq$0.25 dB	$\leq$0.5 dB	$\leq$1 dB	$\leq$2 dB
awgn	1	4p w1	4p w1	4p w1	4p w1	4p w1	4p w1
rayleigh	1	4p w1	4p w1	4p w1	4p w1	4p w1	4p w1
flat gain	1	21p w3	21p w3	21p w3	4p w1	4p w1	4p w1
branch asym.	1	4p w1	4p w1	4p w1	4p w1	4p w1	4p w1
nonlinear bias	1	29p w5	4p w1	4p w1	4p w1	4p w1	4p w1
ISI	3	73p w7	73p w7	57p w5	57p w5	37p w7	21p w3
ISI (complex)	3	73p w7	73p w7	73p w7	73p w7	57p w5	21p w3
ISI + Rayleigh	3	145p w7	73p w7	73p w7	73p w7	37p w7	21p w3
composite	3	145p w7	73p w7	57p w5	21p w3	21p w3	6p w3
coded	code	372p w9	244p w5	244p w5	18p w1	18p w1	18p w1

9.5.1 Results Figures

The Transformer row of Table 9.6 is a single failing initialization rather than a broadly noisy architecture, and the two figures below are the evidence for that reading. Figure 9.3 shows all eight runs: seven are indistinguishable at this scale and one separates progressively as SNR rises, which is why a three-seed sample had roughly a one-in-three chance of missing it entirely. Figure 9.4 shows why the failure is detectable at training time without reference to BER, the failing run's final training loss is eleven times the median, and, equally, why training loss is not a quality signal in the ordinary case.

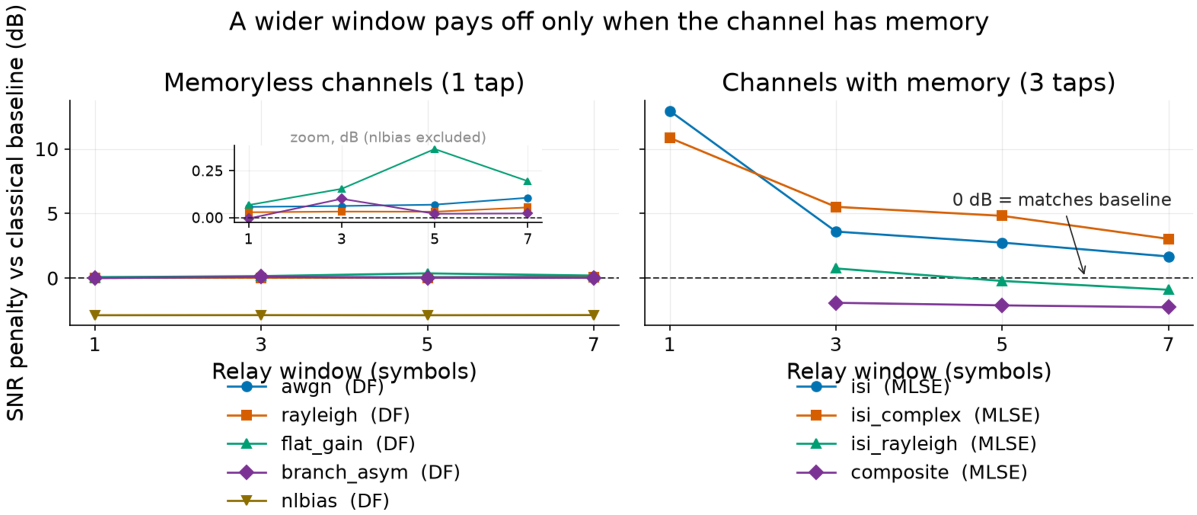

Figure 9.1: SNR penalty against each channel's classical comparator as the relay window widens, for the best configuration at each width. The reader should observe that the two panels behave oppositely: on the memoryless channels the penalty is flat to within a fifth of a decibel across every window (see inset), so the window buys nothing, while on the three-tap channels it falls by 8 to 11 dB between window 1 and window 7. This is the sense in which the relay's window must be matched to the channel's impulse response

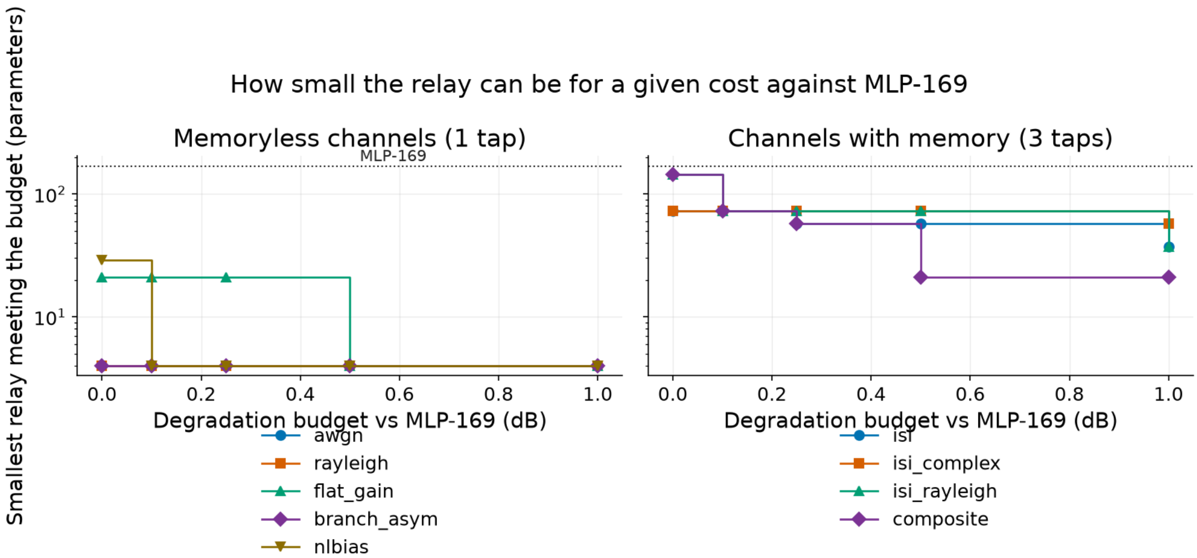

Figure 9.2: Smallest relay meeting a given degradation budget against MLP-169, drawn as a step function because the horizontal axis is a budget the designer chooses rather than a measured quantity. Most memoryless cases reach four parameters once a small degradation is allowed, while strict zero-degradation budgets require larger models for flat gain and nonlinear bias; the three-tap channels need between 21 and 145 depending on the permitted degradation

Table 9.5: Spread in SNR penalty between the best and worst of three initializations at an identical configuration, summarized over all configurations swept on each channel. The reader should observe that on five of the nine channels the median spread is comparable to or larger than the entire effect of varying model size, so a single-initialization size comparison on those channels measures the draw rather than the architecture.

Channel	Configs	Median (dB)	90th pct. (dB)	Max (dB)
rayleigh	17	0.008	0.019	0.028
composite	16	0.036	0.165	0.195
awgn	17	0.055	0.116	0.138
isi_rayleigh	16	0.177	0.588	0.795
isi	18	0.195	0.705	0.996
isi_complex	17	0.332	2.325	3.182
branch_asym	17	0.388	0.892	1.013
flat_gain	17	0.622	1.000	1.250
nlbias	17	1.231	1.242	1.245
All	152	0.180	1.229	3.182

Table 9.6: Spread in SNR penalty against symbol-wise DF between the best and worst of three initializations, for the equal-budget architectures on the canonical channel. The reader should observe the Transformer row: its spread is five times the roughly 0.18 dB that H4's 4.1% high-SNR architecture spread corresponds to, so a single run of that architecture cannot resolve the effect H4 claims. Every other architecture is comfortably inside it

Architecture	Params	Per-seed penalty vs DF (dB)	Spread (dB)
Mamba2-3K	3,004	$+1.113$, $+1.094$, $+1.098$	0.019
Mamba-S6-3K	3,027	$+1.123$, $+1.110$, $+1.082$	0.041
MLP-3K	3,004	$+0.133$, $+0.162$, $+0.189$	0.055
VAE-3K	3,037	$+0.072$, $+0.228$, $+0.142$	0.156
Transformer-3K	3,007	$+2.039$, $+1.147$, $+1.133$	**0.906**

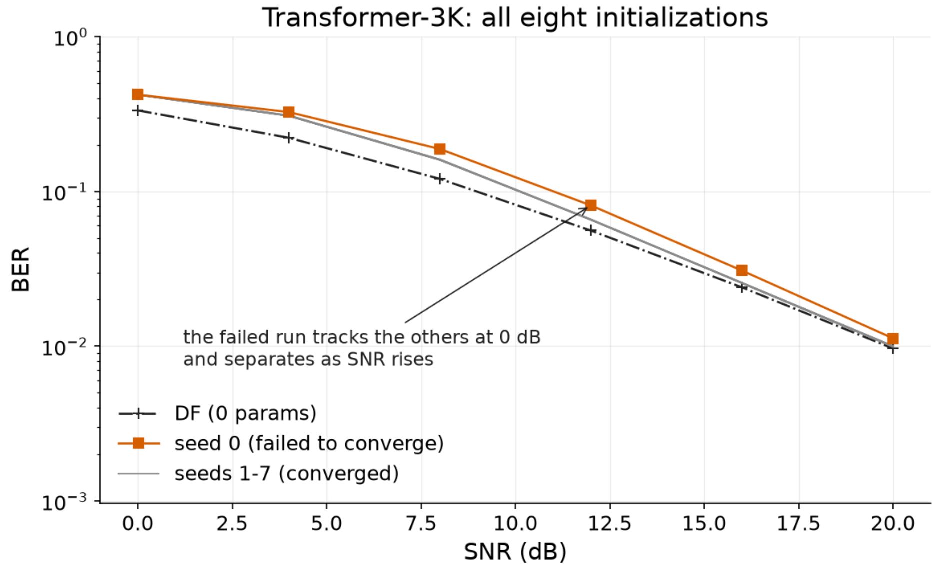

Figure 9.3: Transformer-3K on the canonical channel across eight initializations, against symbol-wise DF. The reader should observe that the failing run is not offset by a constant: it tracks the others at 0 dB and separates as SNR rises, so an evaluation confined to low SNR would not distinguish it from the seven that converge

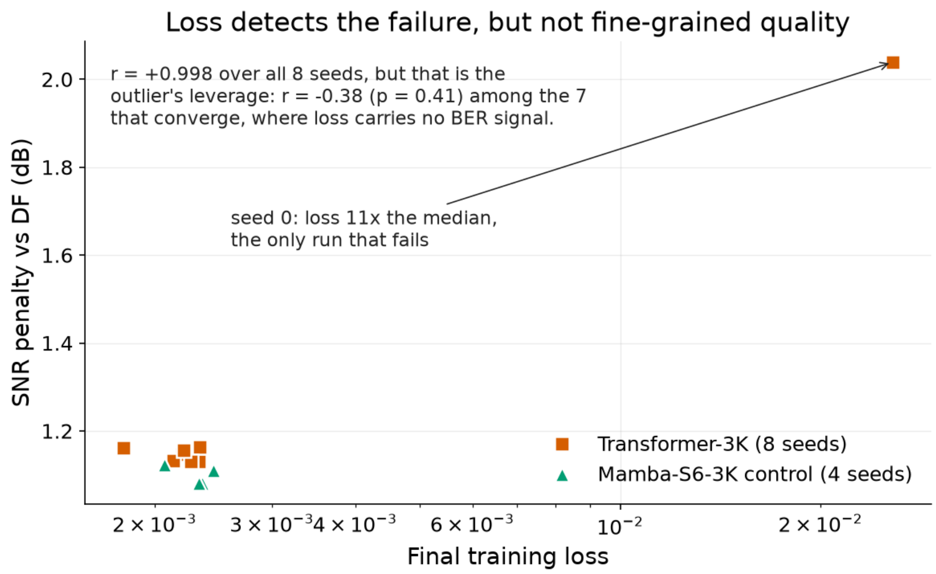

Figure 9.4: Final training loss against SNR penalty for the eight Transformer initializations and the four-seed Mamba-S6 control. The correlation over all eight seeds is $r = +0.998$, but that figure is the outlier's leverage: among the seven runs that converge it is $r = -0.38$ ($p = 0.41$), so loss identifies the failure and carries no information about relative quality once training has succeeded

9.6 Appendix F: Verification of Headline Claims

Each headline claim of Chapter 6 is restated here with the argument or derivation that supports it and the primary reference against which it was checked. The channel used throughout is the normalized 3-tap FIR $\mathbf{h} = [1, 0.7, 0.5]/\|\cdot\| \approx [0.758, 0.531, 0.379]$.

9.6.1 Matched-receiver bounds and the empirical MLSE comparison

The bound here is the bit-wise MAP detector, computed from the per-bit posteriors a BCJR/APP recursion delivers, and the distinction from MLSE is not pedantic: bit-wise MAP minimizes *bit* error probability (for BPSK it coincides with symbol-MAP; for Gray-coded QPSK it is the per-bit marginal of the APPs), so no learned function of the same observation can improve on it on this thesis's BER metric, whereas MLSE minimizes *sequence* error probability and is *not* BER-optimal [28, 29, 27]. Against genie-CSI MLSE the statement that the relay never wins is therefore an *empirical* finding of Chapter 6, not a theorem – a learned relay beating Viterbi on BER would not contradict any optimality result. The BER-optimal comparison at QPSK (Section 6.2.3) confirms it holds there too: BCJR/APP is never worse than MLSE, with the advantage unresolvable above 16 dB; it remains unmeasured at BPSK, though the same I/Q factorisation applies there by construction. The learned-receiver literature is consistent with this reading, its reported wins over "classical" receivers all being wins over mismatched or suboptimal baselines: the sliding bidirectional RNN passes full-CSI Viterbi only when CSI is imperfect [40]; ViterbiNet gains only under CSI uncertainty [37]; DeepRx outperforms LMMSE chains that are not MAP-optimal under the studied impairments [41]; Ye et al.'s OFDM detector [7] beats MMSE under pilot shortage, cyclic-prefix removal and clipping. End-to-end autoencoders [42] beat classical *schemes* by redesigning the transmitter, not as a receiver-only counterexample.

9.6.2 The analytic 0.25 floor and the non-monotonicity of symbol-wise DF

With the normalized taps, the interferer magnitude sum $h_1 + h_2 \approx 0.910$ exceeds the cursor $h_0 \approx 0.758$: the eye is closed. Of the four equiprobable BPSK interferer sign patterns, exactly one (both opposing the cursor) yields a noise-free sample of $0.758 - 0.910 = -0.152 < 0$, a deterministic sign flip, while the remaining three (0.606, 0.910, 1.668) preserve the sign – so a memoryless slicer errs on exactly one pattern in four as noise vanishes, giving the analytic floor of exactly $1/4$. The non-monotonicity follows from the same geometry: at moderate SNR, noise occasionally pushes the flipped sample back across the decision boundary, so the conditional error rate on that pattern rises toward one as SNR grows and total BER climbs toward 0.25 from below – standard behaviour for a closed-eye channel under symbol-by-symbol detection [43, 27].

9.6.3 Complexity: the M^L trellis against a fixed forward pass

MLSE maintains M^{L-1} trellis states and evaluates M^L branch metrics per symbol [28, 27]. The deployed relay, an 11-sample window into 13 hidden units ($11 \to 13 \to 1$, 170 parameters), costs approximately $2(11 \cdot 13 + 13)$ multiply–accumulates plus activations, about 330 floating-point operations per symbol, constant for the deployed architecture – though spanning longer memory generally widens the window (Section 6.5), after which the cost grows roughly linearly in it. For fairness: reduced-state sequence estimation [38] and related decision-feedback trellis methods also break the M^L scaling, so full Viterbi is the steepest classical comparator, not the only one.

9.6.4 The blind regime: CMA converges where decision-directed MLSE does not

The constant modulus algorithm performs blind equalization of constant-modulus signals [44, 45]. The study channel is minimum phase (zeros at $|z| \approx 0.707$), so a short FIR equalizer can approximately invert it; finite equalizer length and noise enhancement leave a residual BER of order 10^{-3} at 20 dB, consistent with the measured 3.3×10^{-3} of Section 6.4, and known ill-convergence of Godard equalizers when too short [46] matches the observed "matches but does not excel" behaviour. Decision-directed blind MLSE, a crude form of per-survivor processing that bootstraps channel estimates from its own decisions [47], has documented failure modes – misconvergence, sign and shift ambiguities, error propagation – matching the instability observed.

9.6.5 The pilot-budget crossover: reliable at ten pilots, collapse at five

Least-squares estimation of three unknown taps is identifiable from five pilots, so the collapse of Section 6.3 is not one of strict non-identifiability: convolution edge effects instead leave a near-square, ill-conditioned system whose least-squares error variance, proportional to $\sigma^2 \operatorname{tr}\big((\mathbf{X}^H\mathbf{X})^{-1}\big)$, grows without useful bound, and Viterbi run with badly wrong taps propagates errors catastrophically – consistent with the Cramér–Rao bound $\sigma^2 L/N_p$ for this estimation problem [48]. The result is specific to the classical LS-plus-Viterbi pipeline: meta-learned demodulators adapt from very few pilots [49], which bounds the claim's scope to the pipeline stated rather than contradicting it.

9.6.6 Genie-CSI MLSE leads the minimal relay by only 1–1.5 dB

A finite-window symbol-wise detector cannot do better than the MAP detector over that same window, which in turn cannot do better than the MAP detector given the whole received sequence: the window truncates the observation, and truncation cannot add information. The ordering is $P_e^{\text{MAP,seq}} \leq P_e^{\text{MAP,window}} \leq P_e^{\text{any windowed detector}}$. The size of the finite-window penalty is channel-dependent and is not derived analytically here. That ordering is all theory supplies here – how large the resulting penalty is on this channel is a quantitative question this thesis does not answer analytically, since no windowed-MAP detector was implemented, so the 1–1.5 dB lost to genie-CSI Viterbi is not decomposed into the window's share and the 170-parameter approximation's. Published learned detectors close the gap further: the sliding bidirectional RNN comes within fractions of a dB of Viterbi [40], and ViterbiNet closes it essentially to zero given sufficient capacity and data [37]. The residual gap in Section 6.2 is thus a property of the deliberately minimal parameter budget, not of learned detection as a method.

אוניברסיטת תל-אביב

הפקולטה להנדסה ע"ש איבי ואלדר פליישמן

בית הספר להנדסת חשמל

ארכיטקטורות למידה עמוקה לתקשורת ממסר דו-קפיצתית: מחקר השוואתי של אסטרטגיות ממסר קלאסיות ומבוססות רשתות נוירונים

חיבור זה הוגש כעבודת גמר לקראת התואר "מוסמך אוניברסיטה"

בהנדסת חשמל ואלקטרוניקה באוניבסיטת תל-אביב

על-ידי

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העבודה נעשתה בבית ספר להנדסת חשמל

בהדרכת ד"ר אנטולי חינה ופרופ' ראג'א ג'יריס

ניסן תשפ"ו

תקציר

ממסר מעביר תעבורה בין מקור ליעד שאין ביניהם נתיב ישיר. תזה זו שואלת שאלה אחת: מתי כדאי להשתמש בממסר *נלמד* במקום בממסר קלאסי, ומה קובע את הגבול ביניהם?

ההשוואה נערכת על תצורה אחת, קבועה מראש: קישור כניסה יחידה ויציאה יחידה (SISO) עם ממסר בודד, דעיכת ריילי מהירה בלתי־תלויה ושווה־התפלגות בשני הדילוגים, פס־בסיס מרוכב, אפנון QPSK מקודד־גריי ותמסורת בלתי־מקודדת, כאשר פונקציית הממסר היא המשתנה היחיד. שמונה אסטרטגיות מוערכות, מהגבר־והעבר (AF) ופענח־והעבר סימבולי (DF) ועד מודלים גנרטיביים ומודלי רצפים, כאומדני מונטה קרלו עם רווחי סמך של 95%.

בערוץ תואם וחסר־זיכרון הממסר הקלאסי מנצח. ממסרים נלמדים עולים על AF ביחס אות־רעש (SNR) נמוך, אך מ־6 dB ומעלה DF משתווה לכל ממסר נלמד או עולה עליו ללא עלות פרמטרים. הקיבולת אינה האילוץ הכובל: סריקת גדלים, בשלושה אתחולים לכל תצורה, מאתרת את הממסר הקטן ביותר שאינו כרוך בעלות מדידה, ומראה שזיכרון הערוץ, ולא מספר הפרמטרים, הוא הקובע אותו. ארבעה פרמטרים מספיקים בערוץ חסר־הזיכרון, בעוד ערוצים בעלי שלושה מקדמים דורשים 73 עד 145 פרמטרים וחלון הפורש את ההפרעה. השגיאה אינה עולה מעבר לפיזור שבין האתחולים עם גידול הקיבולת.

התמונה מתהפכת כאשר הערוץ אינו ידוע. מול מסנן ISI תלת־מקדמי בלתי־ידוע שני הממסרים הקלאסיים חסרי־הזיכרון נכשלים, כאשר שיעור השגיאה של DF עולה עם עוצמת השידור, בעוד ממסר נלמד ממוסגר־חלון בן 170 פרמטרים משקם ממסור אמין עד כדי 1 עד 1.5 dB מגלאי Viterbi שקיבל את המקדמים המדויקים. שני גבולות נמדדים מאתרים את נקודת המעבר. הראשון הוא אמינות אומדן הערוץ: בנקודת עבודה של 10 dB, גילוי קלאסי נעזר־פיילוטים נשאר מוביל בעשרים פיילוטים ומאבד את היתרון בעשרה. השני הוא אריתמטי: גילוי רצף עולה $2M^L$ פעולות כפל־צבירה לכל סימבול מול $2WH + 4H$ של הממסר הקבוע, השניים שווים בסביבות שלושה מקדמים וחצי עבור הארכיטקטורה שנפרסה כאן וגדולים פי 158 בשבעה.

מכאן שהממסר הנלמד לעולם אינו טוב יותר ממקלט קלאסי שניתן לו המודל הנכון. ערכו הוא בכך שהוא ממשיך לפעול כאשר מודל כזה, או אומדן אמין שלו, אינו זמין, ובכך שעלותו אינה גדלה עם זיכרון הערוץ. שתי הקביעות תחומות לצינורות הקלאסיים שמומשו כאן ולמשפחת הפגמים שעליה אומן הממסר.